

Investment in Demand and Dynamic Competition for Customers

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Abstract. The allocation of demand across firms determines aggregate productivity. Firms affect allocations through prices and non-price investment in demand. We develop a framework with search frictions and dynamic customer relationships in which demand investment shapes allocations directly by matching customers to suppliers and indirectly by driving price competition. A quantitative version matches key facts on how firms compete for customers. In equilibrium, markup distortions and business-stealing externalities induce misallocation and demand over-investment. Equilibrium interactions between demand investment and pricing create policy complementarities. Rising demand investment can raise concentration without raising market power, while reducing firms' intangible value through equilibrium effects.

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1 Introduction

The allocation of economic activity across firms is a central determinant of aggregate productivity and output (Restuccia and Rogerson, 2008; Hsieh and Klenow, 2009). In markets with imperfect competition, heterogeneous markups distort both the level of production and the allocation of demand across firms (Baqaee and Farhi, 2020; Edmond, Midrigan and Xu, 2023). Most theories of misallocation treat demand as static, with imperfect competition arising from product differentiation. Recent work instead emphasizes that firms accumulate demand dynamically through non-price actions, e.g. by investing in sales and marketing to build their customer base (Fitzgerald, Haller and Yedid-Levi, 2024; Argente, Fitzgerald, Moreira and Priolo, 2025; Afrouzi, Drenik and Kim, 2025). This *investment in demand* now rivals outlays on physical capital, is the largest component of intangible investment, and reflects efforts to contact and inform customers about firms’ products (He, Mostrom and Sufi, 2025). Firms’ willingness to invest in expanding their customer base aligns with theories in which product-market search frictions generate market power (Menzio, 2024, 2026).

How does investment in demand shape allocative efficiency and market power under product-market search frictions? And what does the rise in demand investment since the 1980s imply for industry concentration, market power, productivity, and intangible capital? We answer these questions in a novel framework with dynamic demand in which firms’ demand investment endogenously shapes market power and pricing. A quantitative version replicates key empirical patterns of how firms compete for demand. The framework captures two sources of misallocation: markup distortions and business-stealing externalities that induce over-investment in demand. The interactions between demand investment and pricing create complementarities between policy instruments that target over-investment in demand and markup distortions. Calibrating the model to the rise in demand investment since the 1980s, we show that declining costs of contacting customers contribute positively to GDP growth, raise concentration without raising market power, and reduce the intangible value of firms’ customer base through equilibrium effects.

We model competition for customers in general equilibrium with households, final good producers, and intermediate good producers. Households choose to supply labor and physical capital. Final good firms combine imperfectly substitutable intermediates into a final good. They are the customers in the market for intermediates. Intermediate producers—the suppliers—are heterogeneous in stochastic productivity, set prices, and invest in contacts with potential customers. The market for intermediates is subject to search frictions: a customer must match with a supplier to procure inputs and can only purchase from suppliers it is in contact with. Upon meeting a supplier, customers observe only its price and decide

whether to start purchasing. Matches are persistent but can be unilaterally dissolved at any time, and customers can be poached by other suppliers. For each intermediate input, a final good firm contracts with at most one supplier. Substitution across intermediate inputs allows variation in demand at the intensive (sales per customer) margin, in addition to variation at the extensive margin (number of customers).

We solve for a Bayesian Markov equilibrium in which customers form beliefs about their suppliers' payoff-relevant state from observed prices. We focus on separating equilibria where prices are strictly decreasing in productivity. Assuming constant returns to scale in the technology for contacting customers, a firm's pricing decision does not depend on its existing customer base. In equilibrium, heterogeneous markups arise from firms' demand investment decisions. When setting prices, firms internalize that their customers may be contacted by competitors. Greater demand investment by rivals increases the elasticity of demand, inducing firms to set lower prices and markups relative to the static optimum. Markups are heterogeneous, reflecting how strongly each firm is threatened by competitors given its production costs.

The framework matches salient empirical facts recently emphasized in work on how firms compete for demand. First, it matches the relationship between markups and firm size (Edmond et al., 2023). Second, it does so by generating a positive relation between sales per customer and markups, but no relation between markups and the number of customers (Afrouzi et al., 2025). Third, it replicates the relative contribution of the number of customers versus sales per customer to overall sales variation across firms (Bernard, Dhyne, Magerman, Manova and Moxnes, 2022; Einav, Klenow, Levin and Murciano-Goroff, 2021), which informs demand elasticities at both margins. Fourth, firms choose to expand their customer base through non-price actions while prices are insensitive to customer base size, consistent with evidence on market share growth (Fitzgerald et al., 2024; Argente et al., 2025). This is despite the number of customers being responsive to prices (Paciello, Pozzi and Trachter, 2019; Roldan-Blanco and Gilbukh, 2021), making prices allocative at the extensive margin.

We assess efficiency by comparing the social planner's allocation to the decentralized equilibrium. The planner faces the same technological constraints and search frictions but can directly assign contact rates, match formation, and quantities within each match. We show analytically that the framework nests standard markup distortions as in, for example, Edmond et al. (2023): the aggregate markup drives a wedge in households' factor supply condition, while dispersion in markups distorts the intensive margin allocation of demand across intermediate inputs. Here, markups arise due to customers' inability to access all suppliers simultaneously. In contrast to the intensive margin, markups do not distort match formation at the extensive margin as in Menzio (2024, 2026): customers always switch to

more productive suppliers when matched, as their value of a match increases monotonically with productivity.

Demand investment is generally inefficient. Two externalities operate in opposite directions. First, suppliers under-invest because they bear the full cost of contacts but do not internalize the benefit of matches to their customers. Second, they over-invest because demand investment diverts customers from competitors, generating business-stealing externalities. An increase in a supplier’s demand investment raises the probability of poaching customers from competitors within the period. In addition, the induced change in the distribution of matches affects competitors’ future customer growth and entry, generating dynamic business-stealing effects only captured in a framework with persistent customer-supplier relationships and supplier-to-supplier transitions.

Quantitatively, business-stealing externalities dominate the benefit of matches to customers, and the social planner would reduce aggregate demand investment by 37% relative to the calibrated equilibrium. Dynamic externalities due to persistent customer relationships are quantitatively more important than static forces. The magnitude and sign of the inefficiency is heterogeneous across suppliers. The positive externality on customers continues to grow in the right tail of the productivity distribution, whereas the business-stealing externality is bounded. Accordingly, while the planner would cut demand investment for most suppliers, it would raise contacts with customers for the 2% most productive firms. Eliminating the demand investment inefficiency raises welfare by about 1.0% in consumption equivalent variation along the transition from the decentralized equilibrium to the planner’s allocation. Both the customer-surplus and business-stealing forces behind this result are in the range of independent estimates from supplier relationships and product markets.

The framework allows us to revisit the quantitative cost of misallocation from markups. Eliminating the aggregate markup raises welfare by 1.9% along the transition. Eliminating distortions from markup dispersion adds another 2.2%. We find small welfare losses for a given distribution of markups because we separate the *source* of market power from the parameter governing its *cost*. In models of endogenous market power based on product differentiation (Kimball, 1995; Atkeson and Burstein, 2008), the elasticity of substitution across products determines both the distribution of markups and the cost of a given dispersion. In our model, markups arise from search frictions and are primarily determined by the elasticity of demand at the extensive margin, the competition for customers. The cost of markup dispersion is instead governed by the elasticity of substitution at the intensive margin, the elasticity of demand per customer, which is identified from variation in sales per customer and empirically small. Separating the source and cost of markups reduces the misallocative effects of markup dispersion, by reconciling an empirically plausible, small

elasticity of substitution across inputs (Carvalho, Nirei, Saito and Tahbaz-Salehi, 2021; Peter and Ruane, 2025) with observed markups. If instead we counterfactually imposed a higher elasticity of substitution consistent with the aggregate markup, the productivity loss from markup dispersion would rise from 1.6% to about 5.4%. Taken together, eliminating aggregate markups, removing markup dispersion, and correcting demand-investment inefficiencies would raise welfare by 5.1% when accounting for transitional dynamics.

We solve for the welfare-maximizing product market policy and uncover a complementarity between a demand-investment tax and a production subsidy. We restrict attention to constant, flat versions of each instrument. The joint optimum features a 61.1% tax on demand investment and a 17.1% production subsidy, raising welfare by 2.5% along the transition. Each instrument on its own is weaker: the optimal demand-investment tax is 33.8% in isolation, while the optimal production subsidy is 13.3% in isolation. The complementarity reflects the two-way interaction between demand investment and markups. A production subsidy raises profits per customer and thus stimulates additional (inefficient) demand investment, limiting the optimal subsidy in isolation. Conversely, a demand-investment tax reduces competitive pressure in prices, which amplifies distortions from markups. Without a subsidy to offset these additional distortions, the optimal tax is substantially lower.

Finally, we use the model to quantify the macroeconomic consequences of the rise in demand investment since the 1980s. We calibrate a change in the technology for contacting customers to match the rise in demand investment and within-industry sales concentration. The implied decline in the cost of contacting customers raises GDP by 4.6% relative to 1980. The accompanying increase in concentration occurs without higher market power and therefore does not raise the misallocative costs of imperfect competition. Aggregate markups change little, reflecting offsetting within-firm and composition effects: stronger competition lowers firms' markups, while reallocation toward more productive, high-markup firms raises the aggregate markup (e.g., Autor, Dorn, Katz, Patterson and Van Reenen, 2020; De Loecker, Eeckhout and Unger, 2020).¹ Despite higher demand investment, the aggregate intangible value of firms' customer base declines, because greater demand investment by competitors raises turnover and lowers the value of each customer relationship in equilibrium. Instead, capitalizing demand investment at a constant depreciation rate yields rising measured intangible values in both the model and the data. At the same time, firm values fan out across firms, consistent with evidence of widening dispersion in firm values and profits (Eeckhout, 2025). These results caution against using industry concentration as a measure of imperfect competition or inferring intangible values by capitalizing spending under fixed depreciation.

¹If part of rising concentration is instead attributed to higher productivity dispersion, the same framework can also accommodate rising aggregate markups.

Related literature. Our analysis builds on the literature on welfare losses from misallocation (Restuccia and Rogerson, 2008; Hsieh and Klenow, 2009), with particular emphasis on distortions arising from markups (Baqae and Farhi, 2020; Peters, 2020; Edmond et al., 2023; Baqae, Farhi and Sangani, 2024a,b; Bornstein and Peter, 2025). Previous work typically generates market power from product differentiation and considers only an intensive margin of demand. A small number of papers have introduced an extensive margin and investment in customers: Cavenaile, Celik, Perla and Roldan-Blanco (2023) develop a theory of customer awareness and targeted advertising. Cavenaile, Celik, Roldan-Blanco and Tian (2025) and Pearce and Wu (2025) allow for an extensive margin under oligopolistic competition for sales per customer. Afrouzi et al. (2025) document that markups covary with the intensive rather than the extensive margin, highlighting the importance of accounting for both separately, and study implications for misallocation under monopolistic competition at the intensive margin. These papers share the assumptions that market power arises from competition among differentiated products and that customers do not decide on matches based on price. In contrast, we allow for competition at the extensive margin among perfect substitutes in a market with search frictions. These departures make the number of suppliers' matches responsive to prices, separate the source and cost of market power, and capture static and dynamic business-stealing externalities—features that are central to understanding market power, the effects of demand investment, and misallocation in our setting.

Our approach to modeling investment in demand and dynamic competition for customers relates to the literature on random search in product markets, building on Butters (1977). Dinlersoz and Yorukoglu (2012) analyze an industry with fully transitory productivity shocks, Greenwood, Ma and Yorukoglu (2024) study digital advertising, and Paciello et al. (2019) show how customers' search effort shapes the pass-through of productivity shocks to prices. Shen (2025) develops an industry-dynamics model with permanent firm heterogeneity in which a rise in marketing rationalizes declining business dynamism, with congestion in matching as a key source of inefficiency. We deviate from previous work by embedding demand investment into a general-equilibrium framework with persistent productivity shocks, an intensive margin of demand, and customer poaching from existing supplier relationships—features that let us quantify both the allocative efficiency of demand investment and its interaction with the misallocation costs of market power. Menzio (2024, 2026) establish efficiency as a benchmark in a static random search model with unit demand and sellers unable to influence the number of customers they contact, as in Burdett and Judd (1983). In related frameworks, Menzio (2023) studies optimal product specialization and Albrecht, Menzio and Vroman (2023) show how endogenous productivity dispersion amplifies market concentration under declining search frictions. Their findings complement our

results on the role of increasing demand investment for market concentration; we focus on implications for markups and the intangible value of firms’ customer base.

Our result of inefficient demand investment relates to a broader literature on efficiency in random search markets (e.g. Mortensen, 1982; Hosios, 1990; Fukui and Mukoyama, 2025), where efficiency of contacts requires that actors are appropriately compensated for their contribution to matching. Work on labor markets typically captures this tradeoff by imposing a matching function and determining payoffs via bargaining or sequential auctions, emphasizing congestion of contacts as the key externality. We instead study efficiency under Poisson matching and price posting, allow existing matches to raise meeting rates, and—unlike related labor search models (Moscarini and Postel-Vinay, 2013; Coles and Mortensen, 2016; Gouin-Bonenfant, 2022; Gottfries and Jarosch, 2025)—account for an intensive margin of demand within a match. Together, these features capture important aspects of product markets, highlight the interaction between contact rates and misallocation from markups, and make business-stealing of existing matches a central driver of inefficiency.

Further related work examines frictions in, or the cost of, attracting customers in business cycle dynamics (Bai, Rios-Rull and Storesletten, 2026; Fernández-Villaverde, Mandelman, Yu and Zanetti, 2024), international trade (Arkolakis, 2010; Drozd and Nosal, 2012), R&D (Cavenaile and Roldan-Blanco, 2021; Ignaszak and Sedláček, 2023), product variety expansion (Baslandze, Greenwood, Marto and Moreira, 2023), firm growth (Roldan-Blanco and Gilbukh, 2021; Boehm, South, Oberfield and Waseem, 2024), and price dynamics (Rudanko, 2025). In a seminal contribution, Gourio and Rudanko (2014) study the role of intangible customer capital for firms’ physical investment dynamics in a directed-search model. Chiavari (2024) emphasizes the role of intangible assets for macroeconomic trends in a customer-capital model in which intangible investment raises firms’ returns to scale in production, as commonly assumed in the literature (Crouzet, Eberly, Eisfeldt and Papanikolaou, 2022, 2024; De Ridder, 2024). In contrast, we focus on the role of intangible investment in demand for competition and industry dynamics, motivated by He et al. (2025), who show that sales and marketing expenditures account for a significant share of intangible investment.

Outline. Section 2 presents the theoretical framework and characterizes an equilibrium. Section 3 describes the quantitative implementation. Section 4 reports the main results on demand investment and misallocation. Section 5 analyzes the macroeconomic effects of rising demand investment. Section 6 concludes.

2 A Theory of Competition for Customers

We study investment in demand and competition for customers in general equilibrium of an economy with households, final good producers, and intermediate good producers. Our focus is on the exchange for intermediate inputs between the two types of firms, which is subject to search frictions and requires suppliers of intermediates to invest in contacting potential customers. This section outlines the environment, the optimization problem for each type of agent, as well as the definition and properties of an equilibrium.

2.1 The Environment

Preferences. Time is discrete and indexed by t . An infinitely-lived representative household values consumption of a final good, C_t , and gets disutility from labor L_t , with preferences

$$(1) \quad \sum_{t=0}^{\infty} \beta^t [u(C_t) - v(L_t)].$$

$u(\cdot)$ is increasing and concave, $v(\cdot)$ is increasing and convex, and $0 < \beta < 1$ discounts time.

Production. The final good is produced by a unit continuum of identical final good firms using intermediate goods indexed by $j \in [0, 1]$.² Each final good firm operates technology

$$(2) \quad Y_t = \left(\int_0^1 y_{jt}^{\frac{\sigma-1}{\sigma}} dj \right)^{\frac{\sigma}{\sigma-1}},$$

where y_{jt} is the quantity of intermediate j used in period t and $\sigma > 1$ is the elasticity of substitution between intermediates.³ Each intermediate j is produced combining physical capital k_{jt} and labor ℓ_{jt} with Cobb–Douglas technology

$$(3) \quad y_{jt} = z_{jt} k_{jt}^{\alpha} \ell_{jt}^{1-\alpha}.$$

z_{jt} is total factor productivity, $\alpha \in (0, 1)$ is the capital share in production, and physical capital depreciates at rate δ_K . Two types of firms operate versions of the intermediate technology with different productivity. Final good producers operate a symmetric *in-house* technology for all intermediate goods j with productivity $z_{jt} = 1 \ \forall j$. In addition, each

²A formulation with households as CES consumers of a bundle of varieties would deliver similar insights. Our choice to model the final good as a bundle of intermediates is reflected in our calibration strategy below, which targets moments for business-to-business transactions.

³The model generally admits a well-defined solution for $\sigma \leq 1$. Infinite markups are not optimal even as $\sigma \downarrow 1$ due to the extensive margin of demand, in contrast to models of monopolistic competition where demand is determined only by a CES-aggregator. We restrict attention to $\sigma > 1$ for comparison.

intermediate input j is produced by a continuum of intermediate good producers, each of which operates the technology for a single intermediate good j with idiosyncratic productivity $z_{jt} = z$. Idiosyncratic productivity z evolves stochastically according to a discrete-time Markov process with conditional CDF $F(z'|z)$ on $(0, \infty)$. We assume that the transition kernel is continuous in both arguments and that for any $z_1 < z_2$ and any z' , $F(z'|z_2) \leq F(z'|z_1)$ with strict inequality for some z' ; i.e., $F(\cdot | z_2)$ first-order stochastically dominates $F(\cdot | z_1)$.

Matching. The economy is subject to search frictions between final good and intermediate good producers. To be able to trade an intermediate input, final good producers (*customers*) must match with an intermediate good producer (*supplier*). Customers can have either no supplier (*unmatched*) or one supplier (*matched*) per intermediate input. We assume that suppliers are always in contact with their customers from the previous period and may contact new customers before production takes place in any period. Contacting new customers is costly for suppliers. To contact $\hat{n}$ new customers, a supplier with n existing customers incurs cost $\hat{s}(\hat{n}, n)$, paid in the final good. We further assume that the cost function satisfies $\hat{s}(\hat{n}, n) = s\left(\frac{\hat{n}}{n}\right)n$, where $s(\cdot)$ is strictly increasing and strictly convex.⁴ Suppliers contact new customers at random and with equal probability, regardless of whether the customer is currently unmatched or already matched. We assume that customers cannot observe the productivity z and existing customers n of a supplier even after being contacted and both customers and suppliers cannot recall previous matches or contacts.

Entry and exit. Intermediate good producers enter and exit the economy. Each period, a fraction $\chi \in (0, 1]$ of previously unmatched customers is contacted by a potential entrant.⁵ After contacting a potential customer, entrants draw their initial productivity from a distribution $F_0(z)$, and choose whether or not to enter the economy with one initial customer. We assume that $F_0(\cdot)$ is continuous and has strictly positive density throughout its support, so that the distribution is connected (i.e., contains no gaps). At the end of each period, intermediate good producers exit exogenously with probability $\delta_F \in [0, 1]$. In addition, intermediate producers may choose to exit endogenously. The customers of exiting suppliers

⁴These properties follow from a meeting technology that is constant returns to scale (CRS) in spending and existing customers. By construction, $\hat{s}(\lambda\hat{n}, \lambda n) = s\left(\frac{\lambda\hat{n}}{\lambda n}\right)\lambda n = \lambda\hat{s}(\hat{n}, n)$, so the cost function is CRS. Moreover, $\hat{s}(\hat{n}, n)$ is increasing and convex in $\hat{n}$ (from the properties of $s(\cdot)$) and decreasing in n , since $\frac{\partial \hat{s}}{\partial n} = s\left(\frac{\hat{n}}{n}\right) - s'\left(\frac{\hat{n}}{n}\right)\frac{\hat{n}}{n} < 0$ by strict convexity of $s(\cdot)$. Equivalently, the cost function can be expressed as a matching technology in n and cost paid $\hat{s}$: $\hat{n}(n, \hat{s}) = n s^{-1}\left(\frac{\hat{s}}{n}\right) \equiv n f\left(\frac{\hat{s}}{n}\right)$, with $f(\cdot)$ strictly concave. The matching technology is CRS since $\hat{n}(\lambda n, \lambda \hat{s}) = \lambda n f\left(\frac{\lambda \hat{s}}{\lambda n}\right) = \lambda \hat{n}(n, \hat{s})$ and increasing in both arguments. The contribution of existing customers to meetings with new customers can be interpreted as the benefit of visibility, word-of-mouth effects, or peer effects in product adoption, consistent e.g. with empirical evidence in Bailey, Johnston, Kuchler, Stroebel and Wong (2022) and Argente, Méndez and Van Patten (2024).

⁵One interpretation is that when a final good firm is unmatched and produces an intermediate input in-house, with probability χ an employee spins off a new company to supply their previous employer.

become unmatched, but cannot immediately be contacted by entrants.

2.2 The Problem of Households

Households consume the final good, choose how much labor L_t to supply, save in physical capital K_t , own all firms and receive profits Π_t . The final good is the numeraire, with its price normalized to one. Households' period t budget constraint is

$$(4) \quad C_t + K_{t+1} = w_t L_t + (1 + r_t) K_t + \Pi_t, \quad \forall t,$$

where w_t is the wage rate and r_t the net return on capital. Households choose $\{C_t, L_t, K_{t+1}\}_{t \geq 0}$ to maximize (1) subject to (4). The optimal solution satisfies

$$(5) \quad v'(L_t) = u'(C_t) w_t,$$

$$(6) \quad u'(C_t) = \beta(1 + r_{t+1}) u'(C_{t+1}).$$

2.3 The Problem of Final Good Producers

Final good producers choose a bundle of intermediates to produce final output with technology (2). Cost minimization delivers the demand for each intermediate input as

$$(7) \quad y_{jt} = \left(\frac{p_{jt}}{P_t} \right)^{-\sigma} Y_t,$$

where p_{jt} is the unit cost of input j , and

$$(8) \quad P_t = \left(\int_0^1 p_{jt}^{1-\sigma} dj \right)^{\frac{1}{1-\sigma}}$$

is the unit cost of the final good. The market for the final good is perfectly competitive and hence $P_t = 1$ and $y_{jt} = p_{jt}^{-\sigma} Y_t$ in any equilibrium.

Cost of intermediates. Final good producers either produce intermediate input j in-house or source it from a supplier they match with. If matched, we assume that the final good firm sources its entire demand for intermediate j at the supplier's price p . The cost of using intermediate j in production of the final good is therefore

$$(9) \quad p_{jt} = \begin{cases} p & \text{if matched with a supplier charging } p \\ \overline{mc}_t & \text{if not matched with a supplier} \end{cases}$$

where from cost-minimization

$$(10) \quad \overline{mc}_t = \left(\frac{r_t + \delta_K}{\alpha} \right)^\alpha \left(\frac{w_t}{1 - \alpha} \right)^{1-\alpha}$$

is the marginal cost of producing one unit of intermediate j in-house. The associated optimal per-unit factor usage is given by $\bar{\ell}_t = \left(\frac{(1-\alpha)(r_t + \delta_K)}{\alpha w_t} \right)^\alpha$ for labor and $\bar{k}_t = \left(\frac{\alpha w_t}{(1-\alpha)(r_t + \delta_K)} \right)^{1-\alpha}$ for capital. As the in-house technology is symmetric across intermediates, factor use and the cost of producing in-house is constant across j .

The static gain to the final good producer per intermediate procured at price p instead of produced in-house is a marginal reduction in total production cost $P_t Y_t$, given by

$$(11) \quad \eta_t(p) = \frac{p^{1-\sigma} - \overline{mc}_t^{1-\sigma}}{\sigma - 1} Y_t.$$

Value of a match. Final good producers are in contact with their previous suppliers for each intermediate and may be contacted by prospective new suppliers before production takes place. Final good firms in an existing match may terminate it and return to in-house production at any time, and they may rematch with a new supplier if contacted, i.e. they cannot commit to a long-term contract. Similarly, unmatched firms may accept or decline a match if contacted. As final good firms cannot observe a supplier's idiosyncratic state, they decide about acceptance and rejection of matches solely based on the price of a supplier, forming expectations about the path of future prices and the value of a match.

Define the value of being in a match with a supplier charging price p at the start of period t before suppliers contact additional customers as $M_t^i(p)$, and the corresponding value of being unmatched as U_t^i , such that

$$(12) \quad M_t^i(p) = M_t(p) + \int_{M_t(p)}^\infty (M - M_t(p)) d\hat{O}_t(M),$$

$$(13) \quad U_t^i = U_t + \int_{U_t}^\infty (M - U_t) d\hat{O}_t(M),$$

where $\hat{O}_t(M)$ is the distribution function of the highest value of a match offered among contacting suppliers, which includes the chance of not being contacted at all. Define the values of being matched with a supplier charging p $M_t(p)$ or being unmatched U_t at the start of production in period t as

$$(14) \quad M_t(p) = \max \left\{ \eta_t(p) + \frac{U_{t+1}^i}{1 + r_{t+1}} + \frac{(1 - \delta_F)}{1 + r_{t+1}} \int_0^\infty (M_{t+1}^i(p') - U_{t+1}^i) dH_{t+1}(p'|p), U_t \right\},$$

$$(15) \quad U_t = \frac{1}{1 + r_{t+1}} \left[U_{t+1}^i + \chi \int_0^\infty (M_{t+1}^i(p) - U_{t+1}^i) dH_{t+1}^0(p) \right],$$

where $H_{t+1}(p'|p)$ is the belief that a supplier charging p in period t will offer a match at or below price p' in $t + 1$, which includes the likelihood of endogenous exit.⁶ $H_{t+1}^0(p)$ is the corresponding belief about entry decisions and prices conditional on entry.⁷ Customers matched with a firm charging p_1 switch to a new supplier charging p_2 during rematching iff $M_t(p_2) \geq M_t(p_1)$. Unmatched customers accept a match iff $U_t \leq M_t(p)$. Matched customers dissolve a match before production iff $M_t(p) < U_t$.

2.4 The Problem of Intermediate Good Producers

We assume all intermediate goods are symmetric and omit the index j from the intermediate good producers' problem. Intermediate good producers are heterogeneous in their productivity z and in the number of customers n purchasing from them in the previous period.

Profit per customer. Intermediate good firms produce output y with technology (3) and their idiosyncratic productivity z . The marginal cost of producing one unit of output at productivity z follows from the cost-minimization problem,

$$(16) \quad mc_t(z) = \frac{1}{z} \left(\frac{r_t + \delta_K}{\alpha} \right)^\alpha \left(\frac{w_t}{1 - \alpha} \right)^{1-\alpha} = \frac{\overline{mc}_t}{z},$$

where $\overline{mc}_t$ captures the common component of marginal cost and optimal per-unit factor demands are $\ell_t(z) = \frac{\bar{\ell}_t}{z}$ and $k_t(z) = \frac{\bar{k}_t}{z}$. Final good producers' problem implies static demand $y_t(p) = p^{-\sigma} Y_t$ per matched customer for any supplier charging price p . Hence, the static profit per customer of an intermediate producer with productivity z and price p is

$$(17) \quad \pi_t(p, z) = (p - mc_t(z)) y_t(p) = \left(p - \frac{\overline{mc}_t}{z} \right) p^{-\sigma} Y_t.$$

Customer growth. We define $i = \frac{\hat{n}}{n}$ as the *contact intensity* at which a firm approaches new customers $\hat{n}$ relative to existing customers n . Let $e_t(p)$ denote the fraction of contacted customers that accept a match with a producer charging price p ; we refer to $e_t(p)$ as the *conversion rate*. Existing customers may also be approached by alternative suppliers. Let $q_t(p)$ denote the fraction of a firm's existing customers that accept a match with a competitor;

⁶Because customers can always dissolve a match after the rematching stage, and since matched and unmatched customers are contacted at the same rate, it follows that $M_t^i(p) \geq U_t^i$ for all p . We therefore abstract from the possibility of dissolving a match before rematching without loss of generality.

⁷For notation, we make the implicit assumption that customers do not recall past prices. This is without loss of generality in the equilibria we consider, as current prices fully reveal suppliers' current idiosyncratic state. Because price and entry/exit choices are sufficient to compute customer values, forming beliefs about choices is equivalent to beliefs of states. As $H_{t+1}(p'|p)$ and $H_{t+1}^0(p)$ include the belief about supplier exit and non-entry, they do not integrate to one but to the unconditional probability of non-exit or entry for given p .

we refer to $q_t(p)$ as the *quit rate*. Both $e_t(p)$ and $q_t(p)$ are equilibrium objects characterized below and depend only on p as customers cannot observe z or n . Contacts with new customers take place before production in each period. Given $e_t(p)$ and $q_t(p)$, i , p , and n , the total number of customers at the time of production and sales is

$$(18) \quad n' = n - q_t(p)n + e_t(p)\hat{n} = (1 - q_t(p) + e_t(p)i)n \equiv \rho_t(p, i)n$$

where $\rho_t(p, i)$ denotes the firm's net customer growth.

Dynamic problem. At the beginning of each period after productivity z is realized, intermediate good producers that decide to operate choose a price p and contact intensity i . We assume intermediate producers cannot commit to long-term contracts and cannot price discriminate among customers, and hence reset a single price every period.⁸ Both p and i affect not only current profits but also future profits through the evolution of the customer base, making both decisions dynamic. With future payoffs discounted at rate r_{t+1} , the intermediate good producers solve the recursive problem⁹

$$(19) \quad \tilde{V}_t(z, n) = \max_{p, i} \left\{ -s(i)n + \rho_t(p, i)n\pi_t(p, z) + \frac{1 - \delta_F}{1 + r_{t+1}} \mathbb{E} \left[\max \{ \tilde{V}_{t+1}(z', \rho_t(p, i)n), 0 \} | z \right] \right\}.$$

We refer to the total cost of contacting customers $s(i)n$ implied by the optimal i as firms' *demand investment*. $\tilde{V}_t(z, n)$ is a contraction mapping for increasing functions and $\pi_t(p, z)$ is strictly increasing in z , such that $\tilde{V}_t(z, n)$ is strictly increasing in z iff $n > 0$. The firm value scales linearly with n such that $\tilde{V}_t(z, n) = nV_t(z)$. The value per customer $V_t(z)$ depends only on the firm's productivity z and is given by

$$(20) \quad V_t(z) = \max_{p, i} \{ -s(i) + \rho_t(p, i)W_t(p, z) \}$$

where we define

$$(21) \quad W_t(p, z) \equiv \left(\pi_t(p, z) + \frac{1 - \delta_F}{1 + r_{t+1}} \mathbb{E} \left[\max \{ V_{t+1}(z'), 0 \} | z \right] \right)$$

as the value per customer at production, comprising current profit and continuation value. As $\tilde{V}_t(z, n)$ is strictly increasing in z , $V_t(z)$ is strictly increasing in z . Hence, suppliers choose to exit or not enter the market at the beginning of period t iff $z < \bar{z}_t$, where $V_t(\bar{z}_t) = 0$. From

⁸These assumptions make the model comparable to setups with static pricing (e.g. Edmond et al., 2023).

⁹Discounting profits at rate r_{t+1} is consistent with firm ownership by households. To ensure that $\tilde{V}_t(z, n)$ is well defined, we assume $\lim_{t \rightarrow \infty} \mathbb{E}_0 [Q_{0,t} \tilde{V}_t(z_t, n_t^{opt}) | z_0, n_0] = 0$, where $Q_{0,t} = \prod_{\tau=1}^t (1 + r_\tau)^{-1}$ and n_t^{opt} is the number of customers under optimal choices. This rules out infinite customer growth with positive probability and must hold in equilibrium, since the total mass of matched customers is bounded above by one.

the property $\tilde{V}_t(z, n) = nV_t(z)$, it follows that the optimal choices of p and i are independent of the number of existing customers n .

2.5 The Allocation of Customers and Aggregation

For each intermediate good, the distribution of matches between customers and suppliers of different productivity evolves endogenously in two stages: First, productivity shocks realize for existing matches; suppliers enter and exit. Second, suppliers contact additional customers and customers decide about matches. We detail both stages below and again omit subscript j as the distribution of matches evolves symmetrically across intermediates.

Market entry and exit. Recall from above that existing suppliers stay in the market and entrants decide to operate iff $z \geq \bar{z}_t$ and that entrants start operating with one initial customer. Let $G_t(z)$ denote the mass of customers matched with suppliers of productivity z or lower during production in period t , and u_t the mass of unmatched customers. As an intermediate step, let $G_t^i(z)$ and u_t^i denote, respectively, the corresponding masses of matched and unmatched customers after the exit and entry of suppliers in period t , defined by

$$(22) \quad G_t^i(z) = \begin{cases} \chi u_{t-1} (F_0(z) - F_0(\bar{z}_t)) + (1 - \delta_F) \int_0^\infty (F(z|\hat{z}) - F(\bar{z}_t|\hat{z})) dG_{t-1}(\hat{z}), & \text{if } z \geq \bar{z}_t, \\ 0, & \text{if } z < \bar{z}_t, \end{cases}$$

$$(23) \quad u_t^i = \left(1 - (1 - F_0(\bar{z}_t))\chi\right)u_{t-1} + (1 - \delta_F) \int_0^\infty F(\bar{z}_t|z) dG_{t-1}(z) + \delta_F(1 - u_{t-1}) = 1 - \lim_{z \rightarrow \infty} G_t^i(z).$$

Distribution of contacts. Suppliers generate a total of $I_t = \int_0^\infty i_t(z) dG_t^i(z)$ contacts with new customers, distributed across different productivity levels with distribution function $D_t(z) = \int_0^z \frac{i_t(s)}{I_t} dG_t^i(s)$. The distribution of the value of new contacts is then given as $\hat{D}_t(M) = \int_0^\infty \mathbb{I}_{M_t(p_t(z)) \leq M} dD_t(z)$. As suppliers contact new customers at random, the number of contacts per customer is distributed according to a Poisson distribution with mean I_t . This means that the probability of a customer being contacted by k additional suppliers is given by $\frac{I_t^k e^{-I_t}}{k!}$ and the probability that the best outside offer for a customer with k contacts has value M or less by $\hat{D}_t(M)^k$. From here we can derive the probability that a customer's best contact is not better than M as

$$\hat{O}_t(M) = \sum_{k=0}^\infty \hat{D}_t(M)^k \frac{I_t^k e^{-I_t}}{k!} = e^{-I_t(1 - \hat{D}_t(M))},$$

and that the highest productivity among contacting suppliers is not higher than z accordingly as $O_t(z) = e^{-I_t(1 - D_t(z))}$.

Rematching between customers and suppliers. Given the distribution of contacts and customers' optimal decision to accept the match with the highest value $M_t(p)$ among contacts, the updated mass $G_t(z)$ of matched customers at firms with productivity less than z and mass of unmatched customers u_t during production in period t are given by

$$(24) \quad G_t(z) = G_t^i(z) + u_t^i (O_t(z) - e^{-I_t}) - G_t^i(z) \sum_{k=0}^{\infty} \frac{I_t^k e^{-I_t}}{k!} \left(1 - \left(1 - \int_z^{\infty} \mathbb{I}_{M_t(p_t(z')) > M_t(p_t(z))} dD(z') \right)^k \right),$$

$$(25) \quad u_t = e^{-I_t} u_t^i = 1 - \lim_{z \rightarrow \infty} G_t(z),$$

where e^{-I_t} is the probability of not being contacted by any supplier.¹⁰

We can now define the customer quit rate $q_t(p)$ and conversion rate $e_t(p)$ as

$$(26) \quad q_t(p) = 1 - \hat{O}_t(M_t(p))$$

$$(27) \quad e_t(p) = \left[u_t^i + \int_0^{\infty} \mathbb{I}_{M_t(p) \geq M_t(p_t(z))} dG_t^i(z) \right] \hat{O}_t(M_t(p)).$$

Aggregation. As intermediate goods are symmetric, the distribution of matched customers $G_t(z)$ and unmatched customers u_t at production for a given intermediate good j corresponds to the distribution across intermediate goods for a given customer. In Appendix A.1 we show that aggregate output of the final good is given by

$$(28) \quad Y_t = \left(u_t \overline{mc}_t^{-\sigma} + \int_0^{\infty} \frac{1}{z} (p_t(z))^{-\sigma} dG_t(z) \right)^{-1} K_t^{\alpha} L_t^{1-\alpha}.$$

2.6 Equilibrium

Because customers in the market for intermediate goods observe only current prices, they must form expectations about suppliers' payoff relevant state z . We therefore adopt a Bayesian equilibrium concept, similar to Coles and Mortensen (2016), and restrict attention to Bayesian equilibria in Markov strategies.

Equilibrium Definition

A Bayesian equilibrium in pure Markov strategies consists of a path for prices $\{w_t, r_t\}_{t=0}^{\infty}$, aggregate quantities $\{\Pi_t, Y_t, C_t, L_t, K_{t+1}\}_{t=0}^{\infty}$; mass of customers across suppliers and mass of unmatched customers $\{G_t^i(z), u_t^i, G_t(z), u_t\}_{t=0}^{\infty}$; quit and conversion rates $\{q_t(p), e_t(p)\}_{t=0}^{\infty}$;

¹⁰We simplify notation by implicitly assuming that $G_t^i(z)$ has no mass points and that no operating supplier offers a match value below U_t . The first assumption holds in all equilibria we consider. The second assumption holds generally: Any supplier posting a price p such that $M_t(p) < U_t$ must anticipate that all previous customers will quit and no new customers will accept such an offer. Hence such a supplier will choose to exit in the beginning of a period and no contact with value below U_t will occur.

suppliers' policies, value function, and exit threshold $\{p_t(z), i_t(z), V_t(z), \bar{z}_t\}_{t=0}^\infty$; customer values $\{M_t^i(p), U_t^i, M_t(p), U_t\}_{t=0}^\infty$ and beliefs $\{H_{t+1}(p'|p), H_{t+1}^0(p)\}_{t=0}^\infty$, such that given initial conditions $G_{-1}(z), u_{-1}, K_0$ for all $t \geq 0$:

1. Given $\{\Pi_t, w_t, r_t\}_{t=0}^\infty$, consumption $\{C_t\}_{t=0}^\infty$, physical capital $\{K_{t+1}\}_{t=0}^\infty$ and labor $\{L_t\}_{t=0}^\infty$ maximize households' utility (1) subject to (4).
2. Given $\{w_t, r_t, q_t(p), e_t(p)\}_{t=0}^\infty$, suppliers' policies $\{p_t(z), i_t(z)\}_{t=0}^\infty$, and value $\{V_t(z)\}_{t=0}^\infty$ solve (20) and $V_t(\bar{z}_t) = 0$.
3. Given $\{G_t^i(z), G_t(z), p_t(z), \bar{z}_t\}_{t=0}^\infty$, customers' beliefs $\{H_{t+1}(p'|p), H_{t+1}^0(p)\}_{t=0}^\infty$ are consistent with Bayes' law. Customers' values $\{M_t^i(p), U_t^i, M_t(p), U_t\}_{t=0}^\infty$ are given by (12), (13), (14) and (15). Customers switch from suppliers charging p to those charging p' iff $M_t(p') \geq M_t(p)$ and from p to unmatched iff $M_t(p) < U_t$.
4. Given suppliers' and customers' choices, the masses of matched and unmatched customers $\{G_t^i(z), u_t^i, G_t(z), u_t\}_{t=0}^\infty$ evolve according to (22), (23), (24) and (25), and the customer quit and conversion rates $\{q_t(p), e_t(p)\}_{t=0}^\infty$ satisfy (26) and (27).
5. Given $w_t, r_t, p_t(z), G_t(z), u_t, L_t$, and K_t , aggregate gross output Y_t is implied by (28).
6. The labor-market clears: $L_t = \bar{\ell}_t \left[u_t \bar{m} \bar{c}_t^{-\sigma} Y_t + \int_0^\infty \frac{1}{z} p_t(z)^{-\sigma} Y_t dG_t(z) \right] \quad \forall t \geq 0$.
7. The capital market clears: $K_t = \bar{k}_t \left[u_t \bar{m} \bar{c}_t^{-\sigma} Y_t + \int_0^\infty \frac{1}{z} p_t(z)^{-\sigma} Y_t dG_t(z) \right] \quad \forall t \geq 0$.
8. The final-goods market clears: $Y_t = C_t + (K_{t+1} - (1 - \delta_K)K_t) + \Gamma_t \quad \forall t \geq 0$, where $\Gamma_t = \int_0^\infty s(i_t(z)) dG_t^i(z)$ is aggregate demand investment.
9. Aggregate profits are given by $\Pi_t = \int_0^\infty \pi_t(p_t(z), z) dG_t(z) \quad \forall t \geq 0$.

In what follows, we will restrict attention to separating equilibria in which prices in the intermediate goods market are strictly decreasing in suppliers' productivity; i.e., $p_t(z_1) > p_t(z_2)$ for $z_1 < z_2$. In addition, we restrict attention to equilibria with monotone beliefs, such that for $p_1 < p_2$ $H_t(p'|p_1) \geq H_t(p'|p_2)$ for any p' .¹¹ Before we proceed, we highlight some useful properties of equilibria in this class.

¹¹As we show below, these equilibria feature efficient transitions of customers among suppliers conditional on contacts. The restriction on decreasing prices ensures comparability with the literature studying market power in models without search frictions (e.g. Edmond et al., 2023). Under our assumptions for $F(z'|z)$ and with decreasing $p_t(z)$, only monotone beliefs are consistent with Bayes' law for prices posted in equilibrium. Our assumption binds only for off-equilibrium beliefs and rules out that prices above the equilibrium range today can be associated with beliefs of lower future prices than those posted in equilibrium.

Properties of Equilibria with Strictly Decreasing Prices and Monotone Beliefs

Any equilibrium in which prices are strictly decreasing in productivity is fully separating by construction. Strict monotonicity of $p_t(z)$ ensures that the inverse $\hat{z}_t(p) = p_t^{-1}(z)$ exists, so customers can unambiguously infer a supplier's productivity type from its price. Define the image of the pricing function as $\mathcal{P}_t = \{p \in \mathbb{R} : \exists z \in [\bar{z}_t, \infty) \text{ for which } p_t(z) = p\}$, where $\bar{z}_t$ is the exit threshold productivity. Consistency of customer beliefs with Bayes' law and the pricing policy requires $H_t(p'|p) = 1 - F(\hat{z}_{t+1}(p')|\hat{z}_t(p))$ for all $p \in \mathcal{P}_t$, $p' \in \mathcal{P}_{t+1}$, and $H_t(p' | p) = \sup_{\hat{p} \in \mathcal{P}_{t+1}: \hat{p} < p'} H_t(\hat{p} | p)$ for $p' \notin \mathcal{P}_{t+1}$. Monotonicity of beliefs further implies that for any $p \notin \mathcal{P}_t$ and $p_1 < p < p_2$ with $p_1, p_2 \in \mathcal{P}_t$, we must have $H_t(p'|p_1) \geq H_t(p'|p) \geq H_t(p'|p_2)$ for any p' . Finally, consistency of beliefs about entrants requires that $H_t^0(p) = 1 - F_0(\hat{z}_t(p))$. Given equilibrium beliefs, customers' value of a match and of being unmatched simplify to

$$(29) \quad M_t(p) = \max \left\{ \eta_t(p) + \frac{U_{t+1}^i}{1 + r_{t+1}} + \frac{1 - \delta_F}{1 + r_{t+1}} \int_{\bar{z}_{t+1}}^{\infty} (M_{t+1}^i(p_{t+1}(z')) - U_{t+1}^i) dF(z'|\hat{z}_t(p)), U_t \right\},$$

$$(30) \quad U_t = \frac{1}{1 + r_{t+1}} \left[U_{t+1}^i + \chi \int_{\bar{z}_{t+1}}^{\infty} (M_{t+1}^i(p_{t+1}(z)) - U_{t+1}^i) dF_0(z) \right]$$

for all $p \in \mathcal{P}_t$. As we focus on equilibria where $p_t(z)$, and hence $\hat{z}_t(p)$, is strictly decreasing, it follows from the properties of $\eta_t(p)$ and $F(z'|z)$ that $M_t(p)$ is strictly decreasing in p for $p \in \mathcal{P}_t$. Further, from monotonicity of beliefs, $M_t(p)$ is strictly decreasing for any p . The following Lemma is an immediate implication.

Lemma 1. *There exists a reservation price $\bar{p}_t \forall t$ such that $M_t(\bar{p}_t) = U_t$, with $M_t(p) > U_t$ iff $p < \bar{p}_t$. In particular, customers quit to being unmatched whenever $p > \bar{p}_t$. Moreover, if customers are simultaneously in contact with two firms of productivities $z_1 < z_2$, then the monotonicity of $p_t(z)$ implies $p_t(z_1) > p_t(z_2)$ and hence $M_t(p_t(z_1)) < M_t(p_t(z_2))$, so customers always move to the more productive supplier.*

Given Lemma 1, the best outside offer for any customer comes from the supplier with the highest productivity, reducing the customer value functions before rematching to

$$(31) \quad M_t^i(p) = M_t(p) + \int_{\hat{z}_t(p)}^{\infty} (M_t(p_t(z')) - M_t(p)) dO_t(z'),$$

$$(32) \quad U_t^i = U_t + \int_{\hat{z}_t(\bar{p}_t)}^{\infty} (M_t(p_t(z')) - U_t) dO_t(z').$$

Accordingly, the equilibrium customer growth rate and flow of customers satisfy

$$(33) \quad \rho_t(p, i) = 1 - q_t(p) + e_t(p)i = \begin{cases} (1 + (u_t^i + G_t^i(\bar{z}_t(p)))i) O_t(\bar{z}_t(p)), & \text{if } p \leq \bar{p}_t, \\ 0, & \text{if } p > \bar{p}_t \end{cases}$$

$$(34) \quad G_t(z) = G_t^i(z) + u_t^i(O_t(z) - e^{-I_t}) - G_t^i(z)(1 - O_t(z)).$$

$\rho_t(p, i)$ is weakly decreasing in p for any given i . The distribution of customers across suppliers is continuous in z if the initial distribution $G_{-1}(z)$ and $O_t(z)$ are continuous.¹²

We impose no restrictions on the pricing function $p_t(z)$ beyond it being strictly decreasing in equilibrium. We can establish additional properties of $p_t(z)$ by showing that any violation leads to profitable deviations for a firm and thus cannot be sustained in equilibrium.

Proposition 1. *Any equilibrium with strictly decreasing prices and monotone beliefs satisfies the following properties:*

- (i) *The price of each firm is bounded above by $p_t(z) \leq p_t^S(z)$, where $p_t^S(z) = \frac{\sigma}{\sigma-1} mc_t(z)$ is the price that maximizes static profits per customer such that $\frac{\partial \pi_t(p, z)}{\partial p} = 0$,*
- (ii) *the firm at the exit threshold sets the reservation price, $p_t(\bar{z}_t) = \bar{p}_t$,*
- (iii) *and the pricing function $p_t(z)$ is continuous.*

Proof. To prove (i), suppose that a firm with productivity z sets a price $p > p_t^S(z)$. By lowering its price, the firm would strictly increase profits per customer $\pi_t(p, z)$ without reducing its growth rate $\rho_t(p, i)$. This strictly raises its total profits, implying that the original price cannot be part of an equilibrium.

To show (ii), suppose that a firm with exit cut-off productivity $\bar{z}_t$ charged a price $p < \bar{p}_t$. Since $p_t(z)$ is strictly decreasing, p would then be the highest price posted in equilibrium. The firm could profitably raise its price infinitesimally without reducing $\rho_t(p, i)$, thereby increasing profits, because $p < p_t^S(\bar{z}_t)$ from property (i). Indeed, while property (i) only establishes the weak inequality $p \leq p_t^S(z)$ for general z , equality cannot hold at the cut-off. If $p = p_t^S(\bar{z}_t)$, then a firm with productivity $\bar{z}_t$ would earn strictly positive static profits. Since the continuation value is an option value and hence non-negative, this would imply $V_t(\bar{z}_t) > 0$, contradicting the definition of $\bar{z}_t$ as the threshold type with $V_t(\bar{z}_t) = 0$. Therefore $p < p_t^S(\bar{z}_t)$. Hence, the original price cannot be part of an equilibrium. Only at $p = \bar{p}_t$ does the firm lose customers to being unmatched if it increases its price further. It follows that

¹²The latter requires continuity of $i_t(z)$ for $z \geq \bar{z}_t$, which follows from continuity of payoffs and interior optimality.

the equilibrium price at the lowest active productivity level, equivalent to the highest price posted in equilibrium, must satisfy $p_t(\bar{z}_t) = \bar{p}_t$.

To prove (iii), suppose for a contradiction that there is a gap: there exists $\tilde{z}$ with $p_2 \equiv \lim_{z \uparrow \tilde{z}} p_t(z) > p_1 \equiv \lim_{z \downarrow \tilde{z}} p_t(z)$, so $(p_1, p_2) \notin \mathcal{P}_t$. These one-sided limits exist since $p_t(\cdot)$ is monotone. First, it must be that the price at the lower end of the gap is strictly below the static optimum, $p_1 < p_t^S(\tilde{z})$; otherwise $p_2 > p_1 = p_t^S(\tilde{z})$ would contradict (i). Second, given that $p_1 < p_t^S(\tilde{z})$, a firm of productivity $\tilde{z}$ posting price p_1 has a profitable deviation: raising its price to some p in between p_1 and p_2 increases its static profits but has no negative effect on the growth rate $\rho_t(p, i)$, since no other firm posts a price in (p_1, p_2) and beliefs are monotone. The same logic applies if firm $\tilde{z}$ posts price p_2 : for any sequence $z_n \downarrow \tilde{z}$ with $p_t(z_n) \uparrow p_1$; the firm z_n for n large still has a profitable deviation by raising its price slightly into (p_1, p_2) . Therefore, such a gap cannot exist, and the pricing function $p_t(\cdot)$ must be continuous. $\square$

Equilibrium Prices and Contact Intensities

Given the properties established above, we can show how competition in the market for intermediates determines suppliers' choices. The optimal price $p_t(z)$ and contact intensity $i_t(z)$ for a supplier with productivity z satisfy the first order conditions

$$(35) \quad \rho_t(p, i) \frac{\partial \pi_t(p, z)}{\partial p} = - \frac{\partial \rho_t(p, i)}{\partial p} W_t(p, z),$$

$$(36) \quad \frac{\partial s(i)}{\partial i} = \frac{\partial \rho_t(p, i)}{\partial i} W_t(p, z),$$

Suppliers' price setting (35) balances the static benefit of raising the price on profits from current customers against the dynamic cost of slower customer growth. The optimal contact intensity (36) trades off the static cost of contacting customers against the value of expanding the customer base. Prices and contact intensities interact through their effect on customer growth $\rho_t(p, i)$ and the value of a customer $W_t(p, z)$. Importantly, suppliers' choices depend on the actions of their competitors through the customer growth rate $\rho_t(p, i)$.

To see how competitors drive suppliers' decisions, we impose the equilibrium conditions derived above. First, for any differentiable $G_t^i(z)$ with derivative $g_t^i(z)$

$$(37) \quad \frac{\partial \rho_t(p, i_t(z))}{\partial p} = [2 + (u_t^i + G_t^i(\hat{z}_t(p)))i_t(z)] i_t(\hat{z}_t(p)) g_t^i(\hat{z}_t(p)) O_t(\hat{z}_t(p)) \frac{\partial \hat{z}_t(p)}{\partial p} < 0,$$

$$(38) \quad \frac{\partial \rho_t(p, i)}{\partial i} = (u_t^i + G_t^i(\hat{z}_t(p))) O_t(\hat{z}_t(p)).$$

Substituting these together with the consistency condition $\hat{z}_t(p_t(z)) = z$ into (35) and (36), we can characterize the optimal choices of suppliers in equilibrium as a system of two differential

equations in $i_t(z)$ and $p_t(z)$

$$(39) \quad \frac{\partial s(i)}{\partial i} = (u_t^i + G_t^i(z)) O_t(z) W_t(p_t(z), z),$$

$$(40) \quad \frac{\partial p_t(z)}{\partial z} = -\frac{2 + (u_t^i + G_t^i(z))i_t(z)}{1 + (u_t^i + G_t^i(z))i_t(z)} i_t(z) g_t^i(z) \cdot \frac{W_t(p_t(z), z)}{\frac{\partial \pi_t(p_t(z), z)}{\partial p}},$$

The boundary condition associated with (40) is $p_t(\bar{z}_t) = \bar{p}_t$.

Condition (39) shows how the optimal contact intensity depends on the distribution of matches and contacts in equilibrium. The benefit of contacting more customers increases in the likelihood of converting contacts into matches. In equilibrium, this conversion rate is given by $(u_t^i + G_t^i(z))O_t(z)$ for a supplier of productivity z . It is the share of customers who are previously unmatched or matched with less-productive firms and are at the same time not contacted by more productive competitors. A supplier of productivity z will find it more beneficial to contact additional customers when more customers are in contact with worse firms, and agree to a match when contacted. On the other hand, if a larger share of contacts is with more productive firms, a supplier will reduce its own investment in demand due to lower effectiveness. This gives more productive firms a stronger incentive to invest in demand. At the level of the individual supplier, the optimal contact intensity increases in the profit made from an additional customer, $W_t(p_t(z), z)$. This suggests that firms charging higher markups per customer should invest more in increasing their demand. We will use this firm-level relationship to discipline the calibration of the model.

Condition (40) determines the optimal pricing policy and shows how idiosyncratic marginal cost pass through into prices. This pass-through is determined by competitive forces. In equilibrium, increasing a supplier's price signals to potential customers a match of lower value and increases the chances of being outbid by a competitor. How strong this concern is differs along the distribution of productivity. From (40), the slope of the price schedule is proportional to the product of the density $g_t^i(z)$ and contact intensity $i_t(z)$ of local competitors at z , scaled by the value per customer relative to the price sensitivity of per-customer profits. The latter governs the relative incentive to increase current profits vs. investing in future customer value. The product of $g_t^i(z)$ and $i_t(z)$ determines the effective amount of *local competition*: how many customers a supplier risks losing when raising prices slightly. When suppliers face many competitors with similar productivity who invest in contacting customers, the risk of being outbid increases substantially after a marginal price increase. This means that local competition is strong and incentivizes suppliers to pass through productivity gains and set prices closer to marginal cost. Conversely, when local competition is weak, either because there are few competitors with similar productivity or because com-

petitors do not invest in contacting customers, suppliers can afford to not pass through productivity gains without losing many potential customers, leading to higher markups.

The pricing mechanism is closely related to the limit-pricing logic common in models of endogenous growth (Grossman and Helpman, 1991; Peters, 2020). There, customers can access all suppliers simultaneously, so the most productive firm sets a price equal to the marginal cost of the second most productive firm and captures the entire market, implying a single active supplier per intermediate. Here, suppliers engage in *limit-pricing in expectation*: they anticipate the most productive competitor in a customer’s choice set and set prices accordingly. Because customers are only in contact with a random subset of suppliers, this expectation-based limit-pricing generates a non-degenerate equilibrium distribution of prices, markups, and productivities. Moreover, the intensity of competition is endogenous, as it depends not only on the distribution of productivity but also on firms’ investment in contacting customers.

3 Calibration and Quantitative Properties

For the remainder of the paper, we study a quantitative version of the framework outlined above. This section describes our calibration strategy and summarizes key quantitative properties of the model, including firms’ decision rules and additional moments for validation.

3.1 Calibration

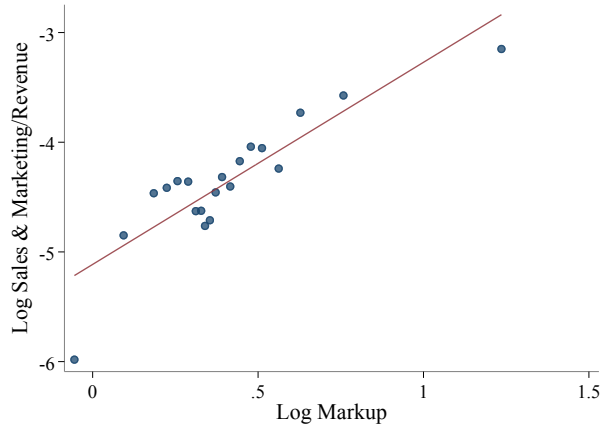
We calibrate the steady state of the model economy at annual frequency and interpret suppliers and the market for intermediate goods as a representative U.S. industry. The calibration uses moments for the period 2010–2020. We first outline how we construct a key target from the data. We then discuss the parameters we fix externally before describing the internal calibration of the remaining parameters.

Demand investment and markups. A key implication of the model is that the incentive to invest in demand increases with profits per customer—i.e., with suppliers’ markups (see (39)). Taking this implication to the data requires empirical measures of demand investment and markups. For demand investment, we use the measure in He et al. (2025), built from the Sales and Marketing Expenditure (S&M) variable in Capital IQ and complemented with information on sales-and-marketing employment and textual analysis of SEC filings. Unlike the residual component of Selling, General, and Administrative expenses (SG&A), this measure predicts future firm value and captures forward-looking investment in demand. In addition, He et al. (2025) document that high S&M-to-sales ratios are prevalent in business-

facing industries and are correlated with the importance of a sales force as well as the need to communicate information to customers, consistent with interpreting demand investment as the cost of contacting customers in the intermediate-goods market. We measure firm-level markups as sales divided by cost of goods sold.¹³

In the data, we document a robust within-industry relationship between the demand-investment-to-revenue ratio and markups (Figure 1). The estimated elasticity is significantly positive and equals 1.84, which we target below.¹⁴ Because we compute markups and demand investment from a selected sample of public firms in Capital IQ, we mirror this selection when computing the corresponding model moments.¹⁵

Figure 1: Demand Investment vs. Markup



Notes: Data from He et al. (2025) and Capital IQ, own calculations. Binscatter of log demand investment to revenue ratio vs. log markup. Adjusted for industry and year fixed effects; cost-weighted.

External parameters and functional forms. We fix a small number of parameters ex ante at standard values. We set the physical capital depreciation rate to $\delta_K = 0.06$ and the annual discount factor to $\beta = 0.96$, implying an empirically reasonable physical capital-to-

¹³For cross-sectional regressions, this approach is equivalent to estimating markups using the production-function approach. The variable-cost elasticity included in production function estimates is commonly assumed constant within industries and absorbed by industry fixed effects. We require only that, conditional on observables, measured and true ratios be proportional in the cross-section of firms, so that any multiplicative measurement error is absorbed by fixed effects. Relatedly, De Ridder, Grassi and Morzenti (2026) argue that cross-sectional variation in production-function markups accurately reflects true markup variation.

¹⁴See Table A1 in Appendix B.1. The documented correlation is consistent with evidence from the universe of U.S. manufacturing establishments in Kehrig and Vincent (2021), who show that low-labor-share establishments spend more on advertising. This aligns with evidence on the importance of customer-related intangibles for private-firm value in Bhandari and McGrattan (2021). Afrouzi et al. (2025) document a similar relationship between markups and total SGA spending in Compustat, instead of the more restrictive measure of demand investment employed here.

¹⁵As in Ottonello and Winberry (2020), we overweight older and larger firms; described in Appendix B.2.

GDP ratio of 2.9. We assume log utility over consumption and labor disutility $v(L) = \omega \frac{L^{1+\frac{1}{\gamma}}}{1+\frac{1}{\gamma}}$. We normalize steady-state labor supply to one, implying $\omega = 0.746$. We set the Frisch elasticity to $\gamma = 1$; it does not affect the baseline steady state but matters for counterfactuals.¹⁶ Finally, we assume that the contact cost function takes the form $s(i) = \frac{\bar{s}}{\psi} i^\psi$.

Internal calibration. We choose nine parameters to match ten moments. Table 1 summarizes the calibration. The model fits all targeted moments closely. Although parameters are jointly determined, we relate each model parameter to one particularly informative moment.

Table 1: Calibrated Parameters and Targeted Moments

Parameter	Value		
Exogenous firm exit rate	δ_F	0.081	
Entrants per unmatched customer	χ	0.078	
Capital elasticity	α	0.327	
Log TFP: unconditional dispersion	σ_z	0.387	
Log TFP: auto-correlation	ρ_z	0.361	
Log TFP: mean	μ_z	0.409	
Intensive-margin elasticity	σ	2.381	
Contact cost: level	$\bar{s}$	39.792	
Contact cost: curvature	ψ	2.895	

Targeted Moment	Model	Data	Source
Firm entry/exit rate	0.081	0.085	BDS
Relative employment of entrants	0.526	0.540	Sterk, Sedláček and Pugsley (2021)
Aggregate labor share	0.614	0.615	BLS
Aggregate markup	*1.250	1.250	Capital IQ
Elasticity of $\frac{\text{demand investment}}{\text{revenue}}$ to markup	*1.849	1.843	Capital IQ
Elasticity of markup to revenue	0.031	0.031	Edmond et al. (2023)
Sales variation: intensive-margin share	0.262	0.260	Bernard et al. (2022)
Customer quit rate (% sales)	0.152	0.150	Dhyne, Duprez and Komatsu (2023)
Top 10% sales share	0.676	0.724	SUSB
Top 1% sales share	0.431	0.423	SUSB

Notes: * indicates moments for which model output is filtered as described in Appendix B.2 (Table A2) to match the Capital IQ sample. The model’s unfiltered economy-wide aggregate markup is 1.197.

We discipline the exogenous firm exit rate δ_F and entrants per unmatched customer χ with the firm entry/exit rate in the Business Dynamics Statistics (BDS) and the average employment of entrants relative to incumbents (Sterk et al., 2021). We target the aggregate labor share by choosing the capital elasticity in production α .

Sales-variation moments discipline the contact technology $s(\cdot)$ and the intensive-margin elasticity of demand σ . The level $\bar{s}$ and curvature ψ of the contact cost jointly determine how

¹⁶We follow Edmond et al. (2023) for comparability.

easily customers reallocate across firms, shaping customer turnover and sales concentration. We target sales concentration in the U.S. from the Statistics of U.S. Businesses (SUSB).¹⁷ For customer turnover, we target the fraction of business-to-business sales in relationships that are terminated at annual frequency. Business-to-business transactions account for more than half of economic activity in the U.S. economy, and our interpretation of demand investment as the cost of contacts to form matches is consistent with the formation of customer-supplier relationships. As turnover in business-to-business transactions is not available at scale for the U.S., we rely on evidence from Belgium in Dhyne et al. (2023). The within-final-good substitution elasticity σ governs substitution across intermediate inputs and thus how much cross-sectional sales variation arises from the intensive margin (sales per customer) versus the extensive margin (number of customers). Consistent with the turnover in customer relationships, we discipline σ using the intensive-margin share of sales variation measured in the same Belgian data as reported by Bernard et al. (2022).¹⁸

Intermediate-good producers' log TFP follows a Gaussian AR(1) process with parameters $(\sigma_z, \rho_z, \mu_z)$. From firms' optimality conditions in (39) and (40), these parameters jointly determine the distribution of markups and their relationship with firm size and demand investment. First, the lower the average productivity μ_z , the more constrained low-productivity suppliers are in their pricing by customers' outside option of in-house technology. A lower μ_z therefore reduces markups for low-productivity firms, strengthening the relationship between markups and firm size; accordingly, μ_z is disciplined by the elasticity of markups with respect to revenue (Edmond et al., 2023). Second, away from the constraint imposed by the outside option, higher dispersion σ_z means that suppliers are further apart in terms of productivity, reducing local competition as implied by (40). Lower competition raises the aggregate markup, so σ_z is disciplined by the aggregate markup in Capital IQ. Third, as variation in markups is driven by variation in productivity z , a higher persistence ρ_z makes markups more persistent. This raises the elasticity of customer value to current z , increasing the elasticity of the demand-investment-to-revenue ratio with respect to markups: high- z (high-markup) firms expect to remain high- z and therefore have a larger forward-looking payoff to investing in demand. Therefore, ρ_z is disciplined by the cross-sectional elasticity of demand investment to markups in Capital IQ as computed above.

¹⁷Further details on moments from the BDS and SUSB are provided in Appendix B.3.

¹⁸Bernard et al. (2022) decompose sales variation into the number of customers, customer fixed effects, and supplier fixed effects. As our model does not feature heterogeneity among customers, we omit customer fixed effects and compute the intensive margin share as the contribution of supplier fixed effects relative to the number of customers. To be robust to customer heterogeneity in the data, we also target sales-weighted customer quit rates instead of the fraction of relationships that are separated.

Sensitivity and robustness. Appendix C confirms the parameter–moment links described above using the local sensitivity analysis of Andrews, Gentzkow and Shapiro (2017). Some calibration moments are less commonly used or harder to measure. Appendix E therefore reports recalibrations that target alternative values for (i) the customer quit rate, (ii) the intensive-margin share of sales variation, including consumer-good-based measures for the U.S. in Einav et al. (2021), and (iii) the level of the aggregate markup. We assess the robustness of our main results to these alternative calibrations.

Additional moments. Table 2 shows that, beyond the calibration targets, the model aligns well with additional empirical moments. First, although we do not target the level of demand investment, the model generates aggregate investment of 0.113 relative to value added, compared to 0.080 in Capital IQ. One interpretation of the higher model-implied value is that Capital IQ’s sales and marketing spending captures most, but not all, demand investment.¹⁹

Second, despite a comparatively low persistence of idiosyncratic productivity, the model generates substantial persistence in firm-level revenue consistent with the data. The high persistence of revenue is driven by the slow turnover of customer relationships, constraining the reallocation of revenue across firms. The difference in persistence of productivity and revenue suggests that customer-base dynamics are an empirically relevant amplification mechanism for firm-level revenue dynamics.

Table 2: Model Validation: Untargeted Moments

	Model	Data	Source
Demand investment over value added	0.113	0.080	Capital IQ
Log revenue auto-correlation	0.990	0.991	Capital IQ
Intermediate input elasticity σ	2.38	0.7–2.5	Peter-Ruane (2025); Carvalho et al. (2021)
Elasticity markup to # customers	0.000	−0.004	Afrouzi et al. (2025)
Elasticity markup to revenue per customer	0.152	0.187	Afrouzi et al. (2025)
Cost pass-through (annual)	0.799	0.59–0.76	Amiti, Itskhoki and Konings (2019)
Customer surplus (ratio to expenditure)	0.30	0.28	Baqae, Burstein, Duprez and Farhi (2025)

Third, we calibrate σ to match the intensive-margin share of total sales variation. Alternatively, one could discipline σ using direct estimates of substitution elasticities across intermediate inputs. Our calibrated value lies at the upper end of the range of firm-level estimates reported in Carvalho et al. (2021) and Peter and Ruane (2025).

Fourth, the model reproduces the central empirical result in Afrouzi et al. (2025): conditional on sales variation at both margins, markups are positively correlated with sales

¹⁹E.g. compensation for some employees building customer relationships might be reported elsewhere, such as in cost of goods sold. We do not target the level of sales and marketing spending in Capital IQ to allow for underreporting of customer investment. As long as underreporting is proportional to true investment, it does not affect the cross-sectional moment used in calibration. See Appendix B.1.

per customer but uncorrelated with the number of customers. This is driven by sales per customer responding contemporaneously to price changes while the number of customers is backward-looking and adjusts slowly. The model also generates average annual marginal-cost-to-price pass-through of 0.799, broadly consistent with estimates in Amiti et al. (2019), who report pass-through coefficients of up to 0.76 using Belgian firm-level data.²⁰

Fifth, the model also matches evidence on the customer surplus generated by supplier relationships. Baqaee et al. (2025) use Belgian data to estimate that the change in customer surplus due to new or separating business-to-business relationships amounts to 28% of spending in these relationships. Their estimates map directly into a weighted average of the static match surplus relative to expenditure $\frac{\eta(p)}{py(p)}$ in the model. The model equivalent to their estimates yields an implied surplus of 30%.²¹ The close fit of the model-implied customer surplus from matching is a key quantitative validation of the welfare analysis below.

3.2 Pricing and Demand Investment

The competition for customers described in this paper speaks to two key findings of the recent literature on sales dynamics: First, variation in sales is largely driven by the extensive margin, the number of customers (Bernard et al., 2022; Einav et al., 2021; Afrouzi et al., 2025; Argente et al., 2025). Second, firms rely heavily on non-price actions to acquire new customers (He et al., 2025), but—conditional on marginal cost—do not use lower prices to attract customers when entering new markets (Argente et al., 2025; Fitzgerald et al., 2024).

While the first fact naturally arises in models with customer dynamics, the second seems at odds with pricing incentives in dynamic models of demand. One potential resolution is that demand does not respond to prices at the extensive margin.²² An alternative explanation is that the economic forces that drive customer acquisition and retention are such that firms optimally choose not to vary prices with the number of customers. Our framework highlights this second explanation. The optimal price of suppliers depends only on productivity z , despite prices being allocative for the number of customers as both the quit and conversion rate vary with a supplier’s price. Prices being independent of the number of customers

²⁰The cited estimate from Afrouzi et al. (2025) is from Table A.7, which estimates the relationship under a rich set of fixed effects to control for marginal cost. Alternative estimates in the same paper show qualitatively similar results, with smaller coefficients for the elasticity of markups to revenue per customer. Our preferred estimate of 0.76 in Amiti et al. (2019) is based on a specification that instruments firm-level cost with exchange rate movements based on exposure to imported intermediates.

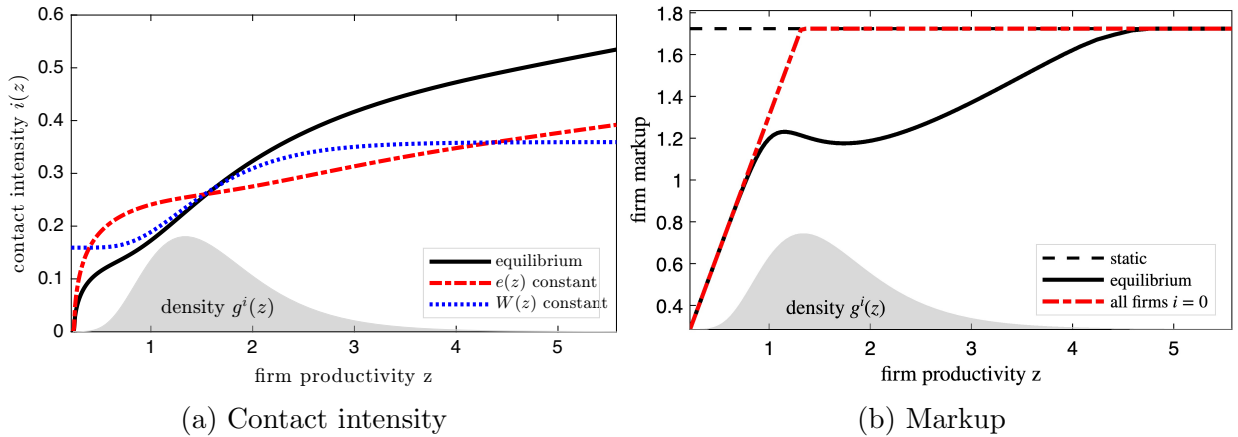
²¹When we isolate customer surplus generated or lost from supplier births and deaths, in line with their IV strategy, the model delivers estimates in the narrow range of 28–30%.

²²See e.g. Einav et al. (2021); Fitzgerald et al. (2024); Afrouzi et al. (2025); Cavenaile et al. (2025). Paciello et al. (2019) and Roldan-Blanco and Gilbukh (2021) provide evidence from consumer goods against this view and show that the extensive margin responds to prices.

follows directly from our restriction that the cost of contacts is constant returns to scale in existing customers, which ensures that suppliers' problem scales linearly in the number of customers. The fact that firms do not vary their price with the number of customers provides a rationale for this restriction and identifies the role of existing customers for the matching technology in our framework. The allocative effect of prices at the extensive margin ensures that firms compete for customers via price. This effect is central to the results discussed below and we highlight its implications in the following sections.

Before turning to the consequences of competition for customers, we build intuition for firms' behavior in the quantitative model. Figure 2 displays suppliers' decision rules. The left panel shows contact intensity $i(z)$ increasing in productivity z . This reflects the two forces in (39): Customer conversion rates $e(z)$ rise with z in equilibrium, as customers choose the most productive firm they are in contact with. In addition, the value of an additional customer at production, $W(p(z), z)$, is also increasing in z . The figure illustrates these channels using counterfactual decision rules that hold fixed, in turn, each of the two margins. The effect of conversion rates is strongest where the customer distribution is dense, i.e. where small productivity gains translate into large increases in customers poached from competitors.

Figure 2: Firm Decision Rules



Notes: The left panel displays suppliers' contact intensity as solid black line. The red dash-dotted line shows a counterfactual contact intensity when holding the conversion rate $e(z)$ constant, while the blue dotted line shows a counterfactual holding constant the value per customer $W(z)$ across firms. The right panel shows suppliers' equilibrium markup as solid black line. The horizontal dashed black line indicates the optimal static markup under CES demand. The red dash-dotted line is the counterfactual markup when all suppliers abstain from contacting customers for one period only ($i(z) = 0$). In both panels, $g(z)$ is the density of customers across suppliers of type z .

The right panel displays suppliers' equilibrium markups as the solid black line. To understand the drivers of equilibrium markups, we consider two counterfactuals. Under CES demand without an outside option, suppliers would charge the constant, static optimal markup

at $\mu = \frac{\sigma}{\sigma-1}$. Constraining suppliers' pricing through the option of in-house production generates markups that initially rise with z and then flatten, even absent poaching or customer acquisition. This is shown by the red dash-dotted line, which displays a counterfactual with $i(z) = 0$ for all firms while imposing that customers quit at prices above $\bar{p}$. The outside option generates markups below one for the lowest-productivity suppliers. Suppliers are willing to temporarily accept losses per customer to maintain the continuation value of a match, a feature that arises directly from the dynamic nature of competition. The equilibrium markup differs from the counterfactual with $i(z) = 0$ because of the threat of customer poaching. When suppliers invest in contacting customers, they intensify local competition, which puts downward pressure on markups. From (40), the intensity of local competition is determined by the product $i(z) \cdot g^i(z)$. The equilibrium markup is non-monotone because suppliers face the most intense local competition where the mass of competitors is dense—in the middle of the productivity distribution. In the far right tail, contact intensities remain high but potential competitors are vanishingly scarce, so the equilibrium markup converges to the static monopoly markup.

4 Competition for Customers and Misallocation

We describe how inefficiencies arise in the decentralized equilibrium and quantify the resulting sources of misallocation under our calibration. We then derive implications for optimal product-market policy, focusing on taxes on demand investment and subsidies to production.

4.1 Misallocation of Demand

We assess the efficiency of the decentralized allocation relative to a social planner facing the same technological constraints and matching frictions. Inefficiencies arise on two margins. First, search frictions in the intermediate-input market prevent customers from accessing all suppliers. These frictions generate positive markups and give rise to the sources of misallocation studied in e.g. Baqaee and Farhi (2020) and Edmond et al. (2023). Second, the distribution of customers across suppliers with different productivities is endogenous and driven by suppliers' contact choices, which reflect suppliers' private benefit and need not coincide with the social value of a contact. We discuss both margins in turn and relegate the detailed setup and optimality conditions of the planner problem to Appendix A.2.

Markups. Aggregate markups act like a wedge between factor payments and marginal products, distorting labor and capital supply and lowering output and welfare—the standard inefficiency from an aggregate markup with endogenous factor supply. Markup dispersion

generates an additional distortion by shifting demand across intermediate inputs at the intensive margin (sales per customer). Quantities respond to relative prices at this margin and dispersion in markups drives a wedge between relative prices and relative marginal products, leading to misallocation of demand across intermediates and lowering aggregate productivity. The strength of this misallocation is governed by the elasticity of substitution across intermediates, σ , which determines customers' response to relative prices. Appendix A.3 characterizes these two standard inefficiencies from markups formally in our setting.

Here, markups and the associated distortions arise from the same friction that makes demand dynamic at the extensive margin: Search frictions make it costly to contact customers, reduce their choice set, and generate market power and a distribution of markups. However, conditional on contacts, markups do not distort the transition of customers across suppliers. At the extensive margin, customers choose the supplier with the lowest price, irrespective of how much lower it is. As equilibrium prices are strictly decreasing in productivity z , customers switch efficiently to more productive suppliers, as in Menzio (2024, 2026). Thus, markups distort quantities within relationships (the intensive margin) but not the direction of customer reallocation across suppliers (the extensive margin).

Allocation of customers. While markups do not distort customers' choice of supplier, inefficiencies still arise at the extensive margin: the distribution of contacts—and ultimately of customers across suppliers—is endogenously determined by suppliers' choice of contact intensities $i(z)$, maximizing suppliers' private profit. Four externalities cause the planner to choose different contact intensities: suppliers do not account for the full *private benefit* as they neglect customer surplus, matches cause *dynamic business-stealing* externalities and *crowd out entry*, and contacts have a *static business-stealing* externality on competitors' matches. The first three enter through the planner's valuation of a match; the last enters directly through the optimality condition for demand investment.

We can show the first three externalities in the planner's value of a match between a customer and a supplier with productivity z at the beginning of period t , $V_t^*(z)$, measured in units of resources, and the corresponding value at production, $W_t^*(z)$. We derive both values in Appendix A.2 as

$$(41) \quad W_t^*(z) = \frac{\partial Y_t}{\partial g_t^*(z)} + \beta \frac{u'(C_{t+1})}{u'(C_t)} \int_0^\infty V_{t+1}^*(\hat{z}) [\hat{f}_{t+1}(\hat{z}|z) - \chi \hat{f}_{t+1}^0(\hat{z})] d\hat{z},$$

$$(42) \quad V_t^*(z) = -s(i_t^*(z)) + (1 - q_t^*(z) + e_t^*(z)i_t^*(z))W_t^*(z) \\ + \int_0^\infty W_t^*(\hat{z})g_t^{i^*}(\hat{z}) \left[-\frac{\partial q_t^*(\hat{z})}{\partial g_t^{i^*}(z)} + \frac{\partial e_t^*(\hat{z})}{\partial g_t^{i^*}(z)} i_t^*(\hat{z}) \right] d\hat{z}.$$

Here $\hat{f}_{t+1}(z'|z)$ and $\hat{f}_{t+1}^0(z)$ denote, respectively, the transition density and entrant density of suppliers adjusted for entry and exit decisions in period $t + 1$, $q_t^*(z)$ and $e_t^*(z)$ are the quit and conversion rates, and $g_t^{i*}(z)$ is the mass of customers matched with suppliers of productivity z after entry and exit in the planner's solution. Comparing these expressions to their equilibrium counterparts in (20) and (21) highlights the externalities.

First, the static benefit of a match to a supplier is the profit per customer, $\pi_t(p_t(z), z)$. We compare this to the static benefit for the planner, $\frac{\partial Y_t}{\partial g_t^*(z)} = \frac{z^{\sigma-1}-1}{\sigma-1}(Z_t^*)^{2-\sigma}(K_t^*)^\alpha(L_t^*)^{1-\alpha}$. The key difference between the two is that the planner takes into account the surplus of a match to the customer, adjusted for the outside option of being unmatched and producing with productivity 1. The absence of customers' surplus from the suppliers' decision induces *under-investment* in demand in equilibrium. The static benefit of a match differs further due to the distortions from markups and differences in aggregate productivity and output.

Second, the dynamic business-stealing externality enters as the integral term in (42), which adjusts the value of a match for its effect on the quit and conversion rates of other firms. Because a larger number of matches reduces the marginal cost of contacting additional customers, firms with more matches will poach more from competitors. This effect on competitors' quit rates thus arises directly from the role of existing customers in the matching technology; its presence is identified by evidence on the joint dynamics of prices and customers. Similarly, increasing matches at productivity z reduces the conversion rate of contacts at firms with lower productivity. Capturing these dynamic externalities requires a framework that can generate the persistence of customer relationships observed in the data.

Third, as entry in our economy is directly linked to the mass of unmatched customers, increasing matches reduces entry in the following periods. The planner's value in (41) internalizes this effect by adjusting the continuation value of a match for the opportunity cost of foregone entry (the $-\chi\hat{f}_{t+1}^0(\hat{z})$ term).

Fourth, the static business-stealing externality directly enters the planner's optimality condition for the contact intensity $i_t^*(z)$, derived in Appendix A.2 as

$$(43) \quad \frac{\partial s(i_t^*(z))}{\partial i} = e_t^*(z)W_t^*(z) + \int_0^\infty W_t^*(\hat{z}) \frac{g_t^{i*}(\hat{z})}{g_t^{i*}(z)} \left[-\frac{\partial q_t^*(\hat{z})}{\partial i_t^*(z)} + \frac{\partial e_t^*(\hat{z})}{\partial i_t^*(z)} i_t^*(\hat{z}) \right] d\hat{z}$$

Comparing this condition to its decentralized-equilibrium counterpart in (36) highlights the static business-stealing externality as the integral on the right. Suppliers do not internalize that by increasing their contact intensity they contact some customers in existing matches, increasing the quit rate at competing firms and destroying the value of the original match. In addition, suppliers do not internalize that by increasing their contact intensity, they also increase the probability that customers with multiple contacts match with them, reducing

the conversion rate and hence the value of other contacts that do not translate into matches.

The crowding out of potential entry as well as the static and dynamic business-stealing externalities imply a lower value of a match or contact to the planner relative to suppliers' private value. These three externalities hence generate *over-investment* in demand in equilibrium. Whether the equilibrium overall features over- or under-investment in demand is therefore ex ante ambiguous. We address this question in the calibrated economy below.

Beyond these four externalities, the planner and equilibrium also differ because the customer distribution feeds back into both the conversion rate $e_t(z)$ and the weights $g^{i*}(z)$ that aggregate firm-level demand investment. Misalignment between private and social match values can additionally distort exit and entry decisions; Appendix A.2 shows that the planner chooses a firm to exit (not enter) whenever $V_t^*(z) < 0$.

4.2 Quantifying Sources of Misallocation

We quantify the sources of inefficiency in the calibrated economy by comparing welfare across four allocations. Let U denote welfare in the decentralized equilibrium and U^* welfare under the full planner solution. Let U^M denote welfare in an economy that holds aggregate TFP and total demand investment fixed at their equilibrium levels, but where the planner chooses other aggregate quantities efficiently (removing the aggregate markup wedge). Finally, let U^Z denote welfare when the planner also allocates quantities within matches efficiently at the intensive margin given the equilibrium distribution of matches and demand investment (removing the loss from markup dispersion). These definitions imply the decomposition

$$(44) \quad \Delta \equiv U^* - U = \underbrace{U^M - U}_{\Delta_{\mathcal{M}} \text{ (agg. markup)}} + \underbrace{U^Z - U^M}_{\Delta_Z \text{ (markup dispersion)}} + \underbrace{U^* - U^Z}_{\Delta_G \text{ (customer allocation)}}.$$

We quantify each term in steady state and including transitions. All transitions start from the steady state of the decentralized equilibrium as initial condition. We report welfare changes between any U_1 and U_2 in terms of consumption equivalents (CEV) as

$$(45) \quad CEV(U_1, U_2) \equiv \exp((1 - \beta)(U_2 - U_1)) - 1.$$

Table 3 compares the decentralized equilibrium to the social planner allocation. Relative to the equilibrium, the planner substantially expands production by increasing factor inputs, raising GDP by 22.8% in the steady state. Welfare increases by 11.0% in steady state terms, but by only 5.1% when accounting for transition dynamics, reflecting the gradual and costly accumulation of the capital stock. At the same time, the planner sharply reduces demand

Table 3: Efficiency Losses of Equilibrium Allocation

		Equil.	Planner	% Δ	% $\Delta_{\mathcal{M}}$	% Δ_Z	% Δ_G
Gross output	Y	1.092	1.286	17.7	20.8	2.7	-5.7
GDP	$Y - \Gamma$	1.000	1.228	22.8	22.7	2.9	-2.8
Demand investment	Γ	0.092	0.058	-36.9	0.0	0.0	-36.9
Tangible capital	K	2.932	4.132	40.9	44.5	3.2	-6.8
Labor	L	0.325	0.353	8.9	10.7	-0.1	-1.7
Consumption	C	0.824	0.980	18.9	18.0	2.9	-2.0
Aggregate TFP	Z	1.640	1.630	-0.6	0.0	1.6	-2.2
Welfare (CEV) – steady state				11.0	8.5	2.8	-0.3
Welfare (CEV) – transition				5.1	1.9	2.2	1.0

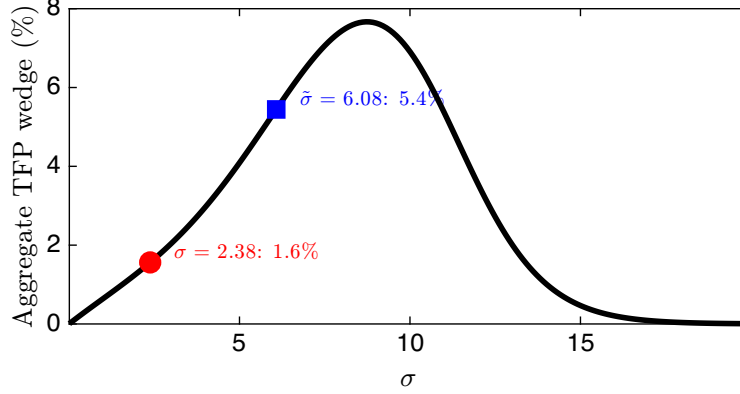
Notes: The top panel shows steady-state quantities; the bottom panel reports welfare gains with and without transitional dynamics. Full transition paths are reported in Appendix D.1. Quantities are normalized so that $GDP = 1$ in the decentralized equilibrium. Column % Δ reports percent changes from the equilibrium to the full social planner allocation. The three rightmost columns decompose efficiency losses into (i) an aggregate markup wedge, (ii) markup dispersion, and (iii) misallocation of customers across firms.

investment by 36.9%. Aggregate TFP changes little, declining slightly by 0.6%. The final three columns of Table 3 decompose the overall effect into the aggregate markup wedge $\Delta_{\mathcal{M}}$, markup dispersion Δ_Z , and misallocation of customers across firms Δ_G .

Cost of aggregate markup. The aggregate markup wedge accounts for a large part of the efficiency losses, as shown in column $\Delta_{\mathcal{M}}$ of Table 3. Eliminating it would raise welfare by 8.5% in steady state but only by 1.9% along the transition, reflecting a gradual build-up of the physical capital stock. The distortion reflects under-investment in capital and inefficiently low labor input in the decentralized equilibrium. This mechanism is standard in the literature and can be replicated in a growth model with aggregate productivity fixed at the market-economy level and consistent calibration of household preferences and technology.

Cost of markup dispersion. Dispersion in intermediate producers' markups, and the resulting misallocation of demand at the intensive margin, lowers aggregate TFP by 1.6% and welfare by 2.2% along the transition (2.8% across steady states). The cost of intensive-margin misallocation depends both on the degree of markup dispersion and how strongly demand responds to relative prices. In many models of imperfect competition, in which heterogeneous markups arise from preferences over differentiated varieties (Kimball, 1995; Atkeson and Burstein, 2008), both markups and the response of demand are governed by the same parameters: the elasticity of substitution across varieties is equivalent to suppliers' price elasticity of demand and calibrated to match markups, but also determines the allocative

Figure 3: Intensive-Margin Misallocation as a Function of Intensive-Margin Elasticity



Notes: The TFP wedge is $100 \times (\frac{Z^*}{Z_M} - 1)$, the percent gain in aggregate TFP from removing markup dispersion, holding fixed the equilibrium distribution of customers $G(z)$ and equilibrium markups $\mu(z)$ while varying only σ .

distortions implied by a given distribution of markups.

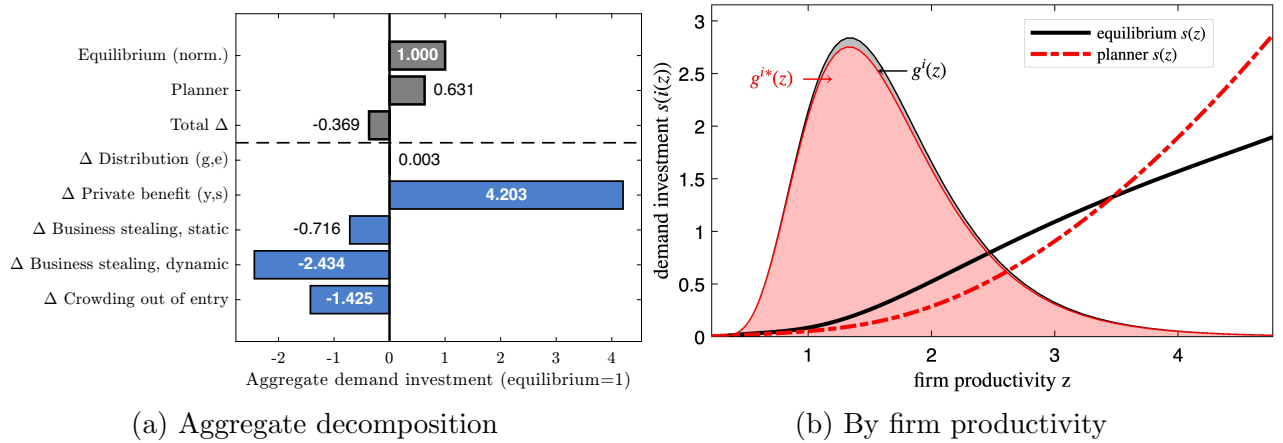
Here, suppliers' relevant demand elasticity—and thus their markup—is determined primarily by the extensive margin through search frictions and demand investment. This allows us to calibrate the intensive-margin elasticity of substitution σ separately. We set $\sigma = 2.38$ to match evidence that only about 26% of total sales variation is attributable to differences in sales per customer. The elasticity required to generate the same aggregate markup via $\mathcal{M} = \frac{\tilde{\sigma}}{\tilde{\sigma}-1}$ is instead $\tilde{\sigma} = 6.08$. The low σ implies that a given dispersion in markups generates relatively little misallocation, since demand responds weakly to price differences at the intensive margin. To further highlight the role of σ , Figure 3 plots the aggregate TFP gain from removing markup dispersion, holding fixed equilibrium markups and the customer distribution while varying σ . Under $\tilde{\sigma} = 6.08$, the efficiency loss of markup dispersion increases more than threefold, from 1.6% to about 5.4%. The relatively modest role of markup dispersion in our baseline is therefore a direct implication of the calibration that separates extensive- and intensive-margin elasticities.²³

Cost of customer misallocation. Relative to the decentralized equilibrium, the social planner reduces aggregate investment in demand by 36.9%, raising welfare by 1.0% along the transition but reducing it by 0.3% in steady state. The difference reflects that excessive demand investment inflates steady-state output but lowers welfare once the cost of building these customer relationships is accounted for.

²³Appendix D.5 shows this logic survives under full recalibration: a static-demand variant (in the spirit of Edmond et al., 2023) has low dispersion and modest misallocation; adding customer dynamics (in the spirit of Afrouzi et al., 2025) raises both; adding price-dependent customer flows (our baseline) sustains high dispersion but lowers the intensive-margin elasticity, keeping misallocation small.

Over-investment in equilibrium is the net effect of two opposing forces. The planner internalizes the full private benefit of a match to both parties, including the ongoing benefit to the customer, which by itself would raise demand investment. This is more than offset by business-stealing externalities—demand investment diverts customers from competitors both through poaching and by crowding out competitors’ matches—and the crowding out of entry. Figure 4a quantifies the contribution of each channel using a Shapley-Owen decomposition, described in detail in Appendix A.4. The remaining terms in the decomposition—differences in the distribution of matches and in exit thresholds—play a negligible role.

Figure 4: Equilibrium vs. Social Planner Demand Investment



Notes: Panel (a) decomposes the aggregate difference in demand investment between the planner and equilibrium into differences due to distribution terms, private benefits, business-stealing (both static and dynamic), and crowding out of entry. Positive bars indicate forces that, in isolation, would lead the planner to choose higher aggregate demand investment than in equilibrium, while negative bars indicate forces that lead the planner to reduce demand investment relative to the equilibrium. Panel (b) plots demand investment per customer $s(i(z))$ by firm productivity in the equilibrium (black solid) and social planner allocation (red dashed), together with the customer distribution.

Independent evidence supports both sides of this tradeoff. The customer surplus ratio validated in Section 3 against Baqaee et al. (2025) confirms that the model captures the magnitude of the surplus that customers derive from supplier matches: the model implies that about 30% of customer expenditure reflects surplus relative to the outside option, compared to 28% in the data. Direct evidence on business stealing is harder to obtain. The model implies that about 71% of newly generated matches reflect displacement of existing relationships rather than net match creation, compared to about 72% in Sinkinson and Starc (2019), and evidence of complete displacement in other settings (Garthwaite, 2014; Hartmann and Klapper, 2018).²⁴ While this evidence on business stealing is derived from studies of indi-

²⁴Sinkinson and Starc (2019) estimate the relative effects of own and rival pharmaceutical advertising on brand sales; in our framework, these map to effects on customer growth.

vidual industries, it suggests that substantial crowding out of competitors through demand investment is a feature of the data. Both validation moments directly discipline static forces. The dynamic channels propagate these forces through persistent customer relationships, with match persistence pinned down by the calibrated quit rate.²⁵

Figure 4b shows that the planner reduces demand investment for most suppliers but not for the most productive, choosing higher contact intensities than in equilibrium for roughly the top 2% of firms. Although business-stealing externalities are present at high productivity levels, they converge to a constant as the density of matches goes to zero. In contrast, customer surplus continues to grow. Suppliers asymptotically capture only part of the joint surplus from a match; the planner internalizes the remaining customer surplus, leading to higher optimal investment at the top of the distribution. The sharp fall in aggregate demand investment reflects the overall dominance of negative externalities on competitors' matches.

4.3 Optimal Demand Investment Taxes and Production Subsidies

The previous section showed that the decentralized equilibrium yields an inefficient allocation of demand relative to the social optimum. Here we examine how much efficiency a government can restore using a restricted set of policy instruments. We consider two tools: First, a flat tax τ_S on demand investment $s(i(z))$, which targets over-investment. Second, a subsidy τ_P on both capital and labor inputs, designed to offset the level distortion from markups. The optimization problem of intermediate-good producers becomes

$$(46) \quad V_t(z) = \max_{p,i} \left\{ -(1 + \tau_S)s(i) + \rho_t(p, i) \left[\pi_t \left(p, \frac{z}{1 - \tau_P} \right) + \frac{1 - \delta_F}{1 + r_{t+1}} \mathbb{E} [\max \{V_{t+1}(z'), 0\} | z] \right] \right\}.$$

We impose government budget clearing period by period, such that

$$(47) \quad \tau_P \overline{m} \overline{c}_t \left(u_t y_t ((1 - \tau_P) \overline{m} \overline{c}_t) + \int_0^\infty \frac{1}{z} y_t(p_t(z)) dG_t(z) \right) = T_t + \tau_S \int_0^\infty s(i_t(z)) dG_t^i(z),$$

where T_t is a lump-sum tax on households. The government chooses constant levels of τ_S and τ_P to maximize welfare, accounting for transitional dynamics. We consider three cases: a demand-investment tax only, a production subsidy only, and both instruments jointly.

Table 4 summarizes the results. The joint optimum features a 61.1% demand-investment tax and a 17.1% production subsidy, raising welfare by 2.5% along the transition, about half of the total welfare improvement attainable by the full planner solution. Under the joint policy, aggregate demand investment falls by 31.6% relative to the baseline.

²⁵Appendix E shows that the main welfare results are robust to alternative quit rate targets.

Table 4: Optimal Demand Investment Taxes and Production Subsidies

		Baseline	τ_P only		τ_S only		Both	
		Value	Value	% Δ	Value	% Δ	Value	% Δ
Demand investment tax (%)	τ_S				33.8		61.1	
Production subsidy (%)	τ_P		13.3				17.1	
Lump-sum tax (% of GDP)	T		16.7		-2.1		17.8	
Gross output	Y	1.092	1.306	19.5	1.035	-5.3	1.265	15.8
GDP	$Y - \Gamma$	1.000	1.193	19.3	0.973	-2.7	1.202	20.2
Demand investment	Γ	0.092	0.113	22.4	0.062	-32.8	0.063	-31.6
Tangible capital	K	2.932	4.052	38.2	2.763	-5.8	4.072	38.9
Labor	L	0.325	0.356	9.5	0.318	-1.9	0.355	9.3
Consumption	C	0.824	0.949	15.2	0.807	-2.1	0.957	16.2
Aggregate TFP	Z	1.640	1.658	1.2	1.605	-2.1	1.606	-2.0
Aggregate Markup	$\mathcal{M}$	1.197	1.194	-0.3	1.204	0.6	1.203	0.5
Welfare (CEV) – steady state				7.0	-0.7		8.0	
Welfare (CEV) – transition				1.3	0.5		2.5	

Notes: We compare steady states for all quantities and report welfare changes with and without transitional dynamics. Transition paths for all variables are provided in Appendix D.2. Baseline refers to the calibrated economy without taxes. % Δ refers to percent changes relative to the baseline. In each of the three tax regimes, we solve for the constant levels of the demand-investment tax and/or production subsidy that maximize welfare taking into account the transition (from the baseline steady state to the steady state with taxes). Government budget clearing is achieved period by period by imposing a lump-sum tax on households. Quantities normalized so that $GDP = 1$ in the baseline.

Complementary policy instruments. Comparing the jointly optimal demand-investment tax and production subsidy to their optimal levels in isolation, we find that the two instruments are complements. The optimal production subsidy is 13.3% in isolation but rises to 17.1% when combined with a demand-investment tax. Likewise, the optimal demand-investment tax is 33.8% in isolation but rises to 61.1% when paired with a production subsidy. The welfare gain of introducing both instruments jointly is 2.5%, exceeding the sum of the gains from each instrument in isolation (1.7%) by about 40%.

The complementarity stems from the two-way interaction between demand investment and markups in equilibrium. Demand investment rises with firms' profits per customer. A production subsidy boosts profits and thereby induces additional (inefficient) demand investment, reflected in a 22.4% increase when the subsidy is introduced alone. Conversely, demand investment intensifies price competition and dampens markup distortions. A tax that raises the cost of demand investment raises markups and worsens related distortions. With a production subsidy in place to offset aggregate markup distortions, the demand-investment tax can be set more aggressively. These findings caution against evaluating different tools for product market interventions in isolation when firms use both prices and

non-price investment to compete for customers. Policies that address distortions at one margin can exacerbate inefficiencies at the other.

Size-dependent policies. The two policies we analyze address average inefficiencies from demand investment and markups but do not differentiate across firms and thus cannot eliminate heterogeneity in distortions. One might consider size-dependent interventions, as often proposed in the literature, to address this heterogeneity. Such policies are difficult to implement in practice, but also lack theoretical grounding in our framework. Here, firm size, measured by e.g. total sales, is not a sufficient statistic for productivity, markups, or contact intensity. Much of the observed variation in sales reflects differences in customer base size, shaped by past productivity and past investment rather than current fundamentals. Consequently, policies conditioned on total sales would be poorly targeted.²⁶ To directly target firm productivity or markups, interventions would need to condition on the number of customers as well—an even less practical basis for policy design. For these reasons, we do not pursue size-dependent policies further.

Robustness. To better understand our findings on efficiency and optimal policy, and to assess the robustness of the quantitative results, we conduct two sets of exercises. First, we implement a comparative static on the contact-cost function. Lower costs of contacting customers intensify competition and compress markups, reducing the welfare loss from both the level and dispersion of markups; higher costs of contacting customers have the opposite effect (Appendix E.2.1, Table A6). Second, we compute the welfare-loss decomposition and the optimal policy under alternative calibrations. All mechanisms discussed in this section, including the complementarity between production subsidies and demand-investment taxes, hold under alternative calibrations (Appendix E.2.2, Table A7).

5 Macroeconomic Effects of Rising Demand Investment

Since the 1980s, both demand investment and industry sales concentration have increased. We study whether changes in competition for customers, driven by technological change in contact costs, can account for these developments by comparing our baseline steady state, calibrated to the 2010s, to an earlier one calibrated to the lower demand investment and concentration of the 1980s. We focus on the implications for aggregate productivity, competition, and the value of firms' customer base.

²⁶Appendix Figure A8 illustrates that size distortions from markup dispersion vary systematically with productivity but show little systematic pattern across the revenue distribution.

5.1 Calibration of Technological Change in Contact Cost

While the precise level and trend in demand investment depend on the measure applied, all indicators point toward an increase in spending as a share of output since the 1980s. Kaplan and Zoch (2020) document an increasing share and average wage of expansionary work, a broader concept including employment in sales and marketing, as well as product development. Bronnenberg, Dube and Syverson (2022) find that the marketing manager share of payroll has increased 25% from 2005 to 2019. Data presented in Greenwood et al. (2024) show that the advertising-to-GDP ratio has increased 35% from 1980 to 2019. The KLEMS database provides a time series of investment in brands and customers at the level of national accounts.²⁷ The ratio of brand value investment to GDP has increased by 15% from 1985 to 2020. Conservatively, we target a 15% increase in aggregate demand investment (measured relative to aggregate output) over time.²⁸ Over the same time period, industry sales concentration has also increased significantly. We target a 25.8% increase in the top 1% sales share, which we calculate as the average increase in the top 4 or top 20 firm sales share (whichever is closer to the top 1% of firms) across sectors reported in Autor et al. (2020).

Table 5 compares the two calibrated steady states. We infer changes in the two parameters ($\bar{s}$, ψ) of the contact cost function to replicate the rise in aggregate demand investment and sales concentration. Holding fixed contact intensities $i(z)$ at the 1980s allocation, average demand investment per customer falls by 31.0% and the P75/P25 ratio falls by 8.3%, implying lower mean contact costs and lower convexity. The two changes are separately identified: $\bar{s}$ mainly affects demand investment, while ψ mainly affects sales concentration.

5.2 The Effects of Rising Demand Investment

At the firm level, the calibrated change in contact technology raises contact intensity for most suppliers, especially the most productive, and reallocates customers toward them.²⁹ Across steady states, this raises aggregate TFP by 3.1% and GDP by 4.6%.

The effect on the aggregate markup is theoretically ambiguous. Higher demand investment intensifies competition for customers and lowers firm-level markups for given productivity, but greater sales concentration reallocates activity toward more productive, high-markup firms. Qualitatively, these opposing forces are consistent with documented changes to firms'

²⁷See Bontadini, Corrado, Haskel, Iommi and Jona-Lasinio (2023) for a description of the KLEMS database and e.g. Corrado, Haskel, Jona-Lasinio and Iommi (2022) for additional analysis on intangible investment using the data. The data can be accessed via the Luiss Lab of European Economics at <https://euklems-intanprod-llée.luiss.it>.

²⁸The sales and marketing spending variable in Capital IQ, which we use to calibrate the baseline 2010s steady state, is not reliable before 2007 as discussed by He et al. (2025).

²⁹Figure A9 in the appendix reports how outcomes and choices change along the productivity distribution.

Table 5: Rise in Demand Investment: 1980s vs. 2010s

		1980s	2010s	% Δ
Demand investment / gross output (target +15%)	$\frac{\Gamma}{Y}$	0.073	0.085	15.0
Top 1% sales share (target +25.8%)		0.342	0.431	25.8
Average demand investment (fixing $i(z)$)		0.326	0.225	-31.0
P75/P25 demand investment (fixing $i(z)$)		3.430	3.146	-8.3
GDP	$Y - \Gamma$	1.000	1.046	4.6
Aggregate TFP	Z	1.590	1.640	3.1
Average customer quit rate	$\mathbb{E}_z[q(z)]$	0.140	0.152	8.8
Aggregate markup	$\mathcal{M}$	1.201	1.197	-0.4
Intangible capital intensity	$\frac{K^V}{K}$	0.557	0.487	-12.5

Notes: Quantities normalized so that $GDP = 1$ in the 1980s steady state. The table compares the calibrated 2010s steady state to another steady state (“1980s”) with the contact cost parameters $(\bar{s}, \psi)$ recalibrated to generate the targeted increase in demand investment and sales concentration. Middle-panel demand-investment measures hold fixed contact intensities $i(z)$ at 1980s levels. % Δ reports percent changes relative to the 1980s steady state. The customer quit rate is computed as sales-weighted average across firms.

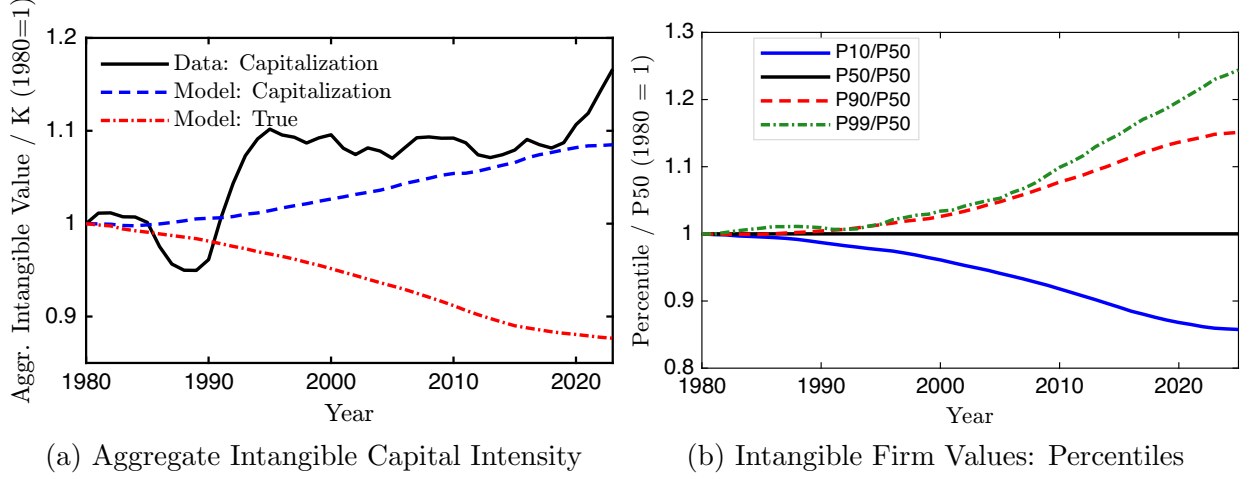
markups and labor shares in the U.S. economy (e.g., Autor et al., 2020; De Loecker et al., 2020). Quantitatively, the within-firm effect slightly dominates the reallocation effect, so the aggregate markup falls by 0.4%. Thus, with endogenous market power due to search frictions, rising concentration need not imply higher markups. When concentration arises from changes in the technology with which customers are matched to suppliers, an increase in concentration can coincide with a market that becomes more competitive.

Rising demand investment also has sharp implications for the value of customer relationships. The intangible intensity of the capital stock—measured as the value of customers $K^V = \int_0^\infty V(z)dG(z)$ relative to physical capital K —falls by 12.5% across steady states, despite the increasing share of demand investment in output. This does not contradict evidence based on perpetual-inventory measures of intangible capital, which capitalize spending using a fixed depreciation rate.³⁰ When we do the same, both the model and the data display roughly a 10% increase in intangible intensity. The transition path is reported in panel (a) of Figure 5, which assumes a linear change in the contact-cost parameters over time and also reports the true intangible capital intensity in the model.³¹

³⁰For applications of the perpetual-inventory method to measure general intangible capital, see Eisfeldt and Papanikolaou (2013), Eisfeldt and Papanikolaou (2014), and Peters and Taylor (2017).

³¹We apply the average customer quit rate in the 1980s economy as the depreciation rate. To be able to go back to the 1980s, we capitalize total SGA spending in the data. As we normalize the 1980s ratio to 1 for all time series, this approximation is valid as long as demand investment has grown in parallel with total SGA spending.

Figure 5: Technological Change in Contact Cost: Effects on Intangible Capital Intensity



Notes: Panel (a) shows the time series of intangible capital intensity $\frac{K^V}{K}$ following the technological change in $s(\cdot)$, using both the capitalization method (model and data) and the true intangible capital intensity in the model. Panel (b) shows the evolution of percentile ratios of firm-level customer-base values, computed as $nV(z)$ over the joint distribution (n, z) . All series are normalized to 1 in 1980.

The difference between capitalized spending and the true intangible value arises because aggregate demand investment endogenously changes the customer quit rate, the depreciation rate of customer capital. Higher demand investment by competitors raises turnover, shortens relationship duration, and, for the median firm, lowers the present value of profits from each customer. This result cautions against mechanically applying the capitalization approach with a fixed depreciation rate to infer the value of firms' customer bases. Our framework suggests that changes in demand investment affect not only the number of customers a firm accumulates but also each customer's equilibrium value.

The effects on intangible values are heterogeneous across firms. While the least productive firms lose customers faster, the most productive firms can poach more easily from competitors. Panel (b) of Figure 5 shows that this leads to a fanning out of firm values: the intangible value of the most productive firms rises relative to the median firm as the economy transitions from the 1980s to the 2010s. This reflects opposite-signed changes in customer values across the productivity distribution: $V(z)$ rises for high-productivity firms but falls for low-productivity firms, consistent with evidence of a widening dispersion in firm values and profits (Eeckhout, 2025).

Robustness. Appendix E.3 shows that the results in this section are qualitatively and quantitatively similar under alternative calibrations. Our benchmark loads the rise in concentration on improved contact technology and implies little net change in the aggregate markup. While empirical estimates vary, an influential literature finds that aggregate markups have

risen meaningfully since the 1980s (De Loecker et al., 2020). Appendix D.4 shows that the model can accommodate an increase in the aggregate markup by attributing part of the rise in concentration to higher firm-level productivity dispersion, σ_z .

6 Conclusion

We have outlined a new quantitative framework to study how investment in demand shapes dynamic competition for customers. We show that the model matches salient empirical facts about how firms compete for customers and highlight two channels through which demand investment affects allocative efficiency. First, demand investment exerts a pro-competitive force by increasing customer turnover and lowering markups, thereby mitigating misallocation from imperfect competition. Second, firms over-invest in contacting customers because they do not internalize business-stealing externalities imposed on competitors. Our results have direct implications for the conduct of competition policy and suggest that instruments aimed at correcting distortions from market power must jointly account for firms’ dual margins of competition—prices and demand investment. The framework also provides a novel perspective on macroeconomic trends, accounting for rising industry concentration without implying greater market power, and highlighting equilibrium forces that may reduce firms’ intangible value despite higher investment.

Our findings relate to a broader agenda for understanding the role of customer-base dynamics in industry competition and macroeconomic outcomes. We provide a framework in which competitive forces are not driven by product differentiation, and we have focused our analysis on implications for misallocation, productivity, and the intangible value of firms. Future work may explore the role of competition for customers in the context of other macroeconomic questions where firms’ margins and profitability are important drivers of outcomes, such as growth or the transmission of monetary policy to prices. Our results also call for richer measurement of customer dynamics and margins of competition beyond prices, both at the micro and macro level. We leave these questions for future research.

References

- AFROUZI, H., A. DRENIK, AND R. KIM (2025): “Concentration, Market Power, and Misallocation: The Role of Endogenous Customer Acquisition,” NBER Working Paper 31415.
- ALBRECHT, J., G. MENZIO, AND S. VROMAN (2023): “Vertical differentiation in frictional product markets,” *Journal of Political Economy Macroeconomics*, 1(3), 586–632.
- AMITI, M., O. ITSKHOKI, AND J. KONINGS (2019): “International shocks, variable markups, and domestic prices,” *The Review of Economic Studies*, 86(6), 2356–2402.

- ANDREWS, I., M. GENTZKOW, AND J. M. SHAPIRO (2017): “Measuring the Sensitivity of Parameter Estimates to Estimation Moments,” *Quarterly Journal of Economics*, 132(4), 1553–1592.
- ARGENTE, D., D. FITZGERALD, S. MOREIRA, AND A. PRIOLO (2025): “How Do Entrants Build Market Share? The Role of Demand Frictions,” *American Economic Review: Insights*, 7(2), 160–76.
- ARGENTE, D., E. MÉNDEZ, AND D. VAN PATTEN (2024): “Cross-Border Product Adoption: Individual Imports, Migrant Networks, and Domestic Retailers,” NBER Working Paper 35104.
- ARKOLAKIS, C. (2010): “Market penetration costs and the new consumers margin in international trade,” *Journal of political economy*, 118(6), 1151–1199.
- ATKESON, A., AND A. BURSTEIN (2008): “Pricing-to-market, trade costs, and international relative prices,” *American Economic Review*, 98(5), 1998–2031.
- AUTOR, D., D. DORN, L. F. KATZ, C. PATTERSON, AND J. VAN REENEN (2020): “The fall of the labor share and the rise of superstar firms,” *The Quarterly Journal of Economics*, 135(2), 645–709.
- BAI, Y., J.-V. RIOS-RULL, AND K. STORESLETTEN (2026): “Demand shocks as technology shocks,” *Review of Economic Studies*, 93(2), 798–832.
- BAILEY, M., D. JOHNSTON, T. KUCHLER, J. STROEBEL, AND A. WONG (2022): “Peer effects in product adoption,” *American Economic Journal: Applied Economics*, 14(3), 488–526.
- BAQAEI, D., A. BURSTEIN, C. DUPREZ, AND E. FARHI (2025): “Consumer Surplus from Suppliers: How Big is it and Does it Matter for Growth?,” *Econometrica*, 93(6), 2043–2081.
- BAQAEI, D. R., AND E. FARHI (2020): “Productivity and misallocation in general equilibrium,” *The Quarterly Journal of Economics*, 135(1), 105–163.
- BAQAEI, D. R., E. FARHI, AND K. SANGANI (2024a): “The darwinian returns to scale,” *Review of Economic Studies*, 91(3), 1373–1405.
- BAQAEI, D. R., E. FARHI, AND K. SANGANI (2024b): “The supply-side effects of monetary policy,” *Journal of Political Economy*, 132(4), 1065–1112.
- BARKAI, S. (2020): “Declining labor and capital shares,” *The Journal of Finance*, LXXV(5).
- BASLANDZE, S., J. GREENWOOD, R. MARTO, AND S. MOREIRA (2023): “The expansion of varieties in the new age of advertising,” *Review of Economic Dynamics*, 50, 171–210.
- BERNARD, A. B., E. DHYNE, G. MAGERMAN, K. MANOVA, AND A. MOXNES (2022): “The origins of firm heterogeneity: A production network approach,” *Journal of Political Economy*, 130(7), 1765–1804.
- BHANDARI, A., AND E. R. MCGRATTAN (2021): “Sweat equity in US private business,” *The Quarterly Journal of Economics*, 136(2), 727–781.
- BOEHM, J., R. SOUTH, E. OBERFIELD, AND M. WASEEM (2024): “The network origins of firm dynamics: Contracting frictions and dynamism with long-term relationships,” Working paper.
- BONTADINI, F., C. CORRADO, J. HASKEL, M. IOMMI, AND C. JONA-LASINIO (2023):

- “EUKLEMS & INTANProd: industry productivity accounts with intangibles,” EUKLEMS & INTANProd database documentation.
- BORNSTEIN, G., AND A. PETER (2025): “Nonlinear pricing and misallocation,” *American Economic Review*, 115(11), 3853–3908.
- BRONNENBERG, B. J., J.-P. DUBE, AND C. SYVERSON (2022): “Marketing Investment and Intangible Brand Capital,” *Journal of Economic Perspectives*, 36(3), 53–74.
- BURDETT, K., AND K. L. JUDD (1983): “Equilibrium price dispersion,” *Econometrica: Journal of the Econometric Society*, pp. 955–969.
- BUTTERS, G. R. (1977): “Equilibrium Distributions of Sales and Advertising Prices,” *Review of Economic Studies*, 44(3).
- CARVALHO, V. M., M. NIREI, Y. U. SAITO, AND A. TAHBAZ-SALEHI (2021): “Supply chain disruptions: Evidence from the great east japan earthquake,” *The Quarterly Journal of Economics*, 136(2), 1255–1321.
- CAVENAILE, L., M. A. CELIK, J. PERLA, AND P. ROLDAN-BLANCO (2023): “A theory of dynamic product awareness and targeted advertising,” Working paper.
- CAVENAILE, L., M. A. CELIK, P. ROLDAN-BLANCO, AND X. TIAN (2025): “Style over substance? Advertising, innovation, and endogenous market structure,” *Journal of Monetary Economics*, 149, 103683.
- CAVENAILE, L., AND P. ROLDAN-BLANCO (2021): “Advertising, innovation, and economic growth,” *American Economic Journal: Macroeconomics*, 13(3), 251–303.
- CHIAVARI, A. (2024): “Customer Accumulation, Returns to Scale, and Secular Trends,” Working paper.
- COLES, M. G., AND D. T. MORTENSEN (2016): “Equilibrium Labor Turnover, Firm Growth, and Unemployment,” *Econometrica*, 84(1), 347–363.
- CORRADO, C., J. HASKEL, C. JONA-LASINIO, AND M. IOMMI (2022): “Intangible capital and modern economies,” *Journal of Economic Perspectives*, 36(3), 3–28.
- CROUZET, N., J. EBERLY, A. EISFELDT, AND D. PAPANIKOLAOU (2024): “Intangible Capital, Firm Scope, and Growth,” Working paper.
- CROUZET, N., J. C. EBERLY, A. L. EISFELDT, AND D. PAPANIKOLAOU (2022): “The economics of intangible capital,” *Journal of Economic Perspectives*, 36(3), 29–52.
- DE LOECKER, J., J. EECKHOUT, AND G. UNGER (2020): “The rise of market power and the macroeconomic implications,” *The Quarterly Journal of Economics*, 135(2), 561–644.
- DE RIDDER, M. (2024): “Market power and innovation in the intangible economy,” *American Economic Review*, 114(1), 199–251.
- DE RIDDER, M., B. GRASSI, AND G. MORZENTI (2026): “The Hitchhiker’s Guide to Markup Estimation: Assessing Estimates from Financial Data,” *Econometrica*, 94(1), 137–168.
- DHYNE, E., C. DUPREZ, AND T. KOMATSU (2023): “The Belgian Business-to-Business Transactions Dataset 2002–2021,” NBB Working Paper 444.
- DINLERSOZ, E. M., AND M. YORUKOGLU (2012): “Information and industry dynamics,” *American Economic Review*, 102(2), 884–913.
- DROZD, L. A., AND J. B. NOSAL (2012): “Understanding international prices: Customers as capital,” *American Economic Review*, 102(1), 364–395.

- EDMOND, C., V. MIDRIGAN, AND D. Y. XU (2023): “How costly are markups?,” *Journal of Political Economy*, 131(7), 1619–1675.
- ECKHOUT, J. (2025): “Presidential Address 2024: The Value and Profits of Firms,” *Journal of the European Economic Association*, 23(1), 52–118.
- EINAV, L., P. J. KLENOW, J. D. LEVIN, AND R. MURCIANO-GOROFF (2021): “Customers and Retail Growth,” NBER Working Paper 29561.
- EISFELDT, A. L., AND D. PAPANIKOLAOU (2013): “Organization capital and the cross-section of expected returns,” *The Journal of Finance*, 68(4), 1365–1406.
- (2014): “The value and ownership of intangible capital,” *American Economic Review*, 104(5), 189–194.
- FERNÁNDEZ-VILLAVERDE, J., F. MANDELMAN, Y. YU, AND F. ZANETTI (2024): “Search complementarities, aggregate fluctuations, and fiscal policy,” *Review of Economic Studies*, 92(4), 2502–2536.
- FITZGERALD, D., S. HALLER, AND Y. YEDID-LEVI (2024): “How exporters grow,” *Review of Economic Studies*, 91(4), 2276–2306.
- FOSTER, L. S., J. C. HALTIWANGER, AND C. TUTTLE (2022): “Rising Markups or Changing Technology?,” NBER Working Paper 30491.
- FUKUI, M., AND T. MUKOYAMA (2025): “Efficiency in Job-Ladder Models,” NBER Working Paper 33910.
- GARTHWAITE, C. L. (2014): “Demand Spillovers, Combative Advertising, and Celebrity Endorsements,” *American Economic Journal: Applied Economics*, 6(2), 76–104.
- GOTTFRIES, A., AND G. JAROSCH (2025): “Dynamic monopsony with large firms and an application to non-competes,” NBER Working Paper 31965.
- GOUIN-BONENFANT, É. (2022): “Productivity dispersion, between-firm competition, and the labor share,” *Econometrica*, 90(6), 2755–2793.
- GOURIO, F., AND L. RUDANKO (2014): “Customer Capital,” *The Review of Economic Studies*, 81(3), 1102–1136.
- GREENWOOD, J., Y. MA, AND M. YORUKOGLU (2024): ““You Will:” A Macroeconomic Analysis of Digital Advertising,” *The Review of Economic Studies*, 92(3), 1837–1881.
- GROSSMAN, G. M., AND E. HELPMAN (1991): “Quality ladders in the theory of growth,” *The review of economic studies*, 58(1), 43–61.
- HARTMANN, W. R., AND D. KLAPPER (2018): “Super Bowl Ads,” *Marketing Science*, 37(1), 78–96.
- HE, B., L. I. MOSTROM, AND A. SUFI (2025): “Investing in Customer Capital,” NBER Working Paper 33171.
- HOSIOS, A. J. (1990): “On the efficiency of matching and related models of search and unemployment,” *The Review of Economic Studies*, 57(2), 279–298.
- HSIEH, C.-T., AND P. J. KLENOW (2009): “Misallocation and manufacturing TFP in China and India,” *The Quarterly journal of economics*, 124(4), 1403–1448.
- HUBMER, J., AND P. RESTREPO (2026): “Not a Typical Firm: Capital–Labor Substitution and Firms’ Labor Shares,” *American Economic Journal: Macroeconomics*, 18(2), 34–71.
- IGNASZAK, M., AND P. SEDLÁČEK (2023): “Customer Acquisition, Business Dynamism and Aggregate Growth,” Working paper.

- KAPLAN, G., AND P. ZOCH (2020): “Markups, Labor Market Inequality and the Nature of Work,” NBER Working Paper 26800.
- KEHRIG, M., AND N. VINCENT (2021): “The Micro-Level Anatomy of the Labor Share Decline,” *The Quarterly Journal of Economics*, 136(2), 1031–1087.
- KIMBALL, M. S. (1995): “The Quantitative Analytics of the Basic Neomonetarist Model,” *Journal of Money, Credit and Banking*, 27(4), 1241–1277.
- MENZIO, G. (2023): “Optimal product design: Implications for competition and growth under declining search frictions,” *Econometrica*, 91(2), 605–639.
- (2024): “Efficient Imperfect Competition with an Application to International Trade,” NBER Working Paper 33253.
- (2026): “Markups: A search-theoretic perspective,” *Journal of Political Economy*, forthcoming.
- MORTENSEN, D. T. (1982): “Property rights and efficiency in mating, racing, and related games,” *The American Economic Review*, 72(5), 968–979.
- MOSCARINI, G., AND F. POSTEL-VINAY (2013): “Stochastic search equilibrium,” *Review of Economic Studies*, 80(4), 1545–1581.
- OTTONELLO, P., AND T. WINBERRY (2020): “Financial Heterogeneity and the Investment Channel of Monetary Policy,” *Econometrica*, 88(6), 2473–2502.
- PACIELLO, L., A. POZZI, AND N. TRACHTER (2019): “PRICE DYNAMICS WITH CUSTOMER MARKETS,” *International Economic Review*, 60(1), 413–446.
- PEARCE, J., AND L. WU (2025): “Market Concentration and Aggregate Productivity: The Role of Demand,” Staff Reports 1159, Federal Reserve Bank of New York.
- PETER, A., AND C. RUANE (2025): “Intermediate Input Substitutability,” NBER Working Paper 31233.
- PETERS, M. (2020): “Heterogeneous markups, growth, and endogenous misallocation,” *Econometrica*, 88(5), 2037–2073.
- PETERS, R. H., AND L. A. TAYLOR (2017): “Intangible capital and the investment-q relation,” *Journal of Financial Economics*, 123(2), 251–272.
- RESTUCCIA, D., AND R. ROGERSON (2008): “Policy distortions and aggregate productivity with heterogeneous establishments,” *Review of Economic dynamics*, 11(4), 707–720.
- ROLDAN-BLANCO, P., AND S. GILBUKH (2021): “Firm dynamics and pricing under customer capital accumulation,” *Journal of Monetary Economics*, 118, 99–119.
- RUDANKO, L. (2025): “Price Setting with Customer Capital: Sales, Teasers, and Rigidity,” *Journal of Political Economy Macroeconomics*, forthcoming.
- SHEN, S. (2025): “Search frictions, customer allocation, and business dynamism,” Working paper.
- SINKINSON, M., AND A. STARC (2019): “Ask Your Doctor? Direct-to-Consumer Advertising of Pharmaceuticals,” *The Review of Economic Studies*, 86(2), 836–881.
- STERK, V., P. SEDLÁČEK, AND B. PUGSLEY (2021): “The Nature of Firm Growth,” *American Economic Review*, 111(2), 547–79.

ONLINE APPENDIX

(Intended for Online Publication Only)

Investment in Demand and Dynamic Competition for Customers

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A Additional Theoretical Results

A.1 Aggregation

This appendix derives the aggregate markup and aggregate productivity in the decentralized equilibrium, taking as given suppliers' price setting and the distribution of customers across suppliers. As the analysis in this section is static, we omit time subscripts for ease of notation.

We define aggregate productivity in the decentralized equilibrium Z^M as the Solow residual of a hypothetical aggregate production function with identical Cobb-Douglas structure as in (3), that is

$$(A.1) \quad Z^M = \frac{Y}{K^\alpha L^{1-\alpha}}$$

where K and L are aggregate capital and labor input in the economy and Y is aggregate output of the final good. We can derive Z^M as

$$\begin{aligned} Z^M &= \frac{Y}{\left(\bar{\ell} u y(\bar{m} \bar{c}) + \int_0^\infty \frac{\bar{\ell}}{z} y(p(z)) dG(z)\right)^{1-\alpha} \left(\bar{k} u y(\bar{m} \bar{c}) + \int_0^\infty \frac{\bar{k}}{z} y(p(z)) dG(z)\right)^\alpha} \\ &= \left(\bar{\ell}^{1-\alpha} \bar{k}^\alpha \left(u \frac{y(\bar{m} \bar{c})}{Y} + \int_0^\infty \frac{1}{z} \frac{y(p(z))}{Y} dG(z)\right)\right)^{-1} \\ (A.2) \quad &= \left(u \bar{m} \bar{c}^{-\sigma} + \int_0^\infty \frac{1}{z} (p(z))^{-\sigma} dG(z)\right)^{-1} \end{aligned}$$

We can further simplify this expression to write Z^M as a function of markups

$$\begin{aligned}
Z^M &= \overline{mc}^\sigma \left(u + \int_0^\infty \frac{1}{z} \left(\frac{\mu(z)}{z} \right)^{-\sigma} dG(z) \right)^{-1} \\
&= \left(\frac{Z^M}{\mathcal{M}} \right)^\sigma \left(u + \int_0^\infty \frac{1}{z} \left(\frac{\mu(z)}{z} \right)^{-\sigma} dG(z) \right)^{-1} \\
\Rightarrow Z^M &= \left(u \left(\frac{1}{\mathcal{M}} \right)^{-\sigma} + \int_0^\infty z^{\sigma-1} \left(\frac{\mu(z)}{\mathcal{M}} \right)^{-\sigma} dG(z) \right)^{\frac{1}{\sigma-1}}
\end{aligned}
\tag{A.3}$$

where $\mu(z) = \frac{p(z)}{\frac{\overline{mc}}{z}}$ is the markup charged by a supplier with productivity z and $\mathcal{M} = \frac{1}{\frac{\overline{mc}}{Z^M}}$ is defined as the aggregate markup in the economy. Aggregate productivity is given by a weighted average of individual productivities, where weights are determined by the relative markups of firms with different productivities.

We can derive a condition for the aggregate markup $\mathcal{M}$ from the labor share of output where

$$\frac{wL}{Y} = \frac{w \frac{Y}{Z^M} \bar{\ell}}{Y} = \frac{w}{Z^M} \frac{(1-\alpha)\overline{mc}}{w} = \frac{1-\alpha}{\mathcal{M}}
\tag{A.4}$$

The same relationship has to hold for any supplier firm, such that

$$\frac{wL}{Y} \mathcal{M} = \mu(z) \frac{w\ell(z)}{p(z)y(z)}
\tag{A.5}$$

$$\Rightarrow \mathcal{M} \frac{p(z)y(z)}{Y} = \frac{\ell(z)}{L} \mu(z)
\tag{A.6}$$

Integrating both sides over the distribution of matches and adding the equivalent condition for the mass of unmatched customers on both sides yields the aggregate markup as

$$\mathcal{M} = u \frac{\bar{\ell}y(\overline{mc})}{L} + \int_0^\infty \frac{\bar{\ell}y(p(z))}{zL} \mu(z) dG(z)
\tag{A.7}$$

where we make use of the fact that we have normalized the price of the final good in (8) to one. As in Edmond et al. (2023), the aggregate markup is given by the input-weighted average of individual markups, evaluated at equilibrium demand for labor and the distribution of matches $G(z)$.³²

³²As the capital-labor-ratio is identical across all firms, we could have arrived at the same expression starting from the capital share of output and results are identical if we weight by capital.

A.2 Social Planner Problem

This appendix outlines and solves the problem of a social planner that maximizes welfare in the economy subject to the same technologies and search frictions as the decentralized equilibrium. In particular, the planner makes choices for households, final good firms and intermediate good firms, taking into account their full effect on aggregate outcomes. We separate the problem of the planner into two stages: First, we solve for the optimal allocation of demand per match across intermediates, taking as given the distribution of matches. As this is a static problem we can separate it from the dynamic considerations of the planner for ease of exposition. Second, we solve for the dynamic problem of the planner that chooses the allocation of customers across suppliers as well as aggregate quantities, taking the optimal static allocation of demand at the intensive margin for a given distribution of customers as given.

Intensive-margin allocation. Consider the optimal allocation of demand a planner would choose at the intensive margin. The problem is static and we omit time subscripts. The planner takes the distribution of matches $G(z)$ and mass of unmatched u as given. As in-house production and intermediate firms operate the same Cobb-Douglas technology with varying productivity, to minimize cost the planner solves

$$(A.8) \quad \min_{y^u, \{y(z)\}_z} uy^u + \int_0^\infty \frac{y(z)}{z} dG(z)$$

$$(A.9) \quad \text{s.t.} \quad \left(u(y^u)^{\frac{\sigma-1}{\sigma}} + \int_0^\infty y(z)^{\frac{\sigma-1}{\sigma}} dG(z) \right)^{\frac{\sigma}{\sigma-1}} = Y$$

This is equivalent to maximizing aggregate productivity $Z = \frac{Y}{K^\alpha L^{1-\alpha}}$. Taking FOCs with respect to $y(z)$ yields

$$(A.10) \quad \frac{1}{z} = y(z)^{-\frac{1}{\sigma}} Y^{\frac{1}{\sigma}} \lambda$$

where λ is the Lagrange multiplier on the constraint. The derivation for y^u is analogous. Multiplying both sides by $y(z)$, integrating over z with $G(z)$ and adding the corresponding condition for y^u yields

$$\begin{aligned} uy^u + \int_0^\infty \frac{y(z)}{z} dG(z) &= \lambda Y^{\frac{1}{\sigma}} \left(u(y^u)^{\frac{\sigma-1}{\sigma}} + \int_0^\infty y(z)^{\frac{\sigma-1}{\sigma}} dG(z) \right) = \lambda Y \\ \Rightarrow K^\alpha L^{1-\alpha} &= \lambda Y \\ (A.11) \quad \Rightarrow \lambda &= \frac{1}{Z} \end{aligned}$$

This yields the optimal relative size of firms under the planner allocation as

$$(A.12) \quad \frac{y(z)}{Y} = \left(\frac{Z}{z} \right)^{-\sigma}$$

and the corresponding aggregate productivity under the optimal allocation as

$$(A.13) \quad \begin{aligned} Z &= \left(u \frac{y^u}{Y} + \int_0^\infty \frac{1}{z} \frac{y(z)}{Y} dG(z) \right)^{-1} \\ \Rightarrow Z &= \left(u + \int_0^\infty z^{\sigma-1} dG(z) \right)^{\frac{1}{\sigma-1}} \end{aligned}$$

Efficient allocation of customers We solve for the dynamic problem of a social planner that chooses the allocation of customers across suppliers as well as aggregate quantities, taking the optimal static allocation of demand at the intensive margin given a distribution of customers as derived above as given. From the effect of a match on aggregate productivity and because of the properties of $F(z'|z)$, it is straightforward to see that given two contacts the planner will always prefer a match with higher productivity. We incorporate this rule of how to transition matches among contacts when setting up the planner problem. The full problem is defined as

$$(A.14) \quad \max_{\{C_t^*, L_t^*, K_{t+1}^*, \{i_t^*(z), g_t^{i*}(z), g_t^*(z), \delta_t^*(z), \delta_t^{E*}(z)\}_{z=0}^\infty\}_{t=0}^\infty} \sum_{t=0}^\infty \beta^t \{u(C_t^*) - v(L_t^*)\}$$

$$(A.15) \quad \text{s.t. } C_t^* + K_{t+1}^* + \int_0^\infty s(i_t^*(z)) g_t^{i*}(z) dz \leq Z_t^* (K_t^*)^\alpha (L_t^*)^{1-\alpha} + (1 - \delta_k) K_t^* \quad \forall t$$

$$(A.16) \quad Z_t^* = \left(u_t^* + \int_0^\infty z^{\sigma-1} g_t^*(z) dz \right)^{\frac{1}{\sigma-1}} \quad \forall t$$

$$(A.17) \quad g_t^*(z) = (1 - q_t^*(z) + e_t^*(z) i_t^*(z)) g_t^{i*}(z) \quad \forall t, z$$

$$(A.18) \quad u_t^* = 1 - \int_0^\infty g_t^*(z) dz \quad \forall t$$

$$(A.19) \quad g_t^{i*}(z) = (1 - \delta_t^*(z)) \int_0^\infty g_{t-1}^*(\hat{z}) f(z|\hat{z}) d\hat{z} + (1 - \delta_t^{E*}(z)) \chi u_{t-1}^* f_0(z) \quad \forall t, z$$

$$(A.20) \quad u_t^{i*} = u_{t-1}^* + \int_0^\infty \delta_t^*(z) \int_0^\infty g_{t-1}^*(\hat{z}) f(z|\hat{z}) d\hat{z} dz - \int_0^\infty (1 - \delta_t^{E*}(z)) \chi u_{t-1}^* f_0(z) dz \quad \forall t$$

$$(A.21) \quad \delta_t^*(z) \geq \delta_F, \quad \delta_t^*(z) \leq 1 \quad \forall t, z$$

$$(A.22) \quad \delta_t^{E*}(z) \geq 0, \quad \delta_t^{E*}(z) \leq 1 \quad \forall t, z$$

$$(A.23) \quad e_t^*(z) = \left[u_t^i + \int_0^z g_t^{i*}(\hat{z}) d\hat{z} \right] e^{-I_t^*(1-D_t^*(z))} \quad \forall t, z$$

$$(A.24) \quad q_t^*(z) = 1 - e^{-I_t^*(1-D_t^*(z))} \quad \forall t, z$$

$$(A.25) \quad I_t^* = \int_0^\infty i_t^*(z) g_t^{i*}(z) dz \quad \forall t$$

$$(A.26) \quad D_t^*(z) = \int_0^z \frac{i_t^*(\hat{z})}{I_t^*} g_t^{i*}(\hat{z}) d\hat{z} \quad \forall t, z$$

for given K_0^* , u_{-1}^* and $\{g_{-1}^*(z)\}_{z=0}^\infty$. $g_{-1}^{i*}(z)$ and $g_{-1}^*(z)$ are the mass of customers matched with suppliers of productivity z at the beginning of the period and during production, respectively. $\delta_t^*(z)$ is the fraction of firms of type z the planner chooses to exit and $\delta_t^{E*}(z)$ the fraction not to enter the economy.

We can make some substitutions to simplify the problem

$$(A.27) \quad \max_{\{C_t^*, L_t^*, K_{t+1}^*, \{i_t^*(z), g_t^{i*}(z), g_t^*(z), \delta_t^*(z)\}_{z=0}^\infty\}_{t=0}^\infty} \sum_{t=0}^\infty \beta^t \{u(C_t^*) - v(L_t^*)\}$$

$$(A.28) \quad \text{s.t. } C_t^* + K_{t+1}^* + \int_0^\infty s(i_t^*(z)) g_t^{i*}(z) dz \leq Y(K_t^*, L_t^*, \{g_t^*(z)\}) + (1 - \delta_k) K_t^* \quad \forall t$$

$$(A.29) \quad Y(K_t^*, L_t^*, \{g_t^*(z)\}) = \left(1 + \int_0^\infty (z^{\sigma-1} - 1) g_t^*(z) dz\right)^{\frac{1}{\sigma-1}} (K_t^*)^\alpha (L_t^*)^{1-\alpha}$$

$$(A.30) \quad g_t^*(z) \leq (1 - q_t^*(z) + e_t^*(z) i_t^*(z)) g_t^{i*}(z) \quad \forall t, z$$

$$(A.31) \quad g_t^{i*}(z) \leq \int_0^\infty g_{t-1}^*(\hat{z}) [(1 - \delta_t^*(z)) f(z|\hat{z}) - \chi(1 - \delta_t^{E*}(z)) f_0(z)] d\hat{z} \\ + \chi(1 - \delta_t^{E*}(z)) f_0(z) \quad \forall t, z$$

$$(A.32) \quad \delta_t^*(z) \geq \delta_F, \quad \delta_t^*(z) \leq 1 \quad \forall t, z$$

$$(A.33) \quad \delta_t^{E*}(z) \geq 0, \quad \delta_t^{E*}(z) \leq 1 \quad \forall t, z$$

$$(A.34) \quad e_t^*(z) = \left[1 - \int_z^\infty g_t^{i*}(\hat{z}) d\hat{z}\right] e^{-\int_z^\infty i_t^*(\hat{z}) g_t^{i*}(\hat{z}) d\hat{z}} \quad \forall t, z$$

$$(A.35) \quad q_t^*(z) = 1 - e^{-\int_z^\infty i_t^*(\hat{z}) g_t^{i*}(\hat{z}) d\hat{z}} \quad \forall t, z$$

for given K_0^* and $\{g_{-1}^*(z)\}_{z=0}^\infty$. Denote as λ_t^{BC} the multiplier on the budget constraint in (A.28), λ_{zt}^i the multiplier on the law of motion for $g^{i*}(z)$ in (A.31), λ_{zt}^ρ the multiplier on the growth rate for $g^*(z)$ in (A.30), and $\lambda_t^{\delta L}$, $\lambda_t^{\delta H}$, λ_t^{EL} , λ_t^{EH} the multipliers on the boundaries for $\delta_t^*(z)$ and $\delta_t^{E*}(z)$ in (A.32) and (A.33). To ease notation, we write the transition probabilities of z including entry and exit as $\hat{f}_{t+1}(z'|z) = (1 - \delta_{t+1}^*(z')) f(z'|z)$ and $\hat{f}_{t+1}^0(z) = (1 - \delta_{t+1}^{E*}(z)) f_0(z)$.

It is useful to first take some relevant derivatives of the auxiliary equations in (A.29), (A.34), and (A.35) that we will use later. These are given as

$$\begin{aligned} \frac{\partial Y_t}{\partial g_t^*(z)} &= \frac{z^{\sigma-1} - 1}{\sigma - 1} \left(1 + \int_0^\infty (z^{\sigma-1} - 1) g_t^*(z) dz\right)^{\frac{2-\sigma}{\sigma-1}} (K_t^*)^\alpha (L_t^*)^{1-\alpha} \\ &= \frac{z^{\sigma-1} - 1}{\sigma - 1} (Z_t^*)^{2-\sigma} (K_t^*)^\alpha (L_t^*)^{1-\alpha} \\ \frac{\partial e_t^*(z)}{\partial g_t^{i*}(z')} &= -e^{-\int_z^\infty i_t^*(\hat{z}) g_t^{i*}(\hat{z}) d\hat{z}} - i_t^*(z') \left[1 - \int_z^\infty g_t^{i*}(\hat{z}) d\hat{z}\right] e^{-\int_z^\infty i_t^*(\hat{z}) g_t^{i*}(\hat{z}) d\hat{z}} \quad \text{iff } z' \geq z, \text{ else } 0 \\ \frac{\partial q_t^*(z)}{\partial g_t^{i*}(z')} &= i_t^*(z') e^{-\int_z^\infty i_t^*(\hat{z}) g_t^{i*}(\hat{z}) d\hat{z}} \quad \text{iff } z' \geq z, \text{ else } 0 \\ \frac{\partial e_t^*(z)}{\partial i_t^*(z')} &= -g_t^{i*}(z') \left[1 - \int_z^\infty g_t^{i*}(\hat{z}) d\hat{z}\right] e^{-\int_z^\infty i_t^*(\hat{z}) g_t^{i*}(\hat{z}) d\hat{z}} \quad \text{iff } z' \geq z, \text{ else } 0 \\ \frac{\partial q_t^*(z)}{\partial i_t^*(z')} &= g_t^{i*}(z') e^{-\int_z^\infty i_t^*(\hat{z}) g_t^{i*}(\hat{z}) d\hat{z}} \quad \text{iff } z' \geq z, \text{ else } 0 \end{aligned}$$

The first order conditions with respect to consumption, capital, and labor yield the familiar optimality conditions

$$(A.36) \quad u'(C_t^*) = \beta u'(C_{t+1}^*) \left[(1 - \delta_k) + \frac{\partial Y_{t+1}}{\partial K_{t+1}^*} \right]$$

$$(A.37) \quad u'(C_t^*) \frac{\partial Y_t}{\partial L_t^*} = v'(L_t^*)$$

The first order conditions with respect to $g_t^{i*}(z)$, $g_t^*(z)$, and $i_t^*(z)$ are given by

$$(A.38) \quad 0 = -\lambda_t^{BC} s(i_t^*(z)) + \lambda_{z,t}^\rho (1 - q_t^*(z) + e_t^*(z) i_t^*(z)) - \lambda_{z,t}^i \\ + \int_0^\infty \lambda_{\hat{z},t}^\rho g_t^{i*}(\hat{z}) \left[-\frac{\partial q_t^*(\hat{z})}{\partial g_t^{i*}(z)} + \frac{\partial e_t^*(\hat{z})}{\partial g_t^{i*}(z)} i_t^*(\hat{z}) \right] d\hat{z}$$

$$(A.39) \quad 0 = \lambda_t^{BC} \frac{\partial Y_t}{\partial g_t^*(z)} - \lambda_{z,t}^\rho + \int_0^\infty \lambda_{\hat{z},t+1}^i [\hat{f}_{t+1}(\hat{z}|z) - \chi \hat{f}_{t+1}^0(\hat{z})] d\hat{z}$$

$$(A.40) \quad 0 = -\lambda_t^{BC} s'(i_t^*(z)) g_t^{i*}(z) + \lambda_{z,t}^\rho e_t^*(z) g_t^{i*}(z) \\ + \int_0^\infty \lambda_{\hat{z},t}^\rho g_t^{i*}(\hat{z}) \left[-\frac{\partial q_t^*(\hat{z})}{\partial i_t^*(z)} + \frac{\partial e_t^*(\hat{z})}{\partial i_t^*(z)} i_t^*(\hat{z}) \right] d\hat{z}$$

Note that while the integrals are over all possible productivities of competitors $\hat{z}$, the derivatives inside the integrals are only non-zero for $\hat{z} \leq z$ as derived above. The notation here is equivalent to having the integral in the first and third line go only over $\hat{z} \leq z$.

Define the value of a match between a customer and a supplier with productivity z at the beginning of the period, measured in units of resources, as $V_t^*(z) \equiv \frac{\lambda_{z,t}^i}{\lambda_t^{BC}}$ and the value of such a match at the production stage accordingly as $W_t^*(z) = \frac{\lambda_{z,t}^\rho}{\lambda_t^{BC}}$. From the first order conditions, the two values are given as

$$(A.41) \quad V_t^*(z) = -s(i_t^*(z)) + (1 - q_t^*(z) + e_t^*(z) i_t^*(z)) W_t^*(z) \\ + \int_0^\infty W_t^*(\hat{z}) g_t^{i*}(\hat{z}) \left[-\frac{\partial q_t^*(\hat{z})}{\partial g_t^{i*}(z)} + \frac{\partial e_t^*(\hat{z})}{\partial g_t^{i*}(z)} i_t^*(\hat{z}) \right] d\hat{z}$$

$$(A.42) \quad W_t^*(z) = \frac{\partial Y_t}{\partial g_t^*(z)} + \beta \frac{u'(C_{t+1}^*)}{u'(C_t^*)} \int_0^\infty V_{t+1}^*(\hat{z}) [\hat{f}_{t+1}(\hat{z}|z) - \chi \hat{f}_{t+1}^0(\hat{z})] d\hat{z}$$

where we have made use of the FOCs for consumption and capital.

The optimal choice of investment for the planner solves

$$(A.43) \quad s'(i_t^*(z)) = e_t^*(z) W_t^*(z) + \int_0^\infty W_t^*(\hat{z}) \frac{g_t^{i*}(\hat{z})}{g_t^{i*}(z)} \left[-\frac{\partial q_t^*(\hat{z})}{\partial i_t^*(z)} + \frac{\partial e_t^*(\hat{z})}{\partial i_t^*(z)} i_t^*(\hat{z}) \right] d\hat{z}$$

Finally, we can derive the optimal choice of $\delta_t^*(z)$ and $\delta_t^{E*}(z)$ from the respective first order conditions and get

$$(A.44) \quad \lambda_{zt}^i \underbrace{\left[\int_0^\infty g_t^*(\hat{z}) f(z|\hat{z}) d\hat{z} \right]}_{>0} = \lambda_t^{\delta L} - \lambda_t^{\delta H}$$

$$(A.45) \quad \lambda_{zt}^i \underbrace{\chi f_0(z) \left[1 - \int_0^\infty g_t^*(\hat{z}) d\hat{z} \right]}_{>0} = \lambda_t^{EL} - \lambda_t^{EH}$$

As $\lambda_t^{BC} > 0$ due to the binding budget constraint, $\lambda_{zt}^i > 0$ iff $V_t^*(z) > 0$. Hence iff $V_t^*(z) > 0$ we get that $\lambda_t^{\delta L} > 0$ and $\lambda_t^{EL} > 0$ such that $\delta_t^*(z) = \delta_F$ and $\delta_t^{E*}(z) = 0$ is optimal. If $V_t^*(z) < 0$, then $\lambda_t^{\delta H} > 0$ and $\lambda_t^{EH} > 0$ and hence $\delta_t^*(z) = \delta_t^{E*}(z) = 1$ and the planner chooses to dissolve the match.

A.3 Markup Distortions Given a Distribution of Customers

This appendix highlights the distortions from the aggregate markup and dispersion in markups in the decentralized equilibrium. Appendix A.2 shows that the FOCs for the planner's choice of labor and capital hold as

$$\begin{aligned} v'(L_t) &= (1 - \alpha) Z_t^M K_t^\alpha L_t^{-\alpha} u'(C_t), \\ u'(C_{t-1}) &= \beta \left(\alpha Z_t^M K_t^{\alpha-1} L_t^{1-\alpha} + 1 - \delta_K \right) u'(C_t). \end{aligned}$$

The aggregate markup for the decentralized equilibrium $\mathcal{M}_t$ is derived in Appendix A.1. Note that $\mathcal{M}_t = \frac{(1-\alpha)Y_t}{w_t L_t} \Rightarrow w_t = \frac{1}{\mathcal{M}_t} (1 - \alpha) Z_t^M K_t^\alpha L_t^{-\alpha}$ and $\mathcal{M}_t = \frac{\alpha Y_t}{(r_t + \delta_K) K_t} \Rightarrow r_t + \delta_K = \frac{1}{\mathcal{M}_t} \alpha Z_t^M K_t^{\alpha-1} L_t^{1-\alpha}$. Substituting into the FOCs for labor and capital in (5) and (6) yields

$$\begin{aligned} v'(L_t) &= w_t u'(C_t) = \frac{1}{\mathcal{M}_t} (1 - \alpha) Z_t^M K_t^\alpha L_t^{-\alpha} u'(C_t), \\ u'(C_{t-1}) &= \beta (1 + r_t) = \beta \left(\frac{1}{\mathcal{M}_t} \alpha Z_t^M K_t^{\alpha-1} L_t^{1-\alpha} + 1 - \delta_K \right) u'(C_t). \end{aligned}$$

Thus, the presence of an aggregate markup distorts the aggregate quantities of inputs into production in the decentralized equilibrium. The aggregate markup acts like a uniform tax on production, distorting labor and capital supply to below their efficient levels.

Aggregate productivity in the decentralized equilibrium Z_t^M for a given distribution of

customers $G_t(z)$ is derived in Appendix A.1 as

$$Z_t^M = \left(u_t \left(\frac{1}{\mathcal{M}_t} \right)^{-\sigma} + \int_0^\infty z^{\sigma-1} \left(\frac{\mu_t(z)}{\mathcal{M}_t} \right)^{-\sigma} dG_t(z) \right)^{\frac{1}{\sigma-1}},$$

Similarly we derive aggregate productivity under the planner allocation Z_t^P for the same distribution of customers $G_t(z)$ in Appendix A.2 as

$$Z_t^P = \left(u_t + \int_0^\infty z^{\sigma-1} dG_t(z) \right)^{\frac{1}{\sigma-1}}.$$

Comparing the two expressions, dispersion in markups distorts the allocation of demand at the intensive margin of demand—sales per customer. At the intensive margin relative price differences matter for quantities purchased, so dispersion in markups misallocates sales and lowers aggregate productivity.

A.4 Decomposing Inefficient Demand Investment

We decompose the planner's value of a match to isolate the forces that drive a wedge into the optimality condition for optimal investment of the planner vs. suppliers in equilibrium. We are applying this decomposition in steady state and will hence omit time subscripts. We first introduce some useful notation. Define the following:

$$\begin{aligned} \rho^*(z) &\equiv 1 - q^*(z) + e^*(z)i^*(z) \\ b^*(z) &\equiv \int_0^\infty W^*(\hat{z}) \frac{g^{i^*}(\hat{z})}{g^{i^*}(z)} \left[-\frac{\partial q^*(\hat{z})}{\partial i^*(z)} + \frac{\partial e^*(\hat{z})}{\partial i^*(z)} i^*(\hat{z}) \right] d\hat{z} \\ B^*(z) &\equiv \int_0^\infty W^*(\hat{z}) g^{i^*}(\hat{z}) \left[-\frac{\partial q^*(\hat{z})}{\partial g^{i^*}(z)} + \frac{\partial e^*(\hat{z})}{\partial g^{i^*}(z)} i^*(\hat{z}) \right] d\hat{z} \\ X^*(z) &\equiv -\chi \int_0^\infty \hat{f}^0(z') V^*(z') dz' \end{aligned}$$

With this notation, define the following recursions

$$\begin{aligned} V_P^*(z) &= -s(i^*(z)) + \rho^*(z) \left[\frac{\partial Y}{\partial g^*(z)} + \beta \int_0^\infty \hat{f}(z'|z) V_P^*(z') dz' \right] \\ V_B^*(z) &= B^*(z) + \rho^*(z) \beta \int_0^\infty \hat{f}(z'|z) V_B^*(z') dz' \\ V_X^*(z) &= \rho^*(z) \beta \left[X^*(z) + \int_0^\infty \hat{f}(z'|z) V_X^*(z') dz' \right] \end{aligned}$$

These three terms represent, for any customer matched with a firm of productivity z at

the beginning of a period before rematching, the expected present discounted value of the private joint surplus of the match to the supplier and the customer directly involved and in all future matches related to this match ($V_P^*(z)$), business stealing from active suppliers ($V_B^*(z)$), and crowding out of entry ($V_X^*(z)$). We can then write the value of a customer that is matched to a firm with productivity z before rematching as

$$V^*(z) = V_P^*(z) + V_B^*(z) + V_X^*(z)$$

To see this, note that

$$\begin{aligned}
V^*(z) &= V_P^*(z) + V_B^*(z) + V_X^*(z) \\
&= -s(i^*(z)) + \rho^*(z) \left[\frac{\partial Y}{\partial g^*(z)} + \beta \int_0^\infty \hat{f}(z'|z) V_P^*(z') dz' \right] \\
&\quad + B^*(z) + \rho^*(z) \beta \int_0^\infty \hat{f}(z'|z) V_B^*(z') dz' \\
&\quad + \rho^*(z) \beta \left[X^*(z) + \int_0^\infty \hat{f}(z'|z) V_X^*(z') dz' \right] \\
&= -s(i^*(z)) + \rho^*(z) \frac{\partial Y}{\partial g^*(z)} + B^*(z) + \rho^*(z) \beta X^*(z) \\
&\quad + \rho^*(z) \beta \int_0^\infty \hat{f}(z'|z) [V_P^*(z') + V_B^*(z') + V_X^*(z')] dz' \\
&= -s(i^*(z)) - \int_0^\infty W^*(\hat{z}) g^{i^*}(\hat{z}) \left[\frac{\partial q^*(\hat{z})}{\partial g^{i^*}(z)} + \frac{\partial e^*(\hat{z})}{\partial g^{i^*}(z)} i^*(\hat{z}) \right] d\hat{z} + \rho^*(z) \frac{\partial Y}{\partial g^*(z)} \\
&\quad + \rho^*(z) \beta \left[\int_0^\infty \hat{f}(z'|z) V^*(z') dz' - \chi \int_0^\infty \hat{f}^0(z') V^*(z') dz' \right] \\
&= -s(i^*(z)) + \int_0^\infty W^*(\hat{z}) g^{i^*}(\hat{z}) \left[-\frac{\partial q^*(\hat{z})}{\partial g^{i^*}(z)} + \frac{\partial e^*(\hat{z})}{\partial g^{i^*}(z)} i^*(\hat{z}) \right] d\hat{z} \\
&\quad + \rho^*(z) \left[\frac{\partial Y}{\partial g^*(z)} + \beta \int_0^\infty (\hat{f}(z'|z) - \chi \hat{f}^0(z')) V^*(z') dz' \right] \\
&= -s(i^*(z)) + \rho^*(z) W^*(z) + \int_0^\infty W^*(\hat{z}) g^{i^*}(\hat{z}) \left[-\frac{\partial q^*(\hat{z})}{\partial g^{i^*}(z)} + \frac{\partial e^*(\hat{z})}{\partial g^{i^*}(z)} i^*(\hat{z}) \right] d\hat{z}
\end{aligned}$$

We can similarly decompose the value of a match at the production stage as

$$W^*(z) = W_P^*(z) + W_B^*(z) + W_X^*(z)$$

where

$$\begin{aligned} W_P^*(z) &= \frac{\partial Y}{\partial g^*(z)} + \beta \int_0^\infty \hat{f}(z'|z) V_P^*(z') dz' \\ W_B^*(z) &= \beta \int_0^\infty \hat{f}(z'|z) V_B^*(z') dz' \\ W_X^*(z) &= \beta \left[X^*(z) + \int_0^\infty \hat{f}(z'|z) V_X^*(z') dz' \right] \end{aligned}$$

To see that this decomposition holds note that

$$\begin{aligned} W^*(z) &= W_P^*(z) + W_B^*(z) + W_X^*(z) \\ &= \frac{\partial Y}{\partial g^*(z)} + \beta X^*(z) + \beta \int_0^\infty \hat{f}(z'|z) [V_P^*(z') + V_B^*(z') + V_X^*(z')] dz' \\ &= \frac{\partial Y}{\partial g^*(z)} + \beta \int_0^\infty (\hat{f}(z'|z) - \chi \hat{f}^0(z')) V^*(z') dz' \end{aligned}$$

Applying the decomposition of $W^*(z)$ into its three components, we can rewrite the optimal investment condition as

$$s'(i^*(z)) = e^*(z) [W_P^*(z) + W_B^*(z) + W_X^*(z)] + b^*(z)$$

We can then quantify the contribution of five margins to aggregate differences between the planner's optimal demand investment and equilibrium investment:

- (i) *Private benefit*: the difference between the private benefit of the match to both the customer and supplier directly involved and in all future matches related to this match ($W_P^*(z)$) relative to the private benefit to the supplier only in equilibrium ($W(p(z), z)$).
- (ii) *Dynamic business-stealing*: business stealing from active suppliers due to the influence of existing matches on the rematching process ($W_B^*(z)$).
- (iii) *Crowding out of entry*: crowding out of potential entry due to a reduction in the number of unmatched customers ($W_X^*(z)$).
- (iv) *Static business-stealing*: business stealing through the effect of investment on the matching efficiency for other suppliers ($b^*(z)$).
- (v) *Distribution effects*: differences in the distribution of customers $g^{i^*}(z)$ that lead to differences in the conversion rate $e^*(z)$ and different weights when aggregating total investment in the economy.

We compute the contribution of each margin through a Shapley–Owen decomposition between the planner’s terms and their equilibrium counterparts (where (ii), (iii), and (iv) are zero in equilibrium). Let $\mathcal{N} \equiv \{1, \dots, 5\}$ index the five margins. For any subset $S \subseteq \mathcal{N}$, let $\Gamma(S)$ denote aggregate demand investment under the counterfactual in which margins in S are set to their planner values and margins in $\mathcal{N} \setminus S$ are set to their equilibrium values. For each such counterfactual, contact intensity $i_S(z)$ is recovered by inverting the demand investment FOC,

$$s'(i_S(z)) = e_S(z) [W_{P,S}(z) + W_{B,S}(z) + W_{X,S}(z)] + b_S(z),$$

and aggregate demand investment is obtained by integrating implied spending with the corresponding distribution of matched customers,

$$\Gamma(S) \equiv \int_0^\infty g_S(z) s(i_S(z)) dz.$$

This notation nests the equilibrium and planner outcomes: $\Gamma(\emptyset) = \Gamma^{eq}$ and $\Gamma(\mathcal{N}) = \Gamma^*$.

For $k \in \mathcal{N}$, define its Shapley–Owen contribution as the average incremental effect of switching margin k from equilibrium to planner across all orderings,

$$\phi_k \equiv \frac{1}{|\mathcal{N}|!} \sum_{\pi} [\Gamma(S_{\pi,k} \cup \{k\}) - \Gamma(S_{\pi,k})], \quad S_{\pi,k} \equiv \{j \in \mathcal{N} : \pi(j) < \pi(k)\},$$

where the sum is over all $|\mathcal{N}|! = 5! = 120$ permutations π of the margins. By construction, the contributions add up:

$$\Gamma(\mathcal{N}) - \Gamma(\emptyset) = \sum_{k \in \mathcal{N}} \phi_k.$$

B Data Appendix

B.1 Capital IQ

Measure of demand investment. Following He et al. (2025), we measure firms’ demand investment using their augmented sales-and-marketing (S&M) measure constructed from Capital IQ line items. As in He et al. (2025), this augmented measure is constructed for a baseline sample of publicly listed firms from Compustat that is matched to Capital IQ identifiers; Capital IQ contributes the underlying line items, and missing values are supplemented using information extracted from firms’ SEC filings. The resulting moments pertain to the Compustat-based public-firm universe (rather than the full set of Capital IQ-covered firms).

Aggregate demand investment. We denote the firm-level measure from He et al. (2025) by SM_f and interpret it as firm f ’s demand investment. The aggregate ratio of SM_f to revenue Rev_f equals $\frac{\sum_f SM_f}{\sum_f Rev_f} = 0.045$. Because our model abstracts from other non-value-added components of revenue, we convert this into a ratio with respect to value added VA_f for comparison to the model analogue. We use an aggregate revenue-to-value-added ratio of $\frac{\sum_f Rev_f}{\sum_f VA_f} = 1.80$. Combining these two ratios, we obtain the aggregate demand-investment share of value added in the data as:

$$\frac{\sum_f SM_f}{\sum_f VA_f} = \frac{\sum_f SM_f}{\sum_f Rev_f} \frac{\sum_f Rev_f}{\sum_f VA_f} = 0.045 \times 1.80 \approx 0.08.$$

We compare this value to the model analogue: aggregate demand investment over value added in the intermediate-good sector, computed in the filtered model sample to mimic selection into the empirical sample (Appendix B.2).

Demand investment and markups. Table A1 displays regressions of the log demand-investment intensity, $\ln\left(\frac{SM_f}{Rev_f}\right)$, on firms’ log markup, $\ln\left(\frac{Rev_f}{COGS_f}\right)$. Our preferred specification is column 4, which includes year and 2-digit NAICS fixed effects (consistent with the model’s lack of aggregate shocks and industry heterogeneity) and weights observations by COGS. Weighting by COGS reduces noise as evidenced by a higher $R^2 = 0.309$ compared to the unweighted regression with the same fixed effects in column 3 ($R^2 = 0.162$). The point estimate of 1.843 implies that a firm with a 1% higher markup has, on average, about a 1.84% higher demand-investment-to-revenue ratio, relative to the industry average in that year. Columns 1 and 2 report results without fixed effects, while columns 5 and 6 add controls for firm size and age. Across specifications, high-markup firms consistently invest

more in their demand.

Table A1: Cross-sectional Relation of Demand Investment and Markups

	(1)	(2)	(3)	(4)	(5)	(6)
$\ln \frac{\text{Rev}_f}{\text{COGS}_f}$	1.295*** (0.256)	2.195*** (0.260)	0.880*** (0.193)	1.843*** (0.379)	1.011*** (0.143)	1.720*** (0.256)
R^2	0.095	0.176	0.162	0.309	0.300	0.339
Fixed Effects	—	—	Year+Ind	Year+Ind	Year+Ind	Year+Ind
Weights	—	Cost	—	Cost	—	Cost
Controls	—	—	—	—	Age+Size	Age+Size

Notes: Data from He et al. (2025), Capital IQ (Compustat-matched public firms). The dependent variable in all regressions is the log of the augmented S&M measure relative to revenue, $\ln \frac{SM_f}{Rev_f}$. 30,674 observations. Robust standard errors clustered at 2-digit NAICS level, in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

B.2 Model Filter to Mimic Selection into Capital IQ

Table A2 shows how model moments vary based on imposing the filter that mimics selection into the publicly-listed firm sample that underlies our Capital IQ-based empirical moments. We filter intermediate good firms (suppliers) along two dimensions, age and size. For age, we follow Ottonello and Winberry (2020) and restrict the sample to firms above seven years old. For size, we assign an inclusion probability to each firm that increases with firm size. Specifically, we use a truncated power function of the form $\min\left(1, \left(\frac{r}{r_0}\right)^\gamma\right)$, where r denotes firm revenue, and the parameters r_0 and γ govern the location and steepness of the cutoff. These parameters are calibrated to match two empirical moments: (i) the average Compustat firm is 480 times larger than the average U.S. firm, and (ii) 15.2% of Compustat firms are smaller than the average U.S. firm. This filter allows the model to reproduce both the heavy right tail and the nontrivial lower tail of the empirical firm size distribution in Compustat.

Given our estimated turnover in the productivity process, the important restriction on the model-based sample is firm size, not firm age. In addition, those intermediate inputs that are produced in-house are produced competitively. This is why the economy-wide markup and demand-investment share of value added are lower in the aggregate model economy compared to the respective moments in the sample of all intermediate good firms.

Table A2: Impact of Filtering on Model Moments

Sample	All firms	Intermediate-good firms only		
Filter	none	none	age	age+size
Elasticity of $\frac{\text{demand investment}}{\text{revenue}}$ to markup	—	2.084	2.079	1.849
Aggregate markup	1.197	1.236	1.236	1.250
Aggregate demand investment over value added	0.092	0.109	0.109	0.113

Notes: Model moments used for calibration or validation in bold.

B.3 BDS and SUSB

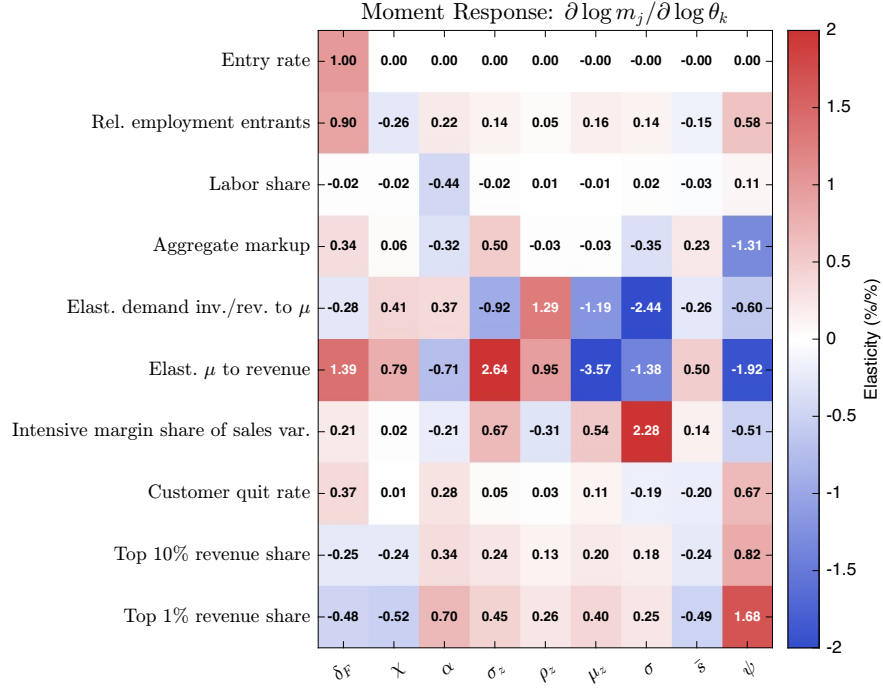
To calibrate entry and exit rates as well as the sales concentration within industries, we rely on data from the Business Dynamics Statistics (BDS) and the Statistics of U.S. Businesses, respectively. For both datasets, we focus on the 2017 wave and exclude agriculture, finance and insurance, as well as real estate (2-digit NAICS 11, 52, and 53). For entry / exit rates, we define entrants as firms of age zero and exits as reported shutdown of firms. We compute entry and exit relative to incumbent firms within each 2-digit NAICS sector and take the simple average across sectors. For sales concentration, we estimate Pareto tails for the distribution of revenues within 5-digit NAICS sectors and take the simple average across sectors.

C Calibration Sensitivity

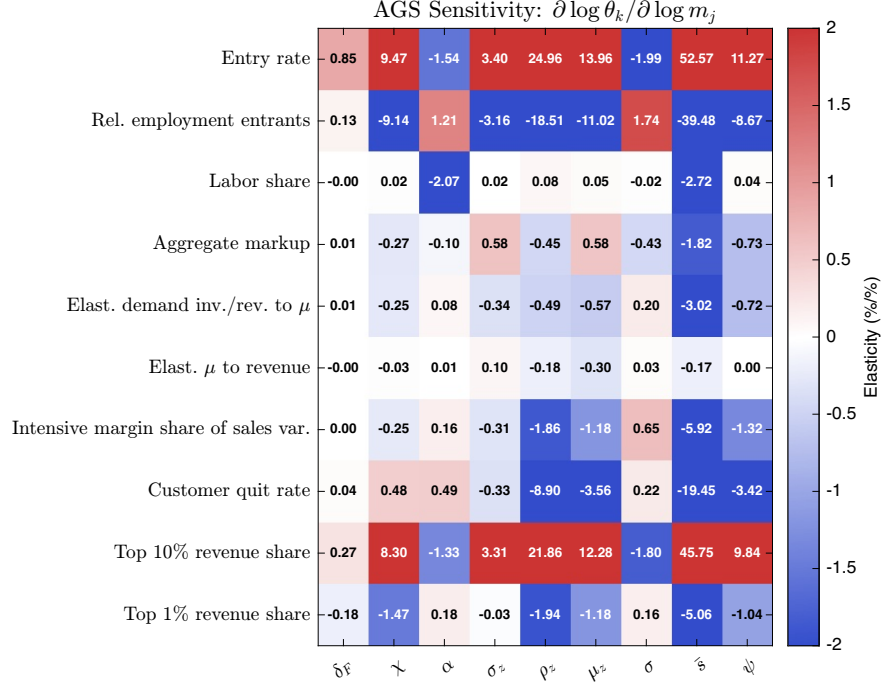
Following Andrews et al. (2017), we assess the local sensitivity of our estimation to the choice of target moments. Let θ denote the vector of calibrated parameters and m the vector of targeted moments. Figure A1 reports two objects computed via centered finite differences. Panel (a) shows the moment-response elasticity $E_{kj} = \partial \log m_j / \partial \log \theta_k$, which summarizes how each model moment moves when a parameter changes. The diagonal aligns with the calibration discussion in the text: each parameter has a natural target moment. In fact, the discussed intuitive parameter–moment assignment maximizes the sum of diagonal elasticities $\sum |E_{\text{diag}}|$ across all $9! = 362,880$ possible assignments (treating the two top revenue share moments as a single concentration slot).

Panel (b) shows the sensitivity matrix $\Lambda = (EE^\top)^{-1}E$, where Λ_{kj} measures how the estimated parameter $\hat{\theta}_k$ responds to a perturbation of data moment m_j in the joint system. Ultimately, what matters is not how parameters respond to moment perturbations locally, but how the paper’s economic conclusions change with the choice of target moments. Appendix E implements this directly by varying the target moments subject to the greatest measurement uncertainty—customer quit rate, intensive-margin share, aggregate markup—and shows that the main quantitative results are stable.

Figure A1: Parameter–Moment Sensitivity



(a) Moment response to parameter: E_{kj}



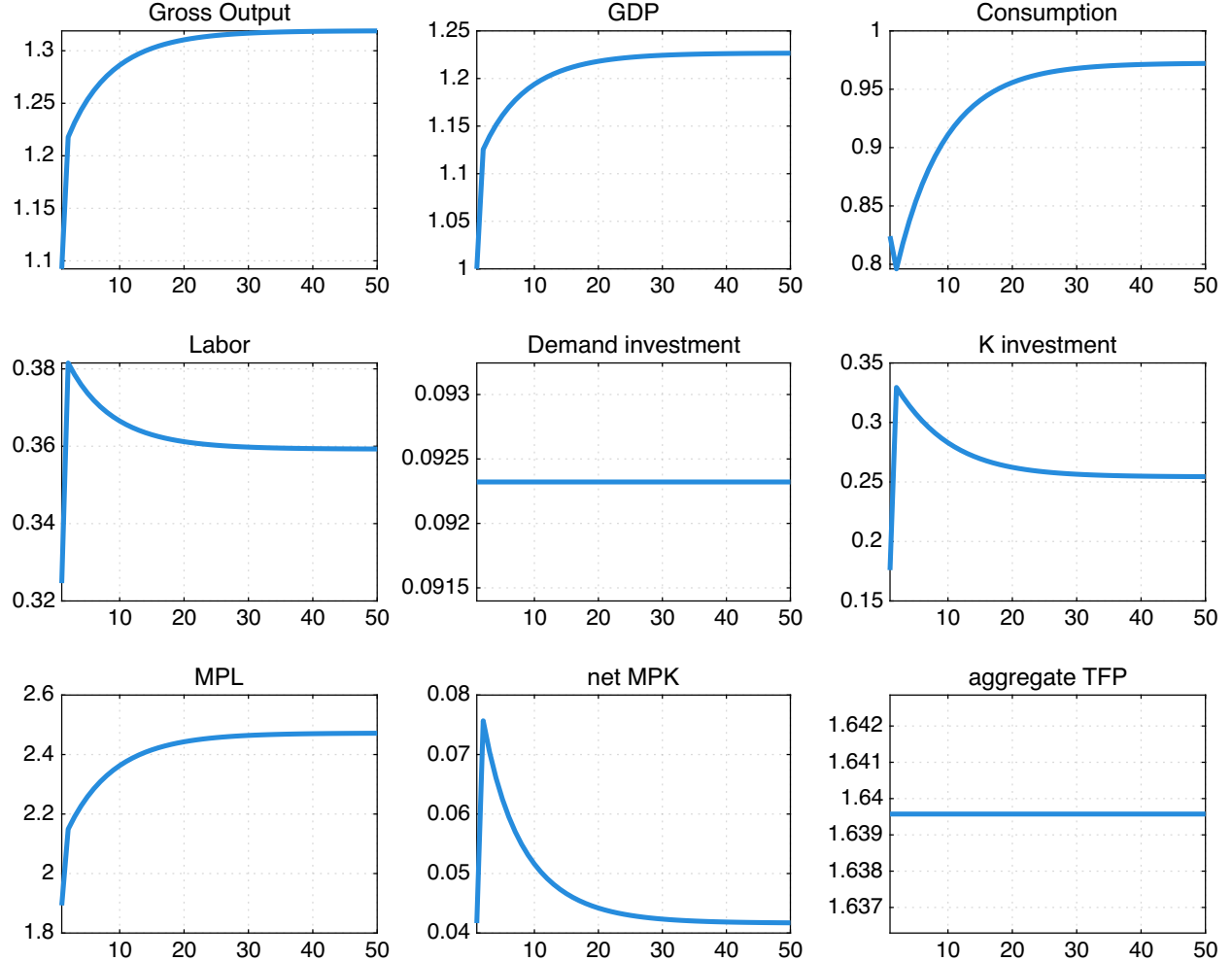
(b) Parameter sensitivity to moment: Λ_{kj} (Andrews et al., 2017)

Notes: Panel (a): $E_{kj} = \partial \log m_j / \partial \log \theta_k$, the elasticity of model moment j to parameter k . Panel (b): $\Lambda_{kj} = [(EE^\top)^{-1}E]_{kj}$, the sensitivity of estimated parameter k to a perturbation of data moment j .

D Additional Quantitative Model Results

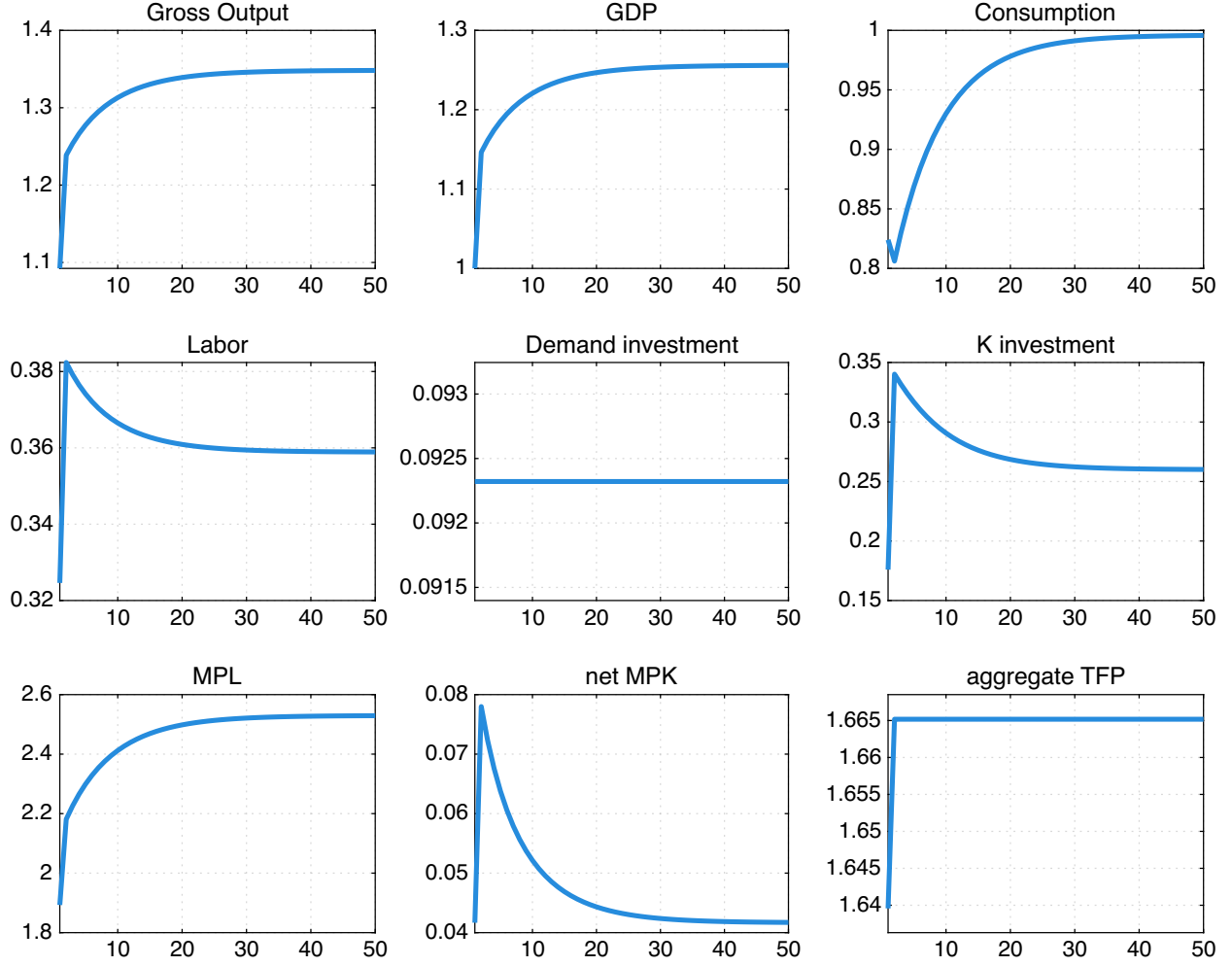
D.1 Social Planner Transition Graphs

Figure A2: Transition: Planner Remedies Only Aggregate Markup



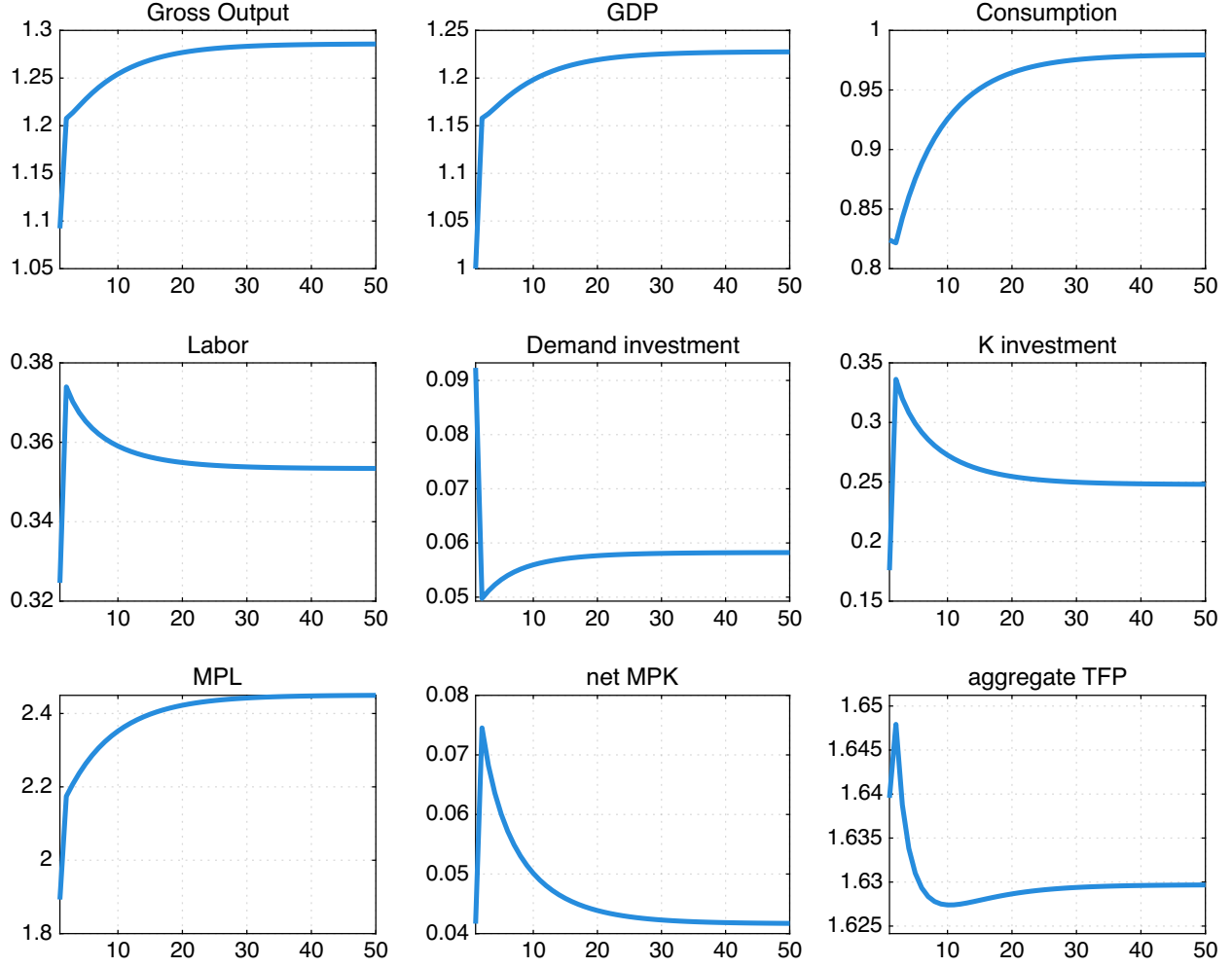
Notes: The planner removes the aggregate markup wedge while holding fixed the equilibrium objects governing demand reallocation: demand investment Γ^M and the implied customer distribution G (extensive margin), and aggregate TFP Z^M (intensive margin). Dispersion in markups across suppliers is therefore not corrected. The planner chooses time-varying (C_t, K_{t+1}, L_t) subject to fixed Γ^M, G, Z^M . Period $t=1$ shows the decentralized equilibrium; at $t=2$ the planner takes over with predetermined K , while (C_t, K_{t+1}, L_t) are already planner-chosen. Quantities normalized so that $GDP = 1$ in the decentralized equilibrium.

Figure A3: Transition: Planner Remedies Aggregate Markup and Markup Dispersion



Notes: The planner removes both the aggregate markup wedge and *dispersion* in markups (intensive-margin misallocation). Selling effort remains fixed at its equilibrium path, so the customer distribution G is fixed. Removing dispersion shifts aggregate TFP from the market value Z^M to the planner value Z^P . The planner chooses time-varying (C_t, K_{t+1}, L_t) subject to fixed Γ^M, G . Period $t=1$ shows the decentralized equilibrium; at $t=2$ the planner takes over with predetermined K , while (C_t, K_{t+1}, L_t) are already planner-chosen. Quantities normalized so that $GDP = 1$ in the decentralized equilibrium.

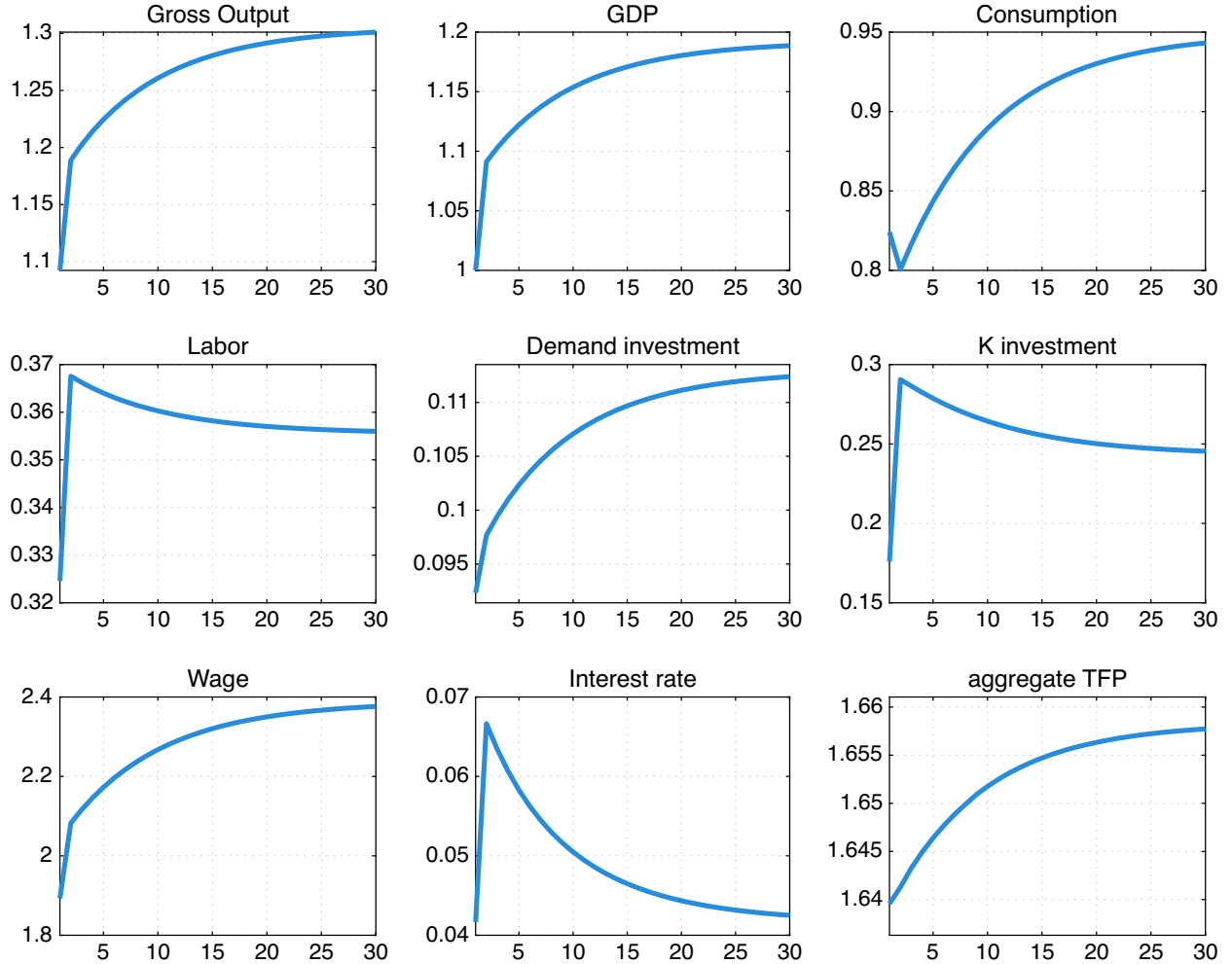
Figure A4: Transition: Full Social Planner (Markup, Dispersion, and Demand Investment)



Notes: The planner removes the aggregate markup and its dispersion *and* chooses contact intensities optimally. Thus, $i_t(z)$, aggregate demand investment Γ_t , and the customer distribution $G_t(z)$ are endogenously time-varying, and aggregate TFP Z_t reflects both intensive-margin reallocation and evolving customer allocation. The planner chooses time-varying $(C_t, K_{t+1}, L_t, i_t(z))$. Period $t=1$ shows the decentralized equilibrium; at $t=2$ the planner takes over with predetermined K , while $(C_t, K_{t+1}, L_t, i_t(z))$ are already planner-chosen. Quantities normalized so that $GDP = 1$ in the decentralized equilibrium.

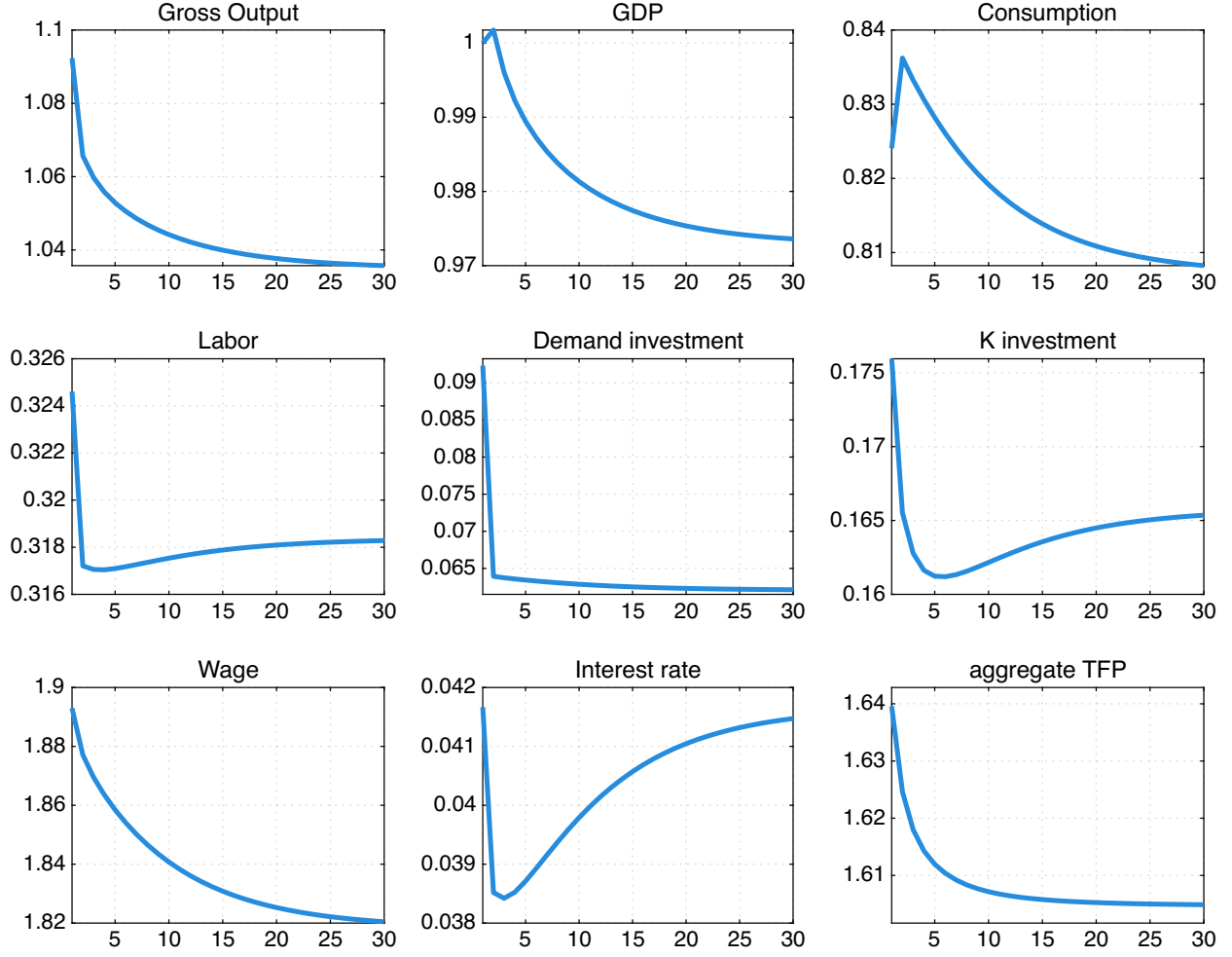
D.2 Optimal Taxation Transition Graphs

Figure A5: Transition: Optimal Production Subsidy Only



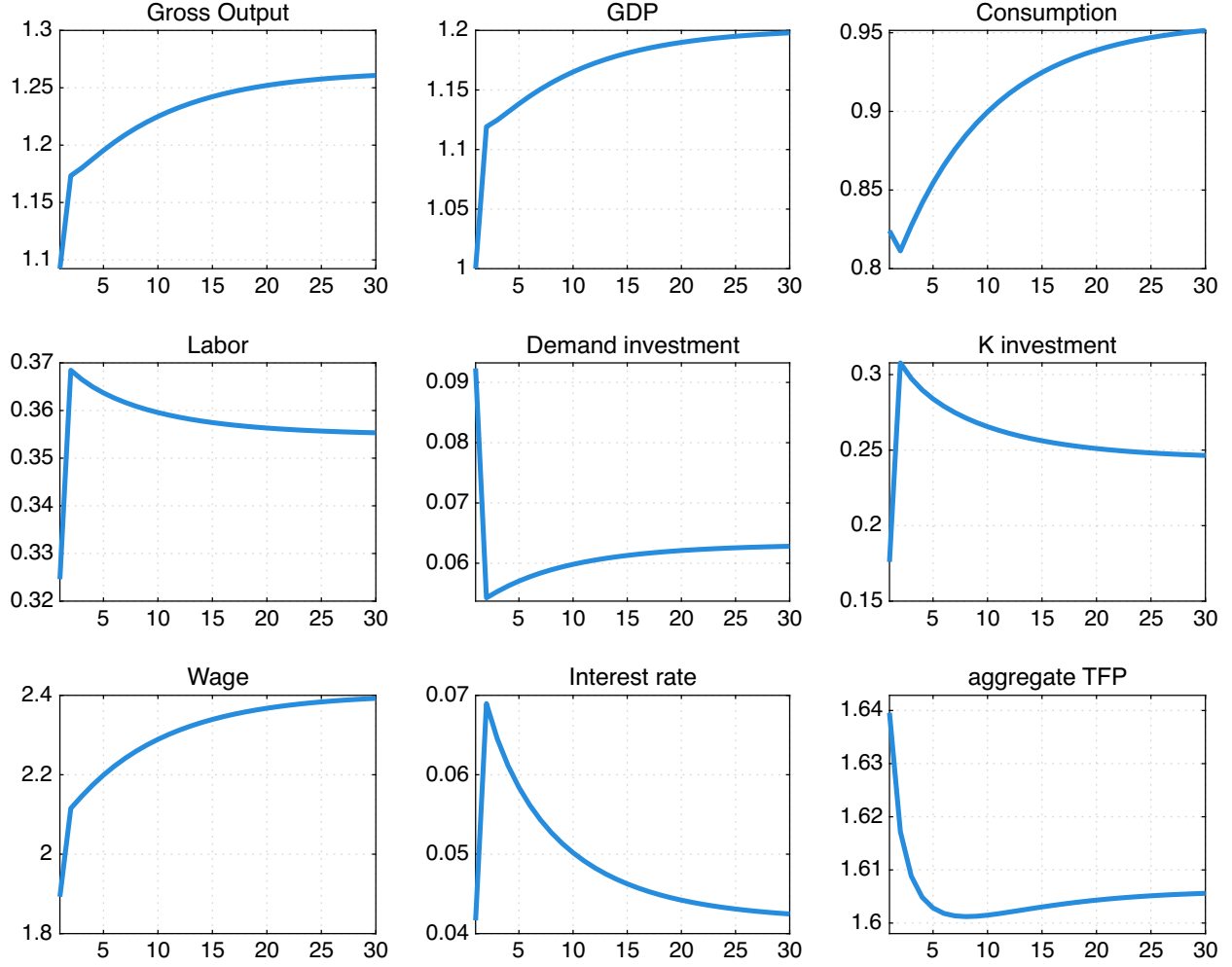
Notes: We solve for a (time-constant) production input subsidy of 13.3%, which maximizes welfare along the transition from the no-tax baseline steady state. The policy is financed by a lump-sum tax (16.7% of GDP in steady state). Period $t=1$ shows the no-tax equilibrium; from $t=2$ onwards the production subsidy is imposed; agents are surprised; we solve for the resulting perfect foresight transition to the new steady state. Quantities normalized so that $GDP = 1$ in the no-tax equilibrium.

Figure A6: Transition: Optimal Demand Investment Tax Only



Notes: We solve for a (time-constant) demand-investment tax of 33.8%, which maximizes welfare along the transition from the no-tax baseline steady state. The tax is rebated as a lump-sum transfer (2.1% of GDP in steady state). Period $t=1$ shows the no-tax equilibrium; from $t=2$ onwards the demand-investment tax is imposed; agents are surprised; we solve for the resulting perfect foresight transition to the new steady state. Quantities normalized so that $GDP = 1$ in the no-tax equilibrium.

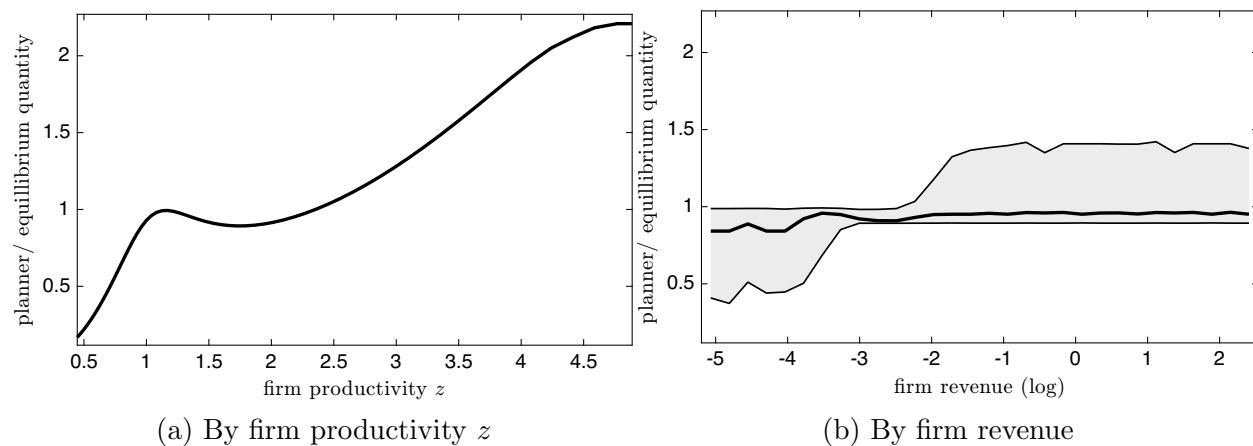
Figure A7: Transition: Optimal Demand Investment Tax and Production Subsidy



Notes: We solve for a (time-constant) demand-investment tax of 61.1% and production subsidy of 17.1%, which maximize welfare along the transition from the no-tax baseline steady state. The policy is financed by a lump-sum tax (17.8% of GDP in steady state). Period $t=1$ shows the no-tax equilibrium; from $t=2$ onwards the demand-investment tax and production subsidy are imposed; agents are surprised; we solve for the resulting perfect foresight transition to the new steady state. Quantities normalized so that $GDP = 1$ in the no-tax equilibrium.

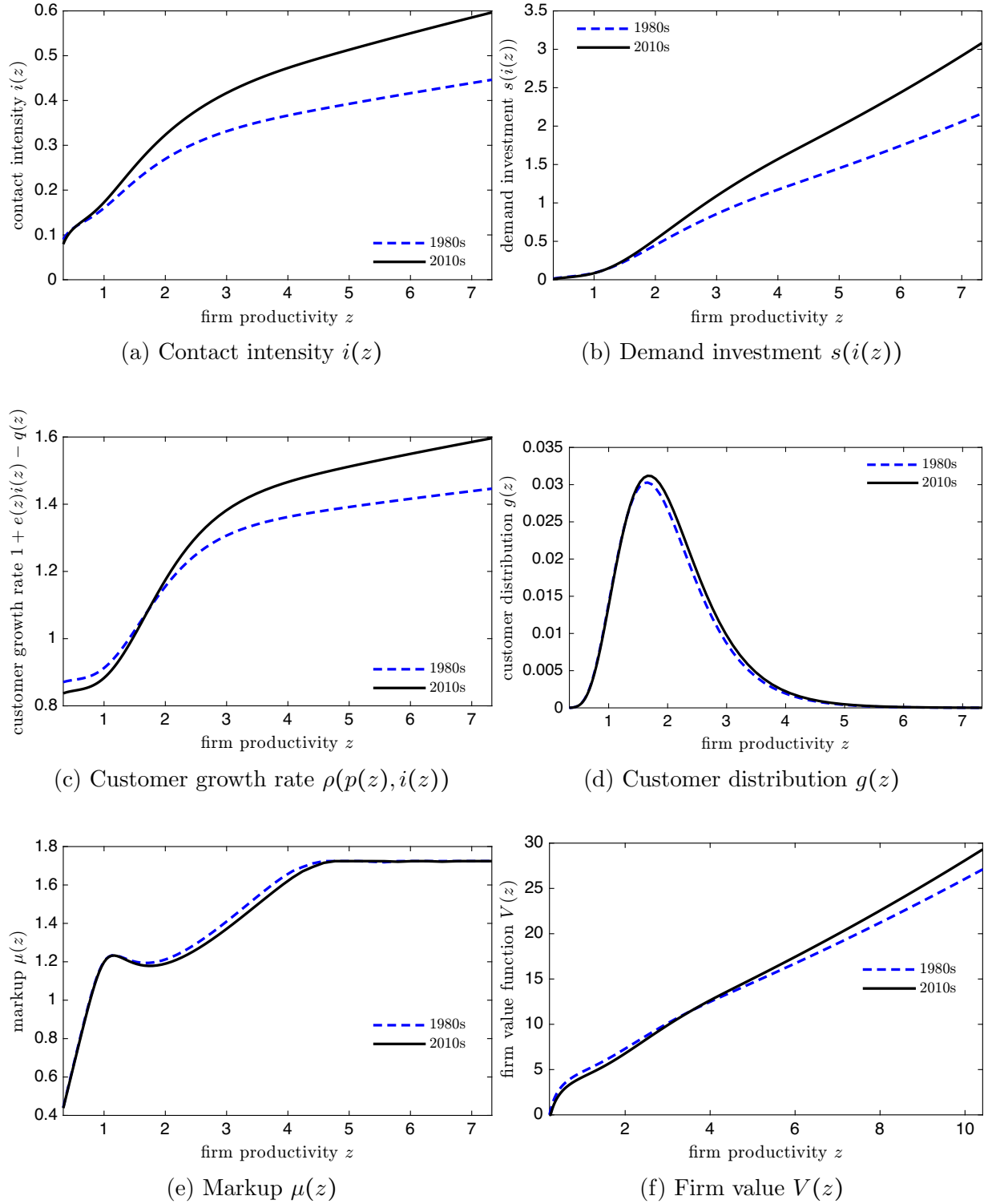
D.3 Additional Model Figures

Figure A8: Size Distortions: Planner vs. Equilibrium by Productivity and Revenue



Notes: Each panel plots the ratio of the planner's to the equilibrium quantity demanded. In panel (a), firms are sorted by productivity z , and the ratio is a deterministic function of z . In panel (b), firms are sorted by revenue; for each revenue level, there is a distribution of ratios. The thick line shows the median and the shaded area the 10th–90th percentile range.

Figure A9: Effects of Contact-Cost Technology Change Across the Firm Distribution



Notes: Each panel compares the 1980s and 2010s steady states implied by the calibrated change in the contact-cost technology $s(i)$.

D.4 Allowing for Rising Aggregate Markups

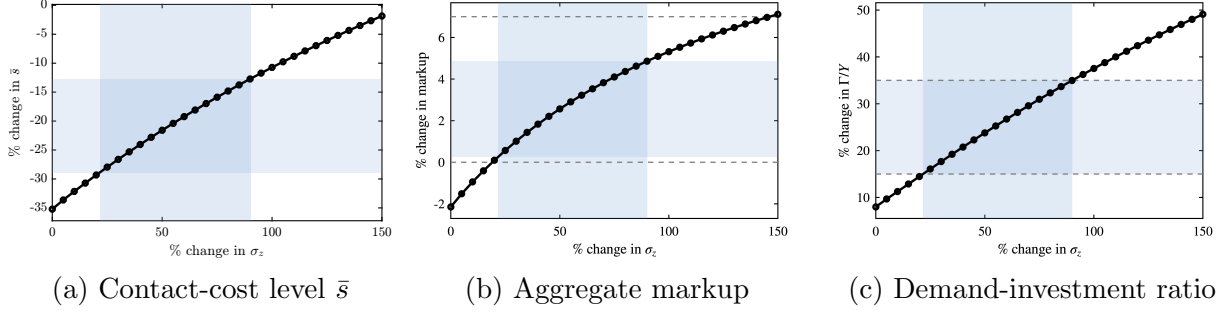
Our benchmark exercise in Section 5 rationalizes the observed rise in sales concentration through an improvement in the contact technology. Our results from this exercise imply little net change in the aggregate markup. Still, the model is not inconsistent with an increase in aggregate measures of markups or profit rates over time, as documented, e.g., by Barkai (2020) and De Loecker et al. (2020). Letting part of the rise in concentration come from a change in the underlying heterogeneity in productivity across firms, their competitive advantage, can generate a rising aggregate markup alongside the rise in concentration. To show this, we consider an alternative exercise in which part of the same rise in concentration reflects higher firm-level productivity dispersion σ_z , which can be interpreted as a reduced-form proxy for skewed technology adoption across firms (e.g., Hubmer and Restrepo, 2026).

We keep the baseline calibration for the 2010s steady state. While the benchmark exercise holds σ_z constant between the two steady states, here we allow it to rise from the 1980s to the 2010s. For each candidate increase on a grid for σ_z , we recenter μ_z so that discretized $E[z]$ remains fixed, and residually solve for the 1980s contact-cost shifter $\bar{s}$ so that the top 1% sales share rises by 25.8%, our target for the increase in concentration in the baseline exercise. For ease of exposition and to emphasize the implications of a change in contact cost vs. productivity dispersion, we keep the curvature parameter ψ fixed throughout this alternative exercise. Higher σ_z reduces competitive pressure: in the pricing ODE (40), the local density term $g^i(z)$ falls, raising firm-level markups at a given level of productivity. In addition, higher σ_z shifts sales toward more productive, higher-markup firms. Figure A10 reports the implied change in $\bar{s}$ (panel a) together with the resulting changes in the aggregate markup (panel b) and demand investment (panel c).

We report results for a range of parameter combinations to account for the uncertainty in estimates of both the rise in demand investment and the rise in markups. We focus on parameterizations that imply a 15–35% increase in the ratio of demand investment to gross output, Γ/Y , the range discussed in Section 5, and a 0–7% increase in the aggregate markup. The upper bound of 7% refers to the increase in the standard cost-weighted Compustat-based markup measure (Edmond et al., 2023). The lower bound of zero change reflects recent evidence that the measured rise attenuates substantially once production technologies are allowed to vary more flexibly across firms within industries over time (Hubmer and Restrepo, 2026; Foster, Haltiwanger and Tuttle, 2022).

Within the intersection of the two-dimensional bounds on empirical targets, the model generates aggregate markup increases of 0.3–4.9%, obtained by jointly raising σ_z by 21.7–90.1% (from a 1980s value between 0.204 and 0.318 to the calibrated 2010s value of 0.387) and

Figure A10: Rising Concentration with Rising Aggregate Markups



Notes: The horizontal axis reports the percent increase in firm-level productivity dispersion, σ_z , from the 1980s to the 2010s. For each value on the horizontal grid, the contact-cost shifter $\bar{s}$ (panel a) is chosen so that the increase in the top 1% sales share is 25.8%, holding ψ fixed. The shaded regions in all panels mark the intersection of the admissible ranges discussed in the text.

lowering $\bar{s}$ by 12.7–28.9%. Because ψ is held fixed, this decline in $\bar{s}$ is directly comparable to the 31.0% decline in average demand investment, evaluated at fixed 1980s contact intensities $i(z)$, reported in Table 5: the range of plausible counterfactuals allowing for changes in σ_z generates 40–90% of the baseline’s decline in contact cost.

D.5 Intensive-Margin Misallocation in Restricted Model Variants

Figure 3 in Section 4 varies the intensive-margin elasticity σ mechanically, holding the baseline customer distribution and the distribution of firm-level markups fixed. The purpose of this exercise is to isolate the role of σ for a given distribution of markups as cleanly as possible. An alternative approach to highlight the importance of separating the source and cost of markups is to compare the baseline model to restricted variants that shut down competition at the extensive margin—and hence return to the close link between the source and cost of markups—and re-estimate the remaining parameters, allowing the equilibrium distributions of customers and markups to adjust. We pursue this approach here to show that our main insights apply when looking at the data fully through alternative models that do not account for extensive-margin competition.

We compare two restricted variants of our model to the baseline. The first, labeled “Static Demand”, shuts down customer-base dynamics entirely, in the spirit of Edmond et al. (2023). As this version does not allow for meaningful firm dynamics, we abstract from productivity shocks, firm entry and exit, and consider a one-period version without loss of generality. The household side remains unchanged. Final-goods producers use a Kimball aggregator to assemble intermediates, generating variable markups. Intermediate-goods firms

have idiosyncratic productivity but each is matched with a unit mass of customers. The second, labeled “Static Pricing”, allows for customer accumulation but makes customer flows independent of prices, in the spirit of Afrouzi et al. (2025). Again, the household side is unchanged and final-goods firms use a Kimball aggregator. Intermediate-goods firms price statically against demand per customer, generating dispersion in markups. Intermediate-goods firms can contact customers with the same technology as in the baseline and contacts are Poisson-distributed. Unlike in the baseline, customers do not choose whom to buy from; instead, we allocate customers at random if they are contacted by multiple firms, shutting down any competition at the extensive margin. In addition, we no longer allow for endogenous quits or poaching from existing matches; unmatched customers accept any match offered and matched customers separate from their supplier at an exogenous rate δ_C .

Table A3: Parameters and Moments Across Recalibrated Model Variants

Parameter		Static Demand	Static Pricing	Baseline
Exogenous firm exit rate	δ_F	–	0.081	0.081
Exogenous customer separation rate	δ_C	–	0.069	–
Entrants per unmatched customer	χ	–	0.054	0.078
Capital elasticity	α	0.264	0.327	0.327
Log TFP: unconditional dispersion	σ_z	–	0.147	0.387
Log TFP: auto-correlation	ρ_z	–	0.888	0.361
Log TFP: mean	μ_z	–	0.154	0.409
Log TFP: Weibull scale	ζ^W	0.036	–	–
Log TFP: Weibull shape	ν^W	0.492	–	–
Kimball super-elasticity ratio (Klenow-Willis)	ε/σ	0.174	0.439	–
Intensive-margin elasticity	σ	9.288	4.862	2.381
Contact cost: level	$\bar{s}$	–	56.410	39.792
Contact cost: curvature	ψ	–	3.852	2.895
Targeted Moment	Data	Static Demand	Static Pricing	Baseline
Firm entry/exit rate	0.081	–	0.081	0.081
Avg. $\Delta \log$ employment entrants vs. age 1–19	0.617	–	0.644	0.643
Aggregate labor share	0.615	0.615	0.615	0.614
Aggregate markup	1.197	1.197	1.195	1.197
Elasticity of $\frac{\text{demand investment}}{\text{revenue}}$ to markup	1.843	–	1.846	1.849
Elasticity of markup to revenue	0.031	0.031	0.031	0.031
Sales variation: intensive-margin share	0.260	(1.000)	0.262	0.262
Customer quit rate (% sales)	0.150	–	0.150	0.152
Top 10% sales share	0.724	0.723	0.665	0.676
Top 1% sales share	0.423	0.422	0.426	0.431
Demand investment / value added, $\Gamma/(Y - \Gamma)$	0.092*	–	0.092	0.092*
Unmatched share u	0.214*	–	0.214	0.214*

Notes: The table follows the structure of Table 1; sources are the same and are therefore omitted. Values marked * are not direct targets in the baseline calibration; they correspond to the baseline model’s implied moments, which are used as targets for the restricted variants.

We calibrate both restricted variants using the same approach as the baseline model. Table A3 shows that each specification matches the moments it targets closely. The Static Demand version is disciplined by the subset of moments that remains well-defined without

customer accumulation and firm dynamics, and uses a Weibull productivity distribution with two shape parameters to match both top sales shares. Absent customer accumulation, our baseline approach of log-normal productivity would severely undershoot the Pareto tail in the sales distribution. The Static Pricing version instead allows for customer accumulation, so the full set of customer-related moments is imposed. We add two additional targets because the model features two more parameters (the exogenous separation rate of customers and the super-elasticity of the Kimball aggregator). To remain consistent across variants, we set model-implied moments from the baseline as additional targets, and include the fraction of unmatched customers and the ratio of demand investment to output in the calibration of the Static Pricing economy.

Table A4 reports the resulting intensive-margin misallocation as a TFP wedge, corresponding to Figure 3. All three model variants target the same two markup moments: the aggregate level and the slope with respect to firm revenue. Overall markup dispersion is not targeted, and varies substantially across models. Two patterns stand out. First, customer accumulation raises implied dispersion for given targets, because heterogeneity in both productivity and customer base generates dispersion in markups conditional on firm size, as emphasized in Afrouzi et al. (2025). Dispersion is highest in the baseline, where dynamic pricing admits a wider range of markups, even temporarily negative profits, consistent with the unconditional distribution in De Loecker et al. (2020). Second, extensive-margin competition reduces the implied misallocation despite the *higher* markup dispersion: relative to Static Demand, the baseline has roughly three times the dispersion but a slightly lower TFP wedge; relative to Static Pricing, it has nearly twice the dispersion but a TFP wedge about 40% lower. Intuitively, the TFP wedge scales with markup dispersion times an effective intensive-margin demand elasticity; once the extensive margin is shut down, that elasticity must rise to match the same aggregate markup, amplifying the allocative cost of any given dispersion. Extensive-margin competition breaks this tight link and separately identifies the source and the cost of markups—the mechanism highlighted in Figure 3.

Table A4: Misallocation Across Recalibrated Model Variants

Variant	Extensive margin	Pricing	Target Markup Level	Target Markup Slope	Markup dispersion	TFP wedge
Static Demand	no	static	1.197	0.031	0.027	1.84%
Static Pricing	yes	static	1.195	0.031	0.053	2.72%
Baseline	yes	dynamic	1.197	0.031	0.091	1.56%

Notes: Markup slope denotes $\partial \log \mu / \partial \log \text{rev}$. Markup dispersion is computed as $\text{sd}(\log \mu)$. TFP wedge denotes the percent gain in aggregate TFP from removing model-implied markup dispersion, computed as $100 \times (Z^*/Z^M - 1)$, holding the customer distribution fixed where applicable.

E Model Robustness

This appendix combines robustness exercises that assess the sensitivity of our quantitative findings to alternative calibration targets. We first outline the alternative calibrations and then report how our main results change across quantifications.

E.1 Alternative Calibration Targets

We recalibrate the baseline steady state while varying one of three data targets at a time. First, we vary the sales-weighted customer quit rate. Moving from static sales variation to persistent customer relationships is a key deviation of our framework from the previous literature on markups and misallocation. To gauge how quickly results change when we move towards static demand, we consider an alternative calibration with a higher quit rate, which corresponds to less persistent customer relationships and higher churn.

Second, we vary the intensive-margin share of sales variation. Again, allowing for variation at both the intensive and extensive margin of demand is a key feature of the model economy. Our baseline target (26%) is based on business-to-business transaction data from Belgium (Bernard et al., 2022). Evidence from settings in which U.S. consumers are the customers points to a lower intensive-margin share of around 15% (Einav et al., 2021; Afrouzi et al., 2025). We consider these estimates as an alternative target. Additionally, we consider a higher intensive-margin share of 35%, moving towards an economy with more flexible demand at the intensive margin.

Third, we vary the level of the aggregate markup in both directions. As considerable uncertainty remains about the level of aggregate markups, we follow Edmond et al. (2023) and consider alternative targets of 1.15 and 1.35 in addition to our baseline of 1.25.

Table A5 reports the implied parameters and shows that each specification matches all targeted moments closely. The implied parameter adjustments follow intuitive comparative statics: a higher quit rate implies lower inferred contact-cost shifter $\bar{s}$; a higher intensive-margin share implies a higher intensive-margin elasticity of substitution σ ; and a higher aggregate markup implies higher productivity dispersion σ_z , in line with our calibration strategy.

Table A5: Robustness: Calibration Parameters and Targeted Moments

	<u>Baseline</u>	<u>Quitrate</u>	<u>Int. Margin</u>		<u>Agg. Markup</u>	
<i>Modified target:</i>		↑ 20%	↓ 15%	↑ 35%	↓ 15%	↑ 35%
Parameters (9)						
Exogenous firm exit rate (δ_F)	0.081	0.085	0.080	0.082	0.080	0.082
Entrants per unmatched customer (χ)	0.078	0.236	0.058	0.113	0.080	0.099
Capital elasticity (α)	0.327	0.326	0.324	0.332	0.344	0.295
Log TFP: unconditional dispersion (σ_z)	0.387	0.464	0.383	0.398	0.279	0.519
Log TFP: auto-correlation (ρ_z)	0.361	0.350	0.222	0.415	0.354	0.509
Log TFP: mean (μ_z)	0.409	0.603	0.360	0.448	0.264	0.688
Intensive-margin elasticity (σ)	2.381	2.380	1.939	2.814	3.042	2.003
Contact cost: level ($\bar{s}$)	39.792	25.056	59.757	30.032	89.442	74.259
Contact cost: curvature (ψ)	2.895	3.074	3.223	2.671	4.000	2.936
Targeted moments (10)						
Firm entry/exit rate	0.081	0.086	0.080	0.082	0.080	0.082
Relative employment of entrants	0.526	0.552	0.521	0.533	0.524	0.534
Aggregate labor share	0.614	0.615	0.616	0.611	0.615	0.618
Aggregate markup	1.250	1.249	1.251	1.250	1.150	1.351
Elasticity of $\frac{\text{demand investment}}{\text{revenue}}$ to markup	1.849	1.838	1.848	1.851	1.848	1.852
Elasticity of markup to revenue	0.031	0.032	0.031	0.031	0.031	0.031
Sales variation: intensive-margin share	0.262	0.266	0.150	0.355	0.263	0.263
Customer quit rate (% sales)	0.152	0.189	0.152	0.153	0.153	0.154
Top 10% sales share	0.676	0.702	0.667	0.691	0.674	0.682
Top 1% sales share	0.431	0.441	0.431	0.430	0.430	0.432

E.2 Robustness: Planner Benchmark and Optimal Policy

E.2.1 Simple Shift in Contact Cost

This section reports a simple robustness check that shifts the contact-cost shifter $\bar{s}$ holding all other parameters fixed at their baseline values, to illustrate how changes in contact cost reshape competition for customers and the welfare cost of misallocation.

Table A6 reports the resulting welfare-loss decomposition (computed along the transition to the efficient allocation). Cutting $\bar{s}$ in half lowers the aggregate-markup component by 10.8% and the markup-dispersion component by 13.0% (relative to the baseline). Doubling $\bar{s}$ raises these components by 10.8% and 22.9%, respectively.

Table A6: Welfare Losses from Misallocation under Alternative Contact Cost

	Total	Aggr. Markup	Markup Dispersion	Customer Misallocation
Baseline calibration	5.14	1.94	2.23	0.97
$\bar{s} - 50\%$	4.66	1.73	1.94	0.99
$\bar{s} + 100\%$	5.72	2.15	2.74	0.82

Notes: The table reports welfare losses along the transition to the efficient steady state (in percent of consumption), decomposed into components due to aggregate markups, markup dispersion, and customer misallocation. The first row corresponds to the baseline calibration. The second and third rows report outcomes when the cost shifter $\bar{s}$ of the contact-cost function is cut in half or doubled, respectively.

E.2.2 Alternative Calibrations

We study the results in Section 4 under the alternative calibrations outlined in Appendix E.1. Two patterns are worth highlighting. First, targeting different values for the customer quit rate and the intensive-margin share of sales variation has relatively little effect on misallocation. Second, targeting a higher aggregate markup has a larger impact on the inefficiency results. In that case, the calibration implies a higher dispersion of productivity (higher σ_z) to reconcile higher markups with observed cross-sectional patterns, and the model features less over-investment in demand: demand investment is more likely to meaningfully improve productivity when competitors are further away. To see this, note that in the limit case where productivity dispersion approaches zero and suppliers become identical, customer poaching is a pure zero-sum game from a social perspective as old and new matches exhibit similar productivity.

As a result, the welfare losses from the level and dispersion of markups increase, while the inefficiency from demand investment declines. This is also reflected in lower optimal demand-investment taxes in the reported policy results. Despite these quantitative shifts, the qualitative insights persist across calibrations, including the complementarity between the production subsidy τ_P and the demand-investment tax τ_S .

Table A7: Robustness: Planner Benchmark and Optimal Policy

	<u>Baseline</u>	<u>Quitrate</u>	<u>Int. Margin Sh.</u>	<u>Agg. Markup</u>		
<i>Modified target:</i>		↑ 20%	↓ 15%	↑ 35%	↓ 15%	↑ 35%
Social Planner: % Δ relative to equilibrium (steady state)						
Gross Output	17.72	20.70	15.83	19.87	11.44	28.10
GDP	22.77	25.84	21.03	24.73	14.24	33.16
Demand investment	-36.92	-27.49	-47.46	-28.14	-46.46	-12.51
Tangible Capital	40.91	46.34	37.52	44.45	24.71	64.41
Labor	8.86	9.76	8.16	9.51	5.62	13.54
Consumption	18.90	21.47	17.55	20.44	11.79	27.54
Aggregate TFP	-0.60	0.12	-0.92	-0.14	-0.35	1.15
Social Planner: % Δ welfare relative to equilibrium (incl. transition)						
Total	5.14	5.75	5.13	5.29	3.10	6.92
<i>Decomposition:</i>						
Aggregate markup	1.94	2.23	1.75	2.12	0.79	3.44
Markup dispersion	2.23	2.37	2.16	2.16	1.52	2.91
Demand investment	0.97	1.15	1.22	1.02	0.79	0.58
Optimal Taxation						
<i>Demand investment tax only</i>						
Optimal demand investment tax	33.82	28.96	47.27	26.55	59.57	7.82
Welfare gain (% Δ)	0.46	0.36	0.69	0.34	0.51	0.05
<i>Production subsidy only</i>						
Optimal production subsidy	13.26	14.50	12.38	14.00	8.66	19.28
Welfare gain (% Δ)	1.28	1.55	1.07	1.47	0.52	2.76
<i>Demand investment tax + production subsidy</i>						
Optimal demand investment tax	61.10	58.10	75.30	53.65	84.58	34.66
Optimal production subsidy	17.05	18.31	16.30	17.66	11.25	22.52
Welfare gain (% Δ)	2.45	2.68	2.47	2.52	1.36	3.48

E.3 Robustness: Macro Effects of Rising Demand Investment

Table A8 shows that the insights from the exercise in Section 5 are similar under the alternative calibrations in Appendix E.1. Across specifications, we infer a decline in average contact costs: holding contact intensities $i(z)$ fixed, average demand investment $s(i(z))$ falls by 25–46%. We also infer a decline in the convexity of the contact cost technology: the P75/P25 ratio of $s(i(z))$, holding fixed $i(z)$, falls by 6–13%. The aggregate implications are similar as well. GDP rises by 3.6–6.0% and aggregate TFP by 2.1–4.4%; the effects are strongest under the higher-markup calibration because it implies greater dispersion in firm productivity (higher σ_z), so a given amount of reallocation toward high-productivity firms translates into larger TFP and GDP gains. Finally, results for the aggregate markup and in-

tangible capital intensity are similar across specifications: aggregate markups fall modestly, and true intangible intensity declines by roughly 10–14% across transitions, while capitalized intangible capital measures (with constant depreciation rates) increase by 6–13%.

Table A8: Robustness: Aggregate Quantitative Exercise

	<u>Baseline</u>	<u>Quitrate</u>	<u>Int. Margin Sh.</u>	<u>Agg. Markup</u>		
<i>Modified target:</i>		↑ 20%	↓ 15%	↑ 35%	↓ 15%	↑ 35%
Targeted moments, % Δ across steady states (1980s vs. 2010s)						
Agg. demand investment / Gross output	15.00	15.00	15.00	15.00	15.00	15.00
Top 1% sales share	25.80	25.80	25.80	25.80	25.80	25.80
Calibrated % in contact technology, for fixed $i(z)$						
Average demand investment	-31.03	-27.99	-40.33	-24.96	-45.83	-25.89
P75/25 demand investment	-8.28	-12.57	-5.77	-10.13	-8.27	-9.09
Untargeted moments, % Δ across steady states (1980s vs. 2010s)						
GDP	4.57	4.85	4.91	4.33	3.64	6.02
Aggregate TFP	3.11	3.52	3.15	3.12	2.08	4.44
Avg. customer quit rate	8.77	7.53	10.51	7.44	9.08	7.47
Aggregate markup	-0.36	-0.14	-0.58	-0.14	-0.59	-0.34
Untargeted moments, % Δ simulated firm panel across transition path						
Intangible capital intensity (true)	-12.45	-10.33	-14.19	-10.95	-12.89	-10.32
Intangible capital intensity (cap.)	8.47	12.84	5.98	9.93	9.19	10.00

F Numerical Model Solution

To ease exposition, we first describe how we solve for a steady state equilibrium, and then describe how we solve for (perfect-foresight) transitions.

F.1 Steady State Equilibrium

We discretize the productivity grid into $N_z = 100$ points $(z_j)_{j=1}^{N_z}$. The steps to solve for a steady state are as follows:

1. Compute r from the Euler equation (6): $r = \beta^{-1} - 1$.
2. Initialize guesses for aggregates $(Y, w, \bar{p})$, value functions $V(z), M(z), U$, and the customer distribution $G(z)$.
3. Jointly solve the FOCs for $p(z)$ and $i(z)$ by integrating the pricing ODE forward.
 - (a) At the lowest grid point, set $p(z_1) \equiv p_1 = \bar{p}$ and $i(z_1) \equiv i_1 = 0$.³³
 - (b) For all $j \geq 2$:
 - i. Given p_{j-1}, i_{j-1} , compute p_j via an explicit Euler step applied to (40). [Details below.]
 - ii. Given p_j and the current guesses for $G(\cdot)$ and V_j (hence W_j^*), compute i_j from (39) using the closed-form inverse of $s(i)$. [Set $i_j = 0$ if the marginal value of an additional contact, the right-hand side of (39), is non-positive.]
4. Given decisions $(p(z), i(z))$, update the value functions $V(z), M(z), U$ and the distribution $G(z)$.
 - (a) Find $\bar{z}$ such that $V(\bar{z}) = 0$.
 - (b) Given $\bar{z}$, update $\bar{p}$ so that $M(\bar{z}) = U$ when firm $\bar{z}$ sets price $\bar{p}$.
5. Update (Y, w) :
 - (a) Using last iteration's Y and the updated firm decisions and distribution, compute labor demand L^{demand} and the capital stock K ; physical investment is $\delta_K K$, aggregate demand investment is $\Gamma = \int s(i(z)) dG(z)$, so consumption is $C = Y - \delta_K K - \Gamma$.
 - (b) Given C , apply the household labor FOC (5) to obtain labor supply L^{supply} .

³³This assumes the grid satisfies $z_1 \leq \bar{z}$.

- (c) Update Y from labor-market clearing $L^{\text{demand}} = L^{\text{supply}}$.
- (d) Update w using the numeraire condition $P = 1$ and (8).

6. Repeat steps 3–5 until convergence.

Solving the pricing FOC forward. Given the boundary condition $p_1 = \bar{p}$, we integrate the ODE in (40) forward on the grid using an explicit (forward) Euler step (a left-endpoint Riemann approximation). Let $\Delta z_j \equiv z_j - z_{j-1}$. For each $j \geq 2$, given (p_{j-1}, i_{j-1}) , evaluate the slope of the price schedule using (40):

$$(F.1) \quad p'(z_{j-1}) = -\frac{2 + (u^i + G^i(z_{j-1})) i_{j-1}}{1 + (u^i + G^i(z_{j-1})) i_{j-1}} \cdot i_{j-1} g_{j-1}^i \cdot \frac{W_{j-1}^*}{\partial \pi(p_{j-1}, z_{j-1}) / \partial p},$$

and take a forward Euler step to evaluate a candidate price

$$p_j^{\text{cand}} = p_{j-1} + \Delta z_j p'(z_{j-1}).$$

In practice, it is helpful to use domain knowledge and enforce monotonicity and the static monopoly bound at each iteration:

$$p_j = \min\{p_j^{\text{cand}}, p^S(z_j), p_{j-1}\},$$

where $p^S(z) = \frac{\sigma}{\sigma-1} \frac{\overline{mc}}{z}$.

Note that with this procedure, $p(z) = \bar{p}$ is implied for all $z \leq \bar{z}$: since $i(z) = 0$ for $z < \bar{z}$ from step 3(b)ii above, the pricing slope in (F.1) evaluates to zero in that region.

Computing model moments. For equilibrium, it suffices to compute the cross-sectional customer distribution $G(z)$. To evaluate firm-level moments, we construct the joint distribution $H(z, n, \text{age})$ of firms by productivity, number of customers, and age. We obtain H by non-stochastic simulation on grids with 200 points for n and 200 points for age.

F.2 Transitional Dynamics

Conceptually, the transition algorithm mirrors the steady-state procedure but replaces the single joint loop with two nested loops: an outer loop for the path of aggregate objects and an inner loop that solves value functions backward in time and the distribution forward. Formally, we solve for a perfect-foresight transition from an *old* steady state at $t = 1$ to a *new* steady state at $t = T$ with a time-varying parameter vector $(\theta_t)_{t=1}^T$ as follows:

1. Initialize paths for all objects by linear interpolation between the old and new steady states, in particular aggregate objects $(r_t, w_t, Y_t, \bar{p}_t)_{t=2}^{T-1}$.
2. For each $t = T-1, T-2, \dots, 2$, iterate backward on the value functions $V_t(z)$, $M_t(z)$, U_t , starting from the new steady state values at $t = T$. Also compute $\bar{z}_t$ such that $V_t(\bar{z}_t) = 0$, and update $\bar{p}_t$ so that $M_t(\bar{z}_t) = U_t$ when firm $\bar{z}_t$ sets price $\bar{p}_t$. Within each t , solve firms' decision rules exactly as in the steady state: integrate the pricing ODE forward along the z -grid via an explicit Euler step and obtain $i_t(z)$ pointwise from (39).
3. For each $t = 2, 3, \dots, T-1$, iterate forward on the distribution $G_t(z)$, starting from $G_1(z) = G^{\text{old}}(z)$, and compute the implied aggregates L_t^{demand} and K_t^{demand} . Set $K_t = K_t^{\text{demand}}$ if $t \geq 3$. [In $t = 2$, $K_t = K^{\text{old}}$.]
4. For each $t = 2, 3, \dots, T-1$, use the household problem to update (r_t, w_t, Y_t) :
 - (a) Updated firm decisions and the distribution imply aggregate demand investment Γ_t , physical investment $Inv_t = K_{t+1} - (1 - \delta_K)K_t$, and consumption $C_t = Y_t - Inv_t - \Gamma_t$.
 - (b) The labor supply condition implies L_t^{supply} given C_t and w_t .
 - (c) Update Y_t from labor-market clearing using L_t^{supply} and L_t^{demand} .
 - (d) Update w_t using the numeraire condition $P_t = 1$ and (8).
 - (e) If $t = 2$, update r_t from capital-market clearing using $K_t^{\text{supply}} = K^{\text{old}}$ and K_t^{demand} .
 - (f) If $t \geq 3$, update r_t from the Euler equation (6) given C_{t-1} and C_t .
5. Repeat steps 2–4 until convergence.

As usual, choose T sufficiently large so that the economy has converged to the new steady state before the terminal date; in practice, set θ_t constant for $t \geq \tau$ with $\tau \ll T$.

F.3 Social Planner

In steady state, we solve for the planner's match value $V^*(z)$ and optimal contact intensity $i^*(z)$ via value function iteration inside a joint fixed-point loop. At each iteration, $i^*(z)$ follows from the planner's optimality condition (43), where the marginal benefit of an additional contact for firm z includes not only the direct expected gain $e^*(z)W^*(z)$ but also externality corrections that capture how firm z 's investment affects other firms' quit rates and conversion rates (business stealing). The planner's match values satisfy (42) and (41).

Given $(V^*(z), i^*(z))$, we update the distribution $g^*(z)$ as in the decentralized equilibrium (entry, separation, productivity transitions, and rematching), which yields aggregate productivity Z^* and aggregate demand investment Γ^* . Given (Z^*, Γ^*) , we update labor from the intratemporal FOC and capital from the steady-state Euler equation and resource constraint. We iterate on $(V^*(z), i^*(z), g^*(z), Z^*, \Gamma^*, K^*, L^*)$ until convergence.

For transition paths, given (Z_t^*, Γ_t^*) the remainder is a standard growth problem. We take $t = 1$ as the market steady state and let the planner take over at $t = 2$ with predetermined capital $K_2 = K^M$, so consumption can jump at $t = 2$. We compute transition paths with a one-dimensional shooting method on the initial consumption jump C_2 , choosing C_2 so that terminal capital satisfies $K_T = K^*$. For any candidate C_2 , we initialize paths for (K_t, C_t, L_t, Z_t) between the market steady state and the planner steady state, then iterate a backward-forward loop: backward in time for $(V_t^*(z), i_t^*(z))$ using the planner Bellman system (with time-varying continuation values), and forward in time for the customer distribution and aggregates. Given the updated distribution, we recompute Z_t and then update (K_t, C_t, L_t) using the Euler equation, the resource constraint, and the intratemporal labor FOC, and repeat until the inner loop converges. We then adjust C_2 via bracketing/bisection until the terminal condition holds.